FINAL MANUSCRIPT

Adaptive Multiscale Singular Packet Decomposition

Exact finite-tree optimization, local rank packets, and residual closure

A geometry-aware image representation with explicit finite optimality domains. The paper separates approximation, optimization, and exact residual accounting, then tests each layer against deterministic and randomized verification suites.

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Abstract

Adaptive Multiscale Singular Packet Decomposition (AM-SPD) combines geometry-dependent supports, local low-rank models, exact finite-tree budget optimization, and a residual archive. A generative packet atlas proposes supports from segmentations, contours, dyadic cells, and residual-driven refinements. On each support, truncated singular-value models select orientation and rank. An integer-budget dynamic program returns the minimum additive squared error for every budget on a declared finite tree. The retained directions are then refitted globally, which cannot increase least-squares error. A finite portfolio of greedy, Lagrange-tree, and exact-budget seeds, together with backward replacement candidates, yields nonregression over the completed candidates. The identity $F = \widehat{F} + R_{\text{archive}}$ preserves exact accounting without conflating lossless closure with sparse modeling. Four principal images and a ten-image stress suite show strong gains in several structured and natural-image cases, but no universal advantage over global SVD. A 36-scenario noise study likewise shows that the deterministic objective is not automatically a denoiser. The implementation suite evaluates 12,154 cases, including 9,960 exhaustive checks of the tree recursion. All optimality claims are restricted to the declared supports, candidate families, budgets, costs, and objective.

Keywords. adaptive image representation, packet decomposition, local low rank, singular-value decomposition, best-basis dynamic programming, residual archive, exact finite-raster reconstruction.

Scope. "Exact" refers to explicitly stated finite optimization or reconstruction domains. It does not mean that every image has a short packet description, that the parameter proxy is a bit rate, or that an unknown continuous boundary is recovered without identifiability assumptions.

Overview and contributions

AM-SPD sits between a fixed transform and an unconstrained learned model. Its spatial supports are generated adaptively, but each accepted finite model remains inspectable. The final representation contains a geometric packet core, supported local singular subspaces, a finite portfolio selector, and an explicit archive for everything not represented by the current model.

The principal contributions are:

1. a countable generative atlas whose finite active frontier is driven by the current residual
2. supported local singular packets with orientation, rank, cost, and genealogy
3. an exact integer-budget dynamic program for every pruning of a declared finite tree
4. global refitting and orthogonal block-gain identities for the selected span
5. finite-portfolio nonregression through retained seeds and backward budget reopening
6. exact residual accounting and a complete pixel-depth finite-raster fallback

The method uses established components in a different assembly. Global low-rank optimality follows the classical SVD theory [1]. Finite tree selection is related to best-basis dynamic programming [2]. The moving packet stage is closer to orthogonal matching pursuit with a generated dictionary [3]. Its boundary motivation is related to Gibbs reconstruction, finite-rate-of-innovation methods, and off-grid recovery [4, 5, 6, 7, 8]. The new claims concern the stated packet construction, budget recursion, portfolio logic, and residual contract rather than a replacement of those theories.

Notation

The image is $F \in [0,1]^{H\times W\times C}$. Spatial atoms are $p_k \in \mathbb{R}^{HW}$, the active span is V_K, the globally refitted approximation is $\widehat{F}_K = P_{V_K} F$, and the residual is $r_K = F - \widehat{F}_K$. A tree node ν has support Ω_ν, local rank

r_ν, orthonormal directions Q_ν, and declared scalar cost C_ν. The residual archive is denoted R_{archive}.

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1 Problem setting and claim boundaries

1.1 From a known edge to an unknown image

The earlier Gibbs construction begins with a decomposition

$$f = s + j,$$

where the smooth packet s is reconstructed spectrally and the discontinuous packet j is synthesized directly. When j is known, the persistent Gibbs term disappears because the jump is no longer approximated by a finite global trigonometric polynomial.

A full image changes the problem. There may be hundreds or thousands of interacting boundaries, textures, gradients, shadows, thin curves, disconnected components, and partially transparent transitions. A fixed finite dictionary cannot contain every possible object. Even a very large dictionary is fragile when the correct packet lies between its entries.

A useful extension is not one universal precomputed dictionary. It is a *generative atlas* whose packets move toward the residual geometry.

The word "infinite" does not mean that infinitely many arrays are stored. It means that a finite grammar can generate a countably infinite refinement tree, while the algorithm evaluates only a finite active frontier at every step.

1.2 Approximation, accounting, and finite closure

The reconstruction has three logically different layers.

1. **Sparse packet approximation.** A finite set of geometric and smooth packets approximates the image.
2. **Exact residual accounting.** The image equals the approximation plus the stored residual at every finite stage.
3. **Exact finite closure.** Pixel-depth dyadic packets span the entire finite raster space.

The first claim is about efficiency and interpretation. The second is an exact identity. The third is a completeness theorem. Confusing them would make the method appear stronger than it is.

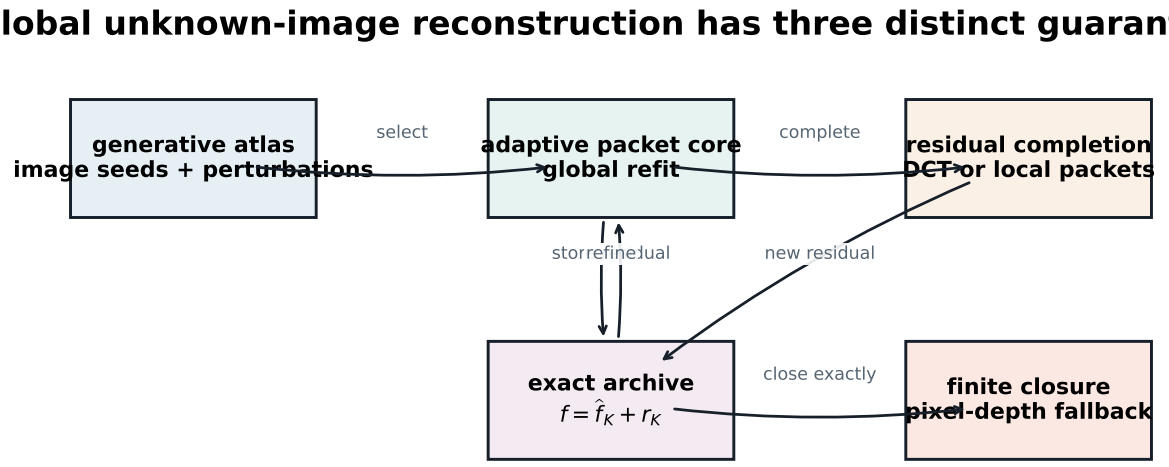


Figure 1: The three reconstruction guarantees are kept separate.

Reading the figure. The geometric core is the compressed interpretable model. The exact archive preserves all information that has not yet entered that model. Pixel-depth refinement is a guaranteed fallback rather than a claim that every image has a short semantic description.

1.3 Boundary-aware assembly

For an image with a jump set Σ, a finite global Fourier approximation uses one smooth basis to represent the fields on both sides of the jump and the discontinuity itself. Oscillatory boundary layers are therefore expected near Σ [4]. Filtering can reduce those oscillations, but it also broadens or biases the transition.

A packet reconstruction instead writes

$$f = \sum_m a_m \mathbf{1}_{\Omega_m} + g,$$

where the geometry packets carry selected discontinuities and g contains the remaining smooth or textured content. When a boundary packet is exact, the jump need not be represented by the smooth residual basis.

Theorem 1.1 (Boundary-preserving assembly on a finite raster)**.** *Let $\Omega_1, \dots, \Omega_M$ be a disjoint partition of a finite raster Ω. For each m, let $A_{m,N}$ act only on the coordinates in Ω_m. Define*

$$R_N f = \sum_{m=1}^{M} \mathbf{1}_{\Omega_m} A_{m,N}(f|_{\Omega_m}).$$

Then

$$\|R_N f - f\|_{\ell_\infty(\Omega)} = \max_m \|A_{m,N}(f|_{\Omega_m}) - f|_{\Omega_m}\|_{\ell_\infty(\Omega_m)}.$$

Thus the assembly step introduces no additional cross-region overshoot. Any remaining error is the local approximation error or an error in the selected partition.

Proof. At every raster coordinate exactly one indicator $\mathbf{1}_{\Omega_m}$ is nonzero. The global error therefore equals the corresponding local error coordinate by coordinate. Taking the maximum gives the identity. □

2 Generative packet atlas

2.1 Seed packets extracted from the image

The atlas begins with a finite collection of imperfect proposals. These are not assumed to be correct objects. They are starting coordinates.

The implementation extracts seeds from:

- Felzenszwalb–Huttenlocher segmentations at three scales [10]
- SLIC superpixels at three resolutions [11]
- connected components above and below channel quantiles
- filled contours derived from Canny edges [9]
- low-frequency DCT atoms
- dyadic quadtree cells
- residual components created during the search

Different seeds fail differently. Superpixels follow local color. Threshold components follow intensity. Contours follow gradient closure. Quadtree cells make no semantic assumption. Their union is a redundant proposal atlas rather than one privileged segmentation.

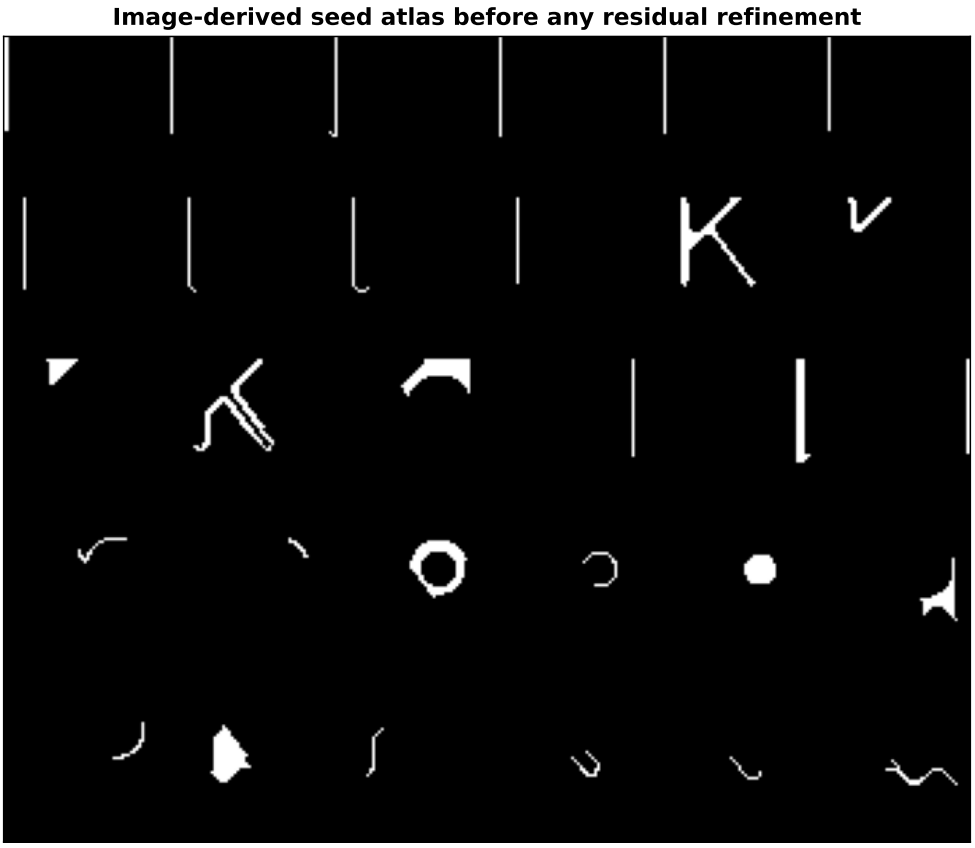


Figure 2: Image-derived seed masks before residual refinement.

Reading the figure. The seeds are intentionally overlapping and imperfect. Their role is to place the generator near useful parts of geometry. The later perturbation and composition stages are what prevent the finite seed set from becoming a hard dictionary boundary.

2.2 Countable generative grammar

Let $\mathscr{P}_0$ denote the finite seed set. At depth d, define a finite family of transforms $\mathcal{T}_d$ containing:

translations of size 2^{-d},
rotations of size 2^{-d},
log-scale changes of size 2^{-d},
radial boundary modes with dyadic amplitudes and phases,
soft signed-distance boundaries,
morphological and local texture refinements,
binary unions, intersections, differences, and symmetric differences.

The generated atlas is

$$\boxed{\mathscr{P}_\infty = \bigcup_{d\geq 0} \mathcal{T}_d \circ \mathcal{T}_{d-1} \circ \cdots \circ \mathcal{T}_1(\mathscr{P}_0).}$$

Every level is finite, so $\mathscr{P}_\infty$ is countable. It is never stored in full.

Proposition 2.1 (Dyadic density of the represented transform parameters)**.** *Let $T_\theta p$ be a packet family whose action is continuous in L^2 for θ in a bounded translation, rotation, or log-scale interval. The dyadic*

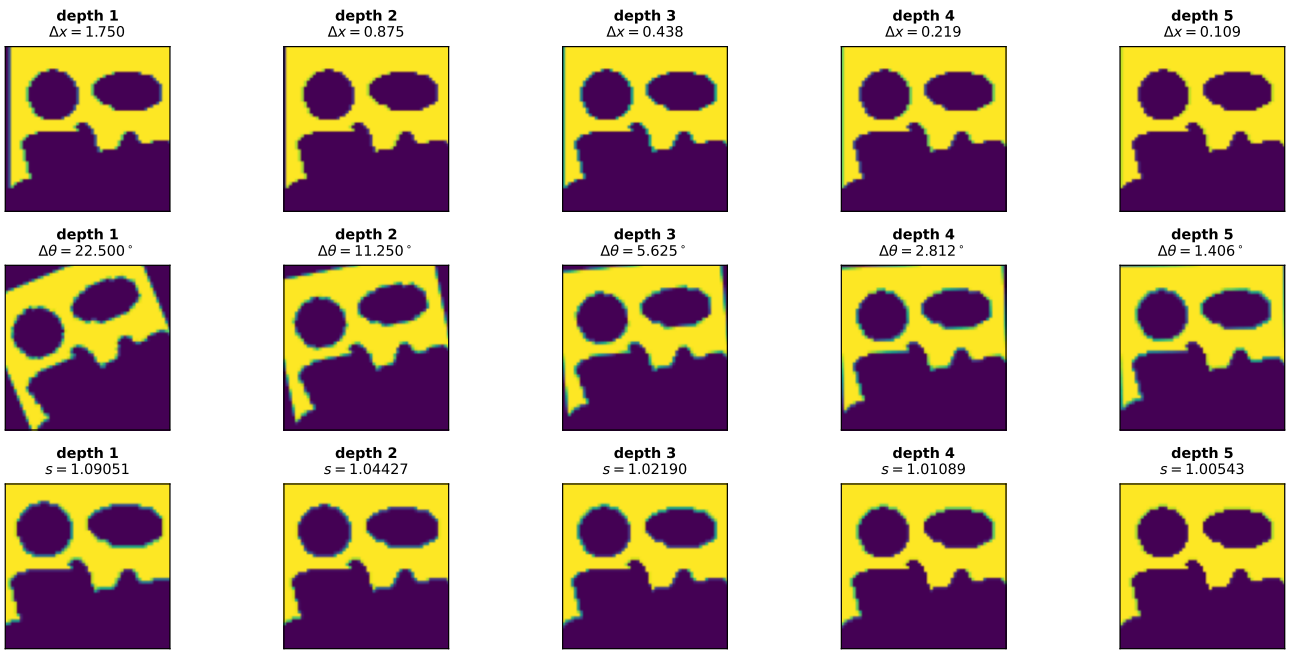


Figure 3: Dyadic perturbations shrink along a lineage.

Reading the figure. The displayed translation, rotation, and scale steps halve with depth. Repeated selection does not cycle through one fixed menu. It opens progressively finer neighborhoods around packets that continue to explain the residual.

parameter schedules are dense in that interval, and the generated samples $T_\theta p$ are dense in the corresponding continuous transformation orbit.

Proof. Dyadic rationals are dense in every bounded real interval. Finite signed sums of increments proportional to 2^{-d} approximate any bounded target parameter. Continuity of the transform action transfers parameter convergence to L^2 convergence of the packet. □

Remark 2.2. The density statement concerns continuously parameterized packets before discontinuous raster thresholding. A hard binary mask on a fixed finite raster has only finitely many distinct arrays, so its rasterized orbit is not a nontrivial continuous manifold.

2.3 Unary geometric refinements

A selected mask can generate:

- translated, rotated, and scaled affine children
- dilated, eroded, and ribbon children
- radial Fourier boundary warps
- signed-distance soft masks at decreasing transition widths
- masked polynomial coordinate fields
- masked local DCT atoms

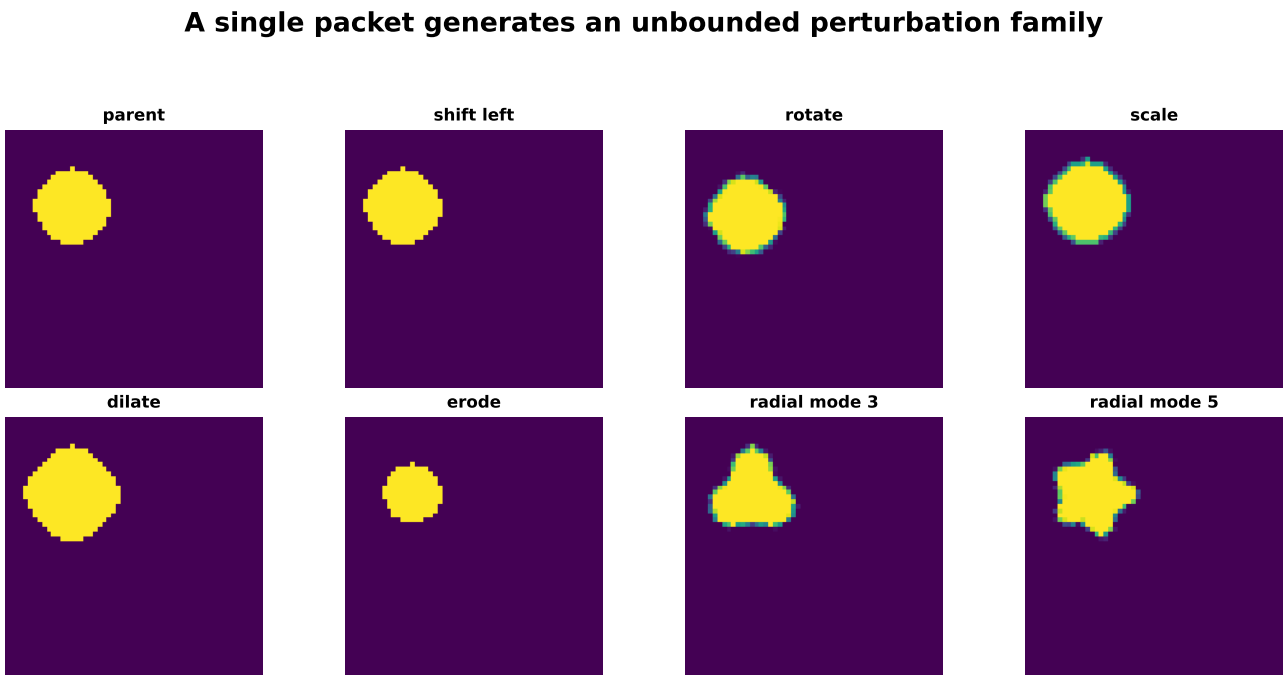


Figure 4: One seed packet generates many geometric descendants.

Reading the figure. The first family already includes motion, scale, morphology, and oscillatory boundary modes. The refined implementation continues these transformations at dyadically decreasing step sizes rather than stopping at the pictured examples.

These children separate two tasks that are often conflated. Boundary perturbations change the geometry. Masked polynomial and cosine atoms change the field inside an existing geometry.

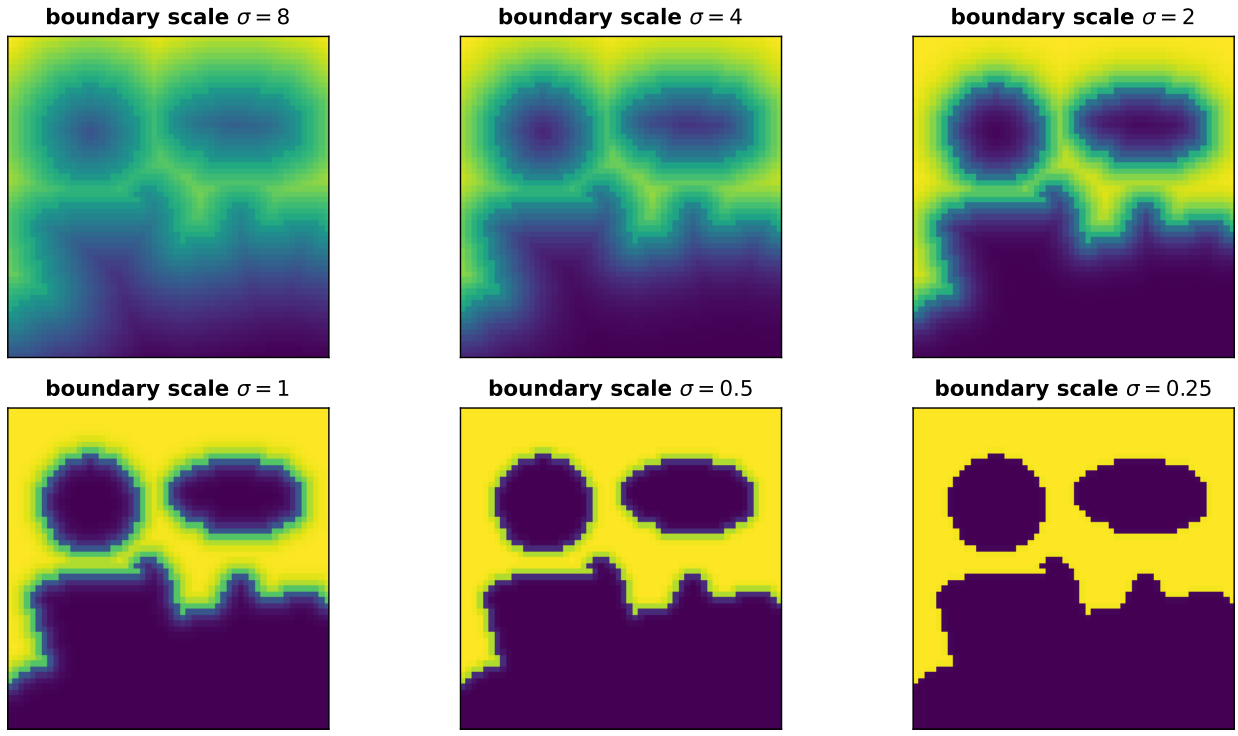


Figure 5: Soft boundaries approach hard packets through decreasing transition widths.
Reading the figure. Soft masks allow subpixel movement and differentiable boundary adjustment. The hard packet is recovered as the transition width tends toward zero on a fixed raster.

2.4 Binary packet composition

Objects are not always perturbations of one connected seed. Text, articulated shapes, repeated motifs, and separated regions may be built from several local pieces. The grammar therefore includes

$$P \cup Q, \qquad P \cap Q, \qquad P \setminus Q, \qquad P \triangle Q.$$

Each combined packet can itself become a new parent and undergo further geometric and field refinement. This creates a compositional tree rather than a list of isolated templates.

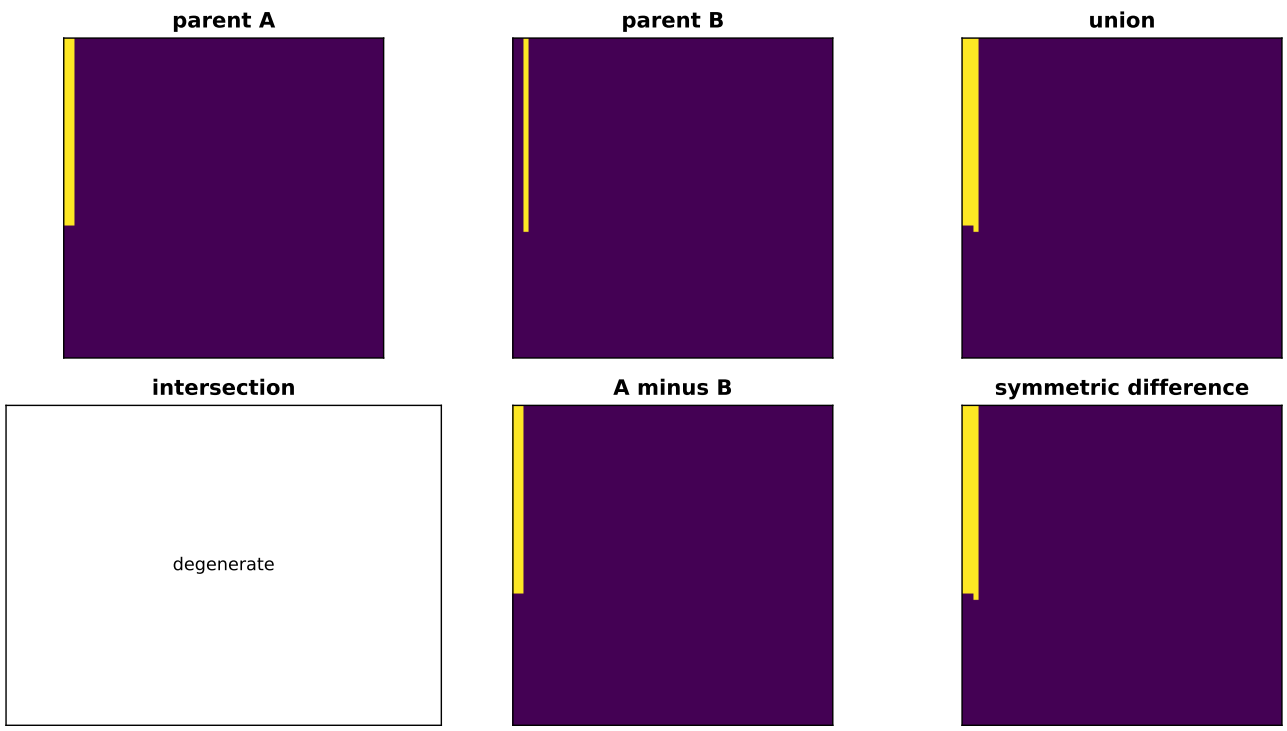


Figure 6: Binary packet composition.

Reading the figure. Two local proposals can be joined, intersected, subtracted, or symmetrically differenced. The resulting packet is eligible for the same endless unary perturbation grammar as every other selected object.

2.5 Residual seeds and the moving atlas

The atlas is not generated independently of the reconstruction. At scheduled iterations the residual magnitude is thresholded at several quantiles. Connected residual components become new seed masks. High-energy quadtree cells are also exposed and selectively subdivided.

Thus the atlas moves according to

$$\boxed{\mathscr{P}_{K+1} = \mathscr{P}_K \cup \mathrm{Descendants}(P_K) \cup \mathrm{ResidualSeeds}(r_K) \cup \mathrm{DyadicRefinement}(r_K).}$$

The packet generator and reconstruction are coupled. The current error determines where the coordinate system becomes more detailed.

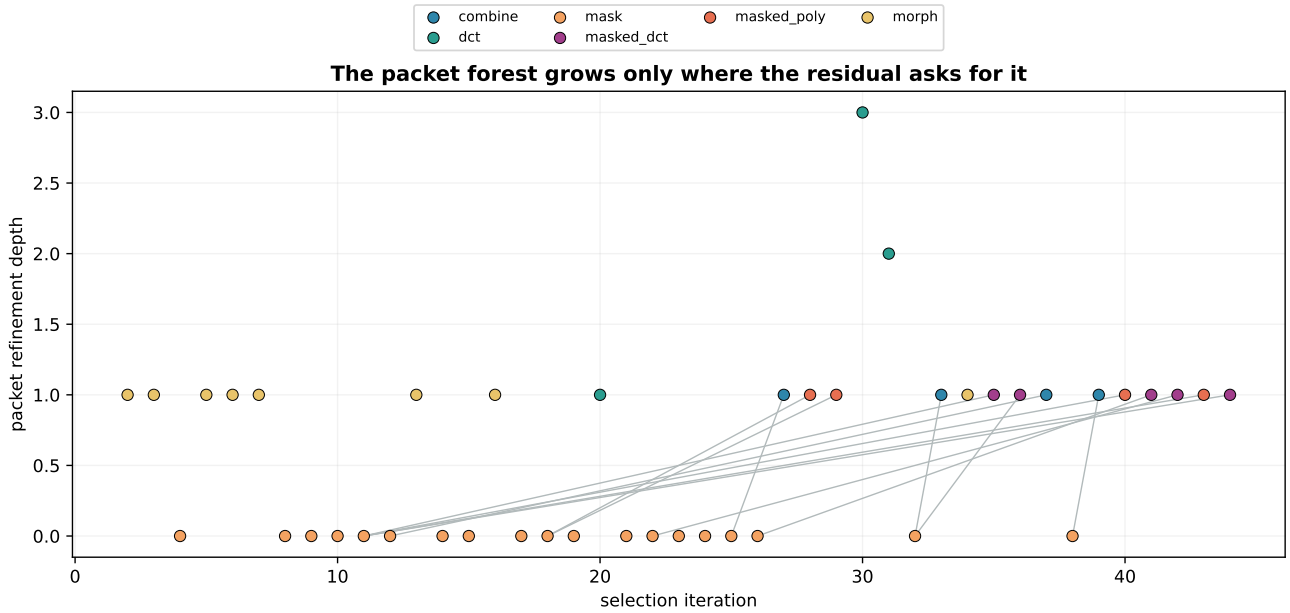


Figure 7: A selected packet genealogy.

Reading the figure. The search is not a flat ranking of a static dictionary. Children exist because a parent was useful. New residual components can also enter as roots, which lets the atlas recover from poor initial proposals.

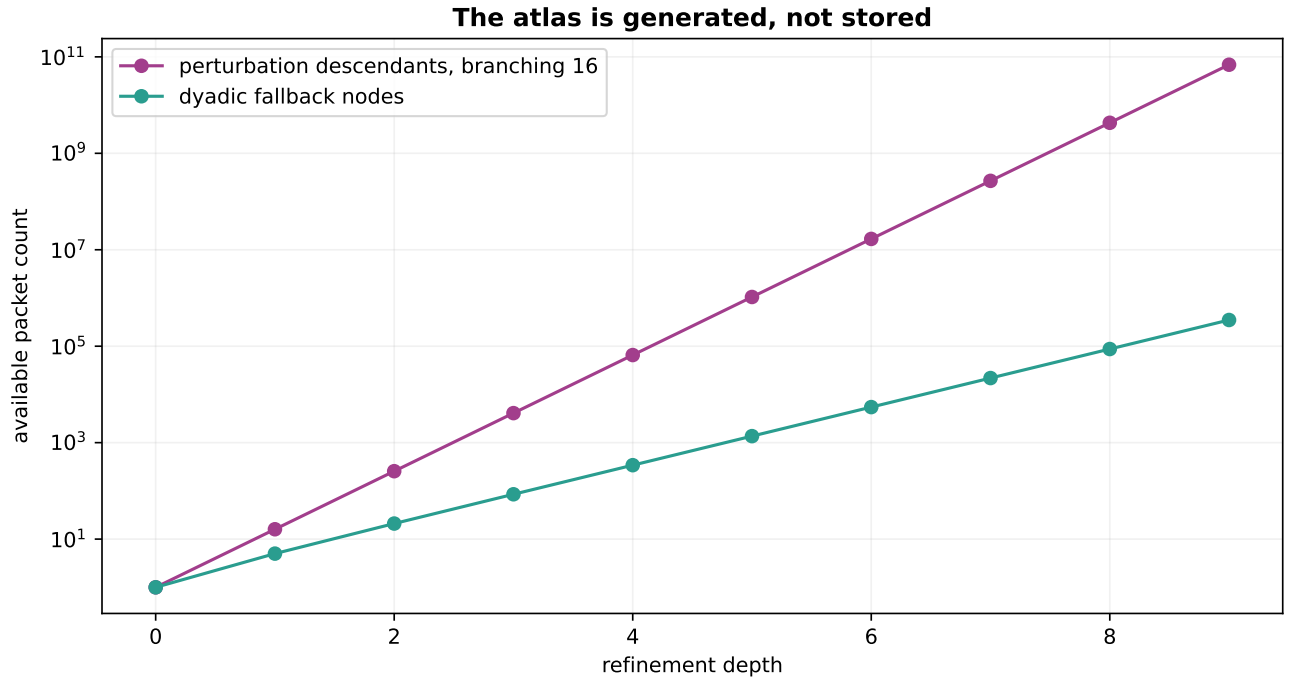


Figure 8: Potential family size under repeated refinement.

Reading the figure. The number of possible descendants grows without requiring their storage. The algorithm sees only a bounded frontier, while the fallback dyadic tree guarantees that unexplained regions remain reachable.

3 Global synthesis and exact accounting

3.1 Shared spatial packets with multichannel coefficients

A spatial atom $p \in \mathbb{R}^{HW}$ is shared across color channels. Its coefficient is a vector

$$\gamma_p \in \mathbb{R}^C.$$

The reconstruction from active atoms $p_1, \ldots, p_K$ is

$$\widehat{f}_K(x) = \sum_{k=1}^{K} p_k(x)\gamma_k.$$

This separates object support from color or feature values. One mask can represent a colored region without requiring one duplicate geometry packet per channel.

3.2 Global least-squares refitting

Let

$$A_K = [p_1 \ \cdots \ p_K]$$

and let $F \in \mathbb{R}^{HW \times C}$ be the flattened image. The active coefficients are

$$\Gamma_K = \arg\min_{\Gamma} \|F - A_K\Gamma\|_F^2 .$$

The mathematical statements use the orthogonal least-squares projector. The recovered reference implementation solves a Gram system with a ridge of 10^{-9} for conditioning and falls back to `lstsq`. The exact identities below therefore describe the ideal projection, while the floating-point implementation is checked to numerical tolerance.

Theorem 3.1 (Monotone residual under nested global refitting)**.** *If*

$$V_K = \operatorname{span}\{p_1, \ldots, p_K\} \subseteq V_{K+1},$$

and $\widehat{F}_K$ is the orthogonal least-squares projection of F onto $V_K \otimes \mathbb{R}^C$, then

$$\left\|F - \widehat{F}_{K+1}\right\|_F \leq \left\|F - \widehat{F}_K\right\|_F .$$

Proof. The minimization domain at stage K is contained in the domain at stage $K+1$. The minimum over the larger set cannot be larger. □

The measured residual sequences were nonincreasing to numerical precision in every tested reconstruction.

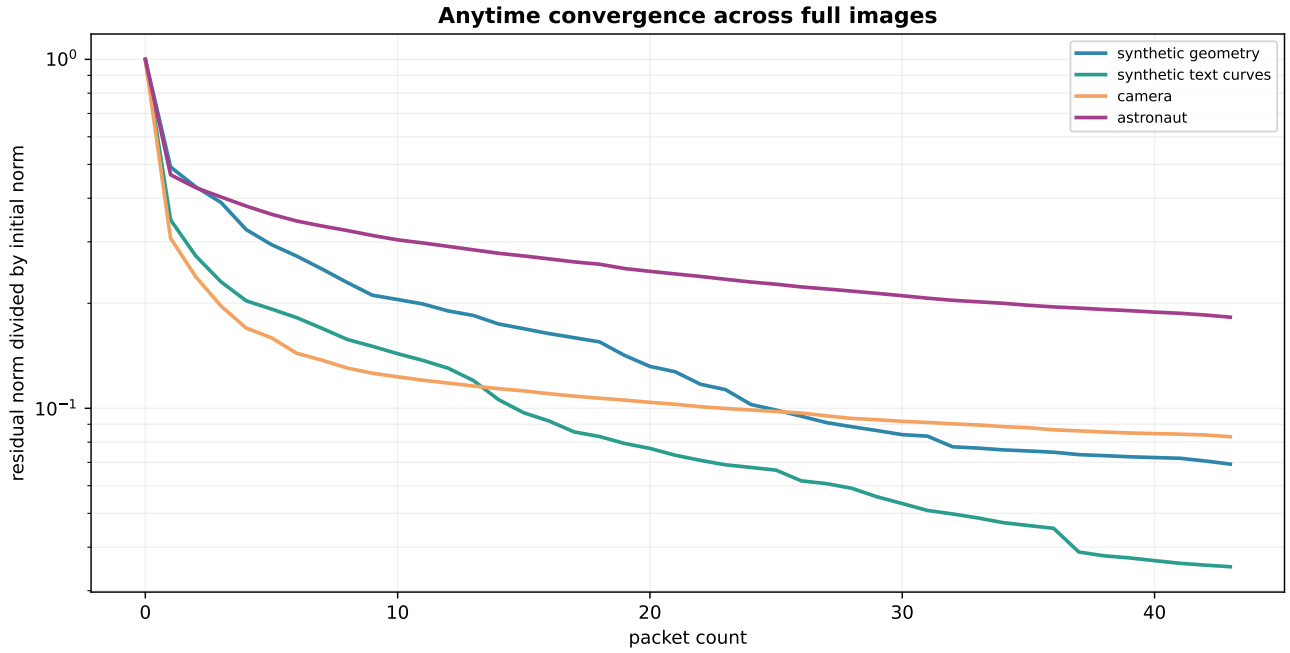


Figure 9: Residual norm during global packet selection.

Reading the figure. Every accepted packet enlarges the active span. All coefficients are refitted together, so earlier packets can change their amplitudes when a better explanation enters.

3.3 Orthogonal candidate gain

Let r_K be the current multichannel residual and let p be a candidate. Orthogonalize the candidate against the active span:

$$p_\perp = (I - P_{V_K})p.$$

When $p_\perp \neq 0$, the exact reduction in squared residual is

$$\boxed{G(p \mid V_K) = \frac{\left\| p_\perp^\mathsf{T} r_K \right\|_2^2}{\|p_\perp\|_2^2}.}$$

The score used by the implementation is

$$\text{score}(p) = \frac{G(p \mid V_K)}{1 + \lambda\, \text{complexity}(p)}.$$

For an exact least-squares residual, this is the exact global projection gain after removing the represented part of the candidate. With the reference ridge solve, it is evaluated as a numerically accurate approximation and is checked by the verification suite.

3.4 Lazy finite-frontier search

At each iteration the candidate pool may be much larger than the evaluation budget. The implementation retains a deterministic shallow frontier and samples from the deeper frontier. This keeps the search moving while preserving access to recently generated fine descendants.

Algorithm 3.2 (Lazy generative packet reconstruction)**.** Given an image f, a maximum selected-packet count K, and a finite candidate-evaluation budget:

1. Generate the image-derived seed atlas and a constant packet.

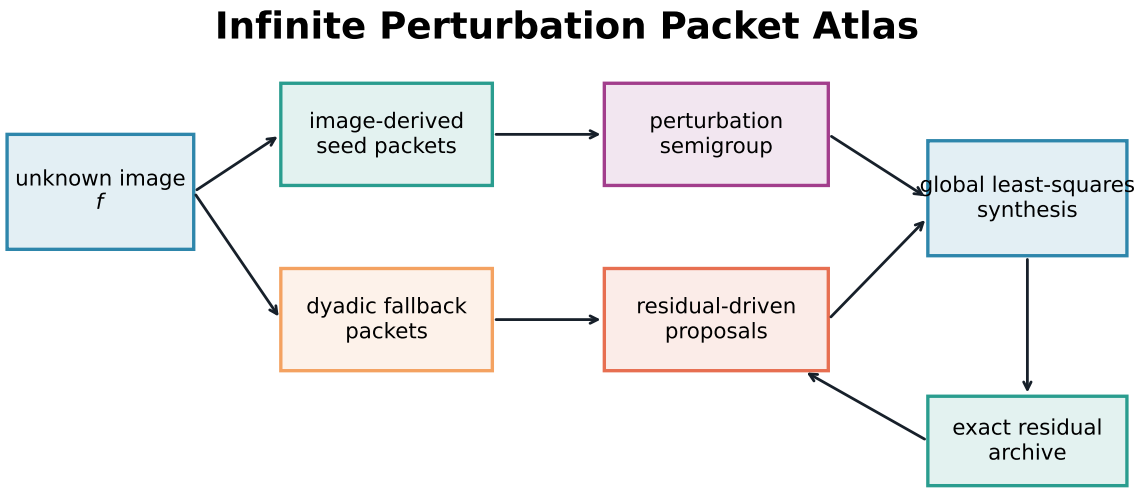


Figure 10: Architecture of the infinite perturbation packet atlas.
Reading the figure. Image-derived seeds and residual proposals feed a lazy generator. The selected packet forest is globally refitted. The residual is both an exact archive and the signal that decides where new coordinates should appear.

2. Fit the active coefficients globally.
3. Generate residual seeds and high-energy quadtree packets on schedule.
4. Evaluate a finite frontier using the exact orthogonal gain divided by packet complexity.
5. Select the highest-scoring nonduplicate atom.
6. Globally refit every active coefficient.
7. Generate dyadically finer descendants and compositional children of the selected packet.
8. Store the exact residual.
9. Repeat until the packet budget or residual tolerance is reached.
10. Optionally complete the residual using sparse DCT packets, local refinement, or pixel-depth packets.

3.5 Residual accounting identity

At every finite step define

$$r_K = f - \widehat{f}_K.$$

Then

$$\boxed{f = \widehat{f}_K + r_K.}$$

The identity is elementary, but it prevents an important category error. Approximation never destroys the omitted information. Compression, interpretation, and exact storage remain distinct operations.

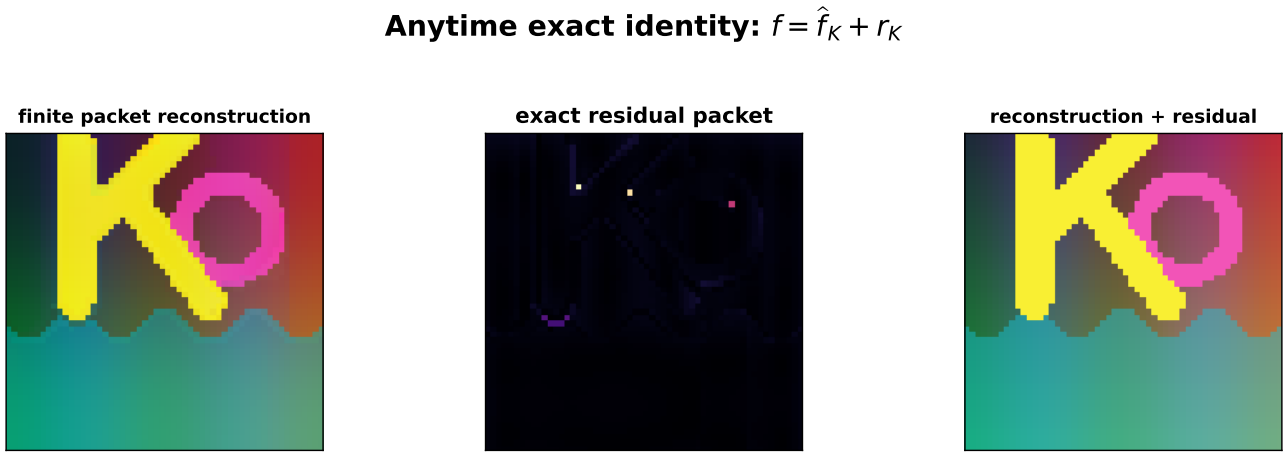


Figure 11: The exact archive identity.

Reading the figure. The reconstruction is the current model. The residual is not discarded. It is an exact complementary packet that can be stored, transmitted, further modeled, or used to generate the next local geometry proposals.

3.6 Residual completion

The benchmark uses a second stage

$$\widehat{f}_{K,L} = \Pi_{[0,1]} \left(\widehat{f}_K + D_L r_K \right),$$

where $D_L r_K$ is the best L-coefficient DCT component of the residual and $\Pi_{[0,1]}$ clips to the physical image range.

Proposition 3.3 (Orthogonal residual completion)**.** *If D_L is an orthogonal projector onto an L-dimensional residual subspace, then*

$$\|r_K - D_L r_K\|_2 \leq \|r_K\|_2 .$$

Proposition 3.4 (Bounded synthesis is nonexpansive)**.** *For every truth image $f \in [0,1]^{H \times W \times C}$,*

$$\left\| \Pi_{[0,1]}(g) - f \right\|_2 \leq \|g - f\|_2 .$$

Proof. Coordinatewise projection onto a closed interval is the metric projection onto a closed convex set. Metric projections in Euclidean space are nonexpansive relative to every point in the set. □

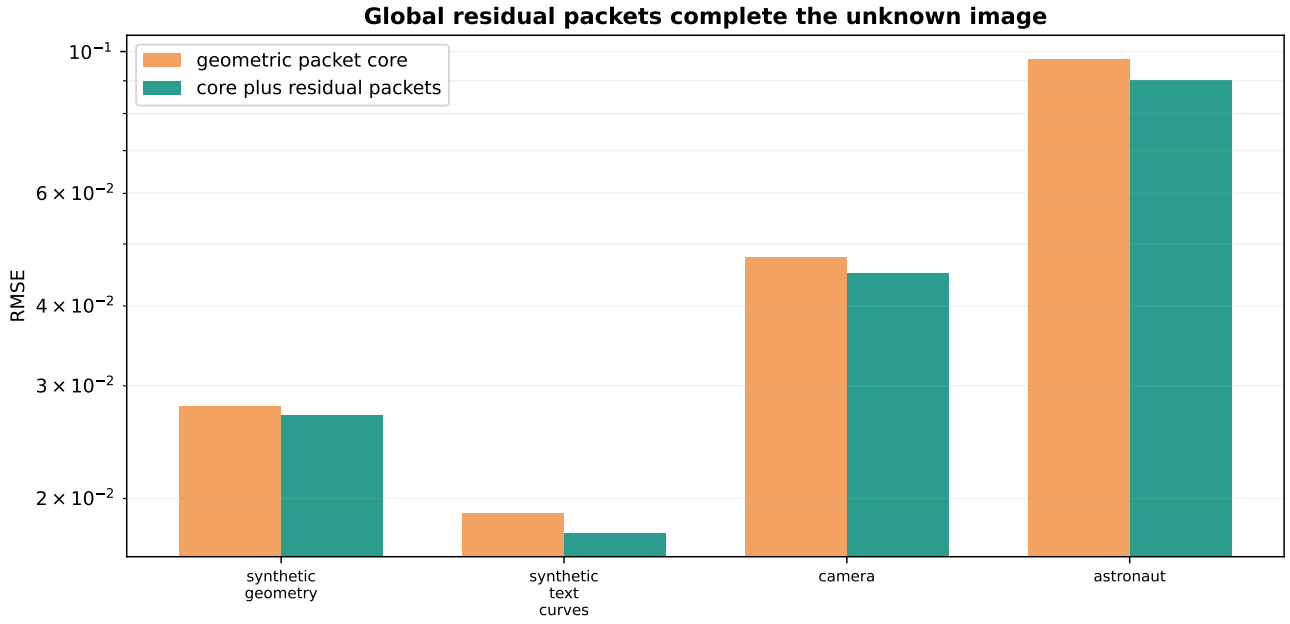


Figure 12: Geometric core and residual-completed reconstruction.

Reading the figure. The geometric forest explains boundaries and regions. The residual DCT stage captures remaining global smooth or textured structure. The two layers remain separately inspectable.

3.7 Native edge synthesis

A step packet reconstructed from a constant atom and its exact region mask has no representation-induced overshoot. The verification suite reconstructs a two-dimensional step to a maximum error of order 10^{-15}.

Theorem 3.5 (Exact native packet synthesis)**.** *Suppose*

$$f = g + a\mathbf{1}_\Omega$$

with an exact packet $\mathbf{1}_\Omega$ in the active span and g in the selected smooth span. Then the least-squares reconstruction equals f exactly. In particular, it introduces no Gibbs overshoot along $\partial\Omega$.

3.8 Error under approximate geometry

Let

$$j = a\mathbf{1}_\Omega, \qquad \widehat{j} = \widehat{a}\mathbf{1}_{\widehat{\Omega}}.$$

For $1 \le p < \infty$,

$$\boxed{\left\| j - \widehat{j} \right\|_{L^p} \le |a - \widehat{a}|\, \mu(\Omega)^{1/p} + |\widehat{a}|\, \mu(\Omega \triangle \widehat{\Omega})^{1/p}.}$$

Proof. Add and subtract $\widehat{a}\mathbf{1}_\Omega$. The amplitude term is supported on Ω. The geometry term has magnitude $|\widehat{a}|$ on the symmetric difference. □

This inequality states the honest boundary of unknown-packet reconstruction. If the geometry is not exact, the remaining boundary error is controlled by amplitude error and symmetric-difference area. It is not renamed as an exact solution.

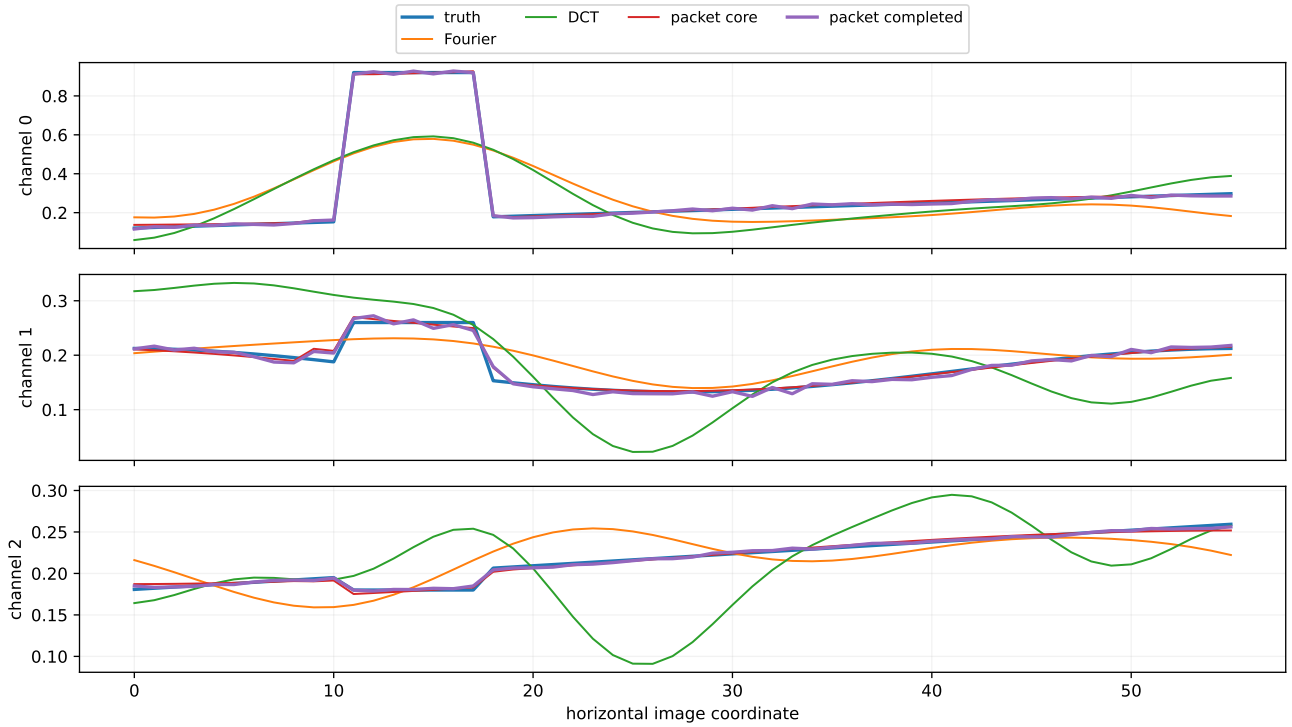


Figure 13: A boundary cross-section through global and packet reconstructions.

Reading the figure. The global spectral approximation spreads its error around the discontinuity. A native packet keeps the discontinuity in the geometry coordinate. Approximate packet boundaries produce localized geometry residuals rather than a universal oscillatory tail.

3.9 Finite-raster closure

For a finite $H \times W$ raster, let e_{yx} be the indicator of pixel (y, x). Then

$$\{e_{yx} : 0 \le y < H,\ 0 \le x < W\}$$

is the standard basis of $\mathbb{R}^{HW}$. With one coefficient vector per pixel it spans $\mathbb{R}^{HW \times C}$.

Theorem 3.6 (Exact finite-image closure)**.** *Every finite image can be reconstructed exactly from pixel-depth spatial packets and one coefficient vector per color channel.*

Proof. For every pixel, select its standard indicator and use the image value at that pixel as the channel coefficient. Summing the packets returns every coordinate exactly. □

The implementation reaches these packets through dyadic quadtree refinement or by direct residual completion. This proves completeness, not semantic sparsity.

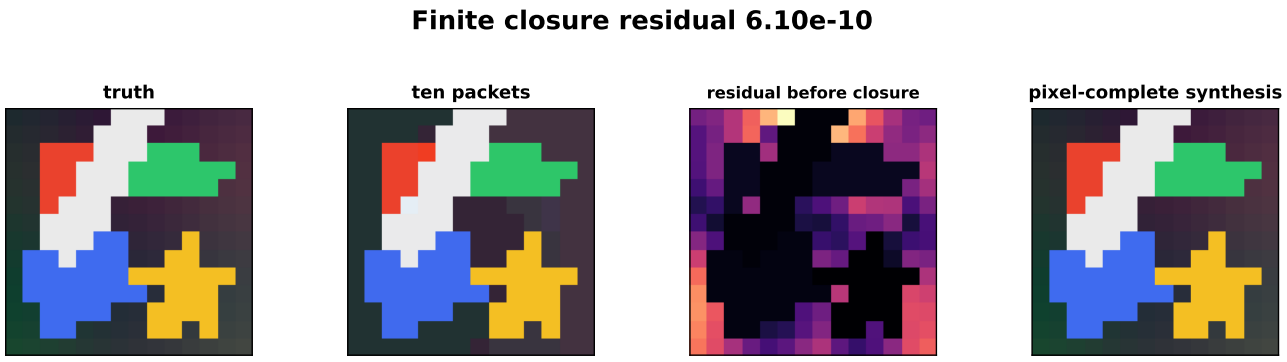


Figure 14: Pixel-depth completion on a small image.

Reading the figure. The first ten packets leave a visible residual. The coordinate fallback closes that residual to numerical solve tolerance. This is the finite exact endpoint of the atlas.

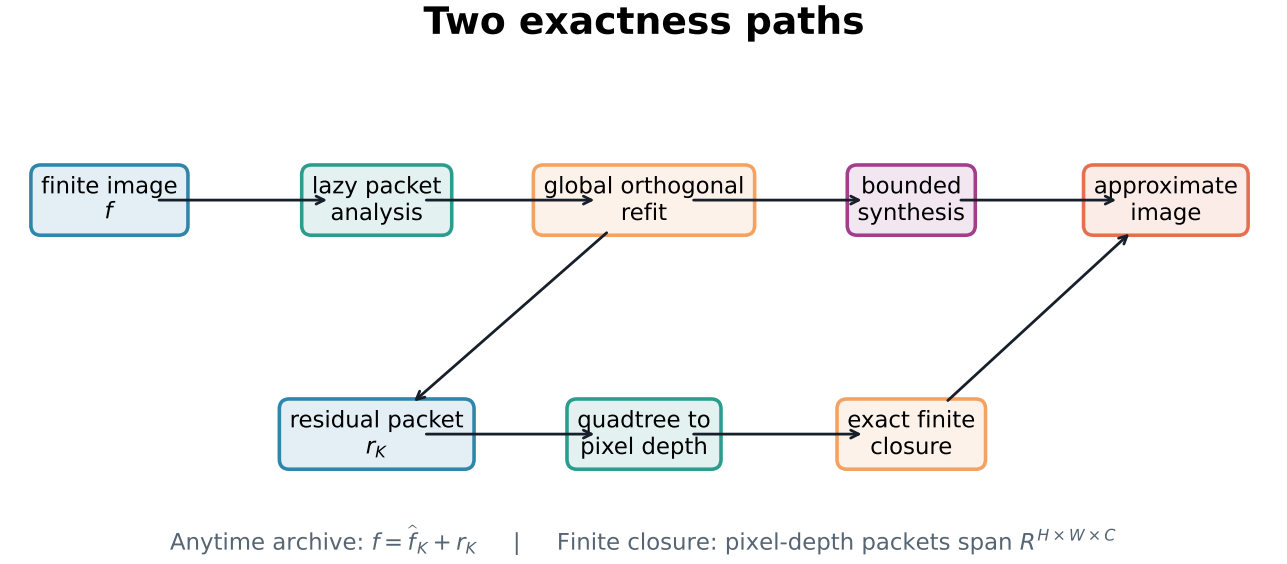


Figure 15: Two exactness paths.

Reading the figure. The archive path is exact at every stage because it stores the residual. The synthesis path becomes exact after the packet system reaches a complete finite basis.

4 Singular packet representation

4.1 Relation to global SVD

The resemblance is real, but the two constructions begin at different places.

A global image SVD starts from one fixed matrix representation of the image and finds the best approximation of a chosen rank. The perturbation atlas starts from a generative family of local geometric and smooth packets. It discovers a useful span before asking for canonical coordinates inside that span.

The original packet search is therefore closer to orthogonal matching pursuit with a moving dictionary than to SVD [3]. Singular Packet Decomposition enters after packet discovery and again during residual refinement.

The concise answer is: the moving atlas is not itself SVD. Once a packet span has been selected, its canonical orthogonal form is exactly an SVD. Local SVD modes can then be fed back into the moving atlas as new packets.

4.2 Canonical singular packets of a selected atlas

Let

$$P_K = \begin{bmatrix} p_1 & \cdots & p_K \end{bmatrix} \in \mathbb{R}^{N \times K}$$

be the selected spatial packet matrix, where $N = HW$. Let

$$A_K \in \mathbb{R}^{K \times C}$$

contain the channel coefficients. The packet-core reconstruction is

$$\widehat{F}_K = P_K A_K.$$

Take the thin SVD

$$P_K = U_K \Sigma_K V_K^{\mathsf{T}}.$$

Define

$$B_K = \Sigma_K V_K^{\mathsf{T}} A_K.$$

Then

$$\boxed{\widehat{F}_K = U_K B_K.}$$

Definition 4.1 (Singular packet)**.** A column of U_K is a *singular packet* of the selected atlas. It is an orthonormal coordinate direction obtained from the geometry-aware packet span rather than from the image matrix alone.

Theorem 4.2 (Exact singular-packet reparameterization)**.** *The singular-packet representation $U_K B_K$ is exactly equal to the original globally refitted packet reconstruction $P_K A_K$.*

Proof. Substitute the SVD of P_K:

$$P_K A_K = U_K \Sigma_K V_K^{\mathsf{T}} A_K = U_K B_K.$$

□

The computation gives

$$\boxed{\max_{\text{four cases}} \|P_K A_K - U_K B_K\|_\infty = 5.75 \times 10^{-14}.}$$

This residual is floating-point roundoff.

4.3 Energy ordering within a selected packet span

Write the rows of B_K as b_j^{T}. Since the columns of U_K are orthonormal,

$$\|\widehat{F}_K\|_F^2 = \sum_j \|b_j\|_2^2.$$

For a subset J of singular coordinates,

$$\left\| \widehat{F}_K - \sum_{j \in J} u_j b_j^{\mathsf{T}} \right\|_F^2 = \sum_{j \notin J} \|b_j\|_2^2.$$

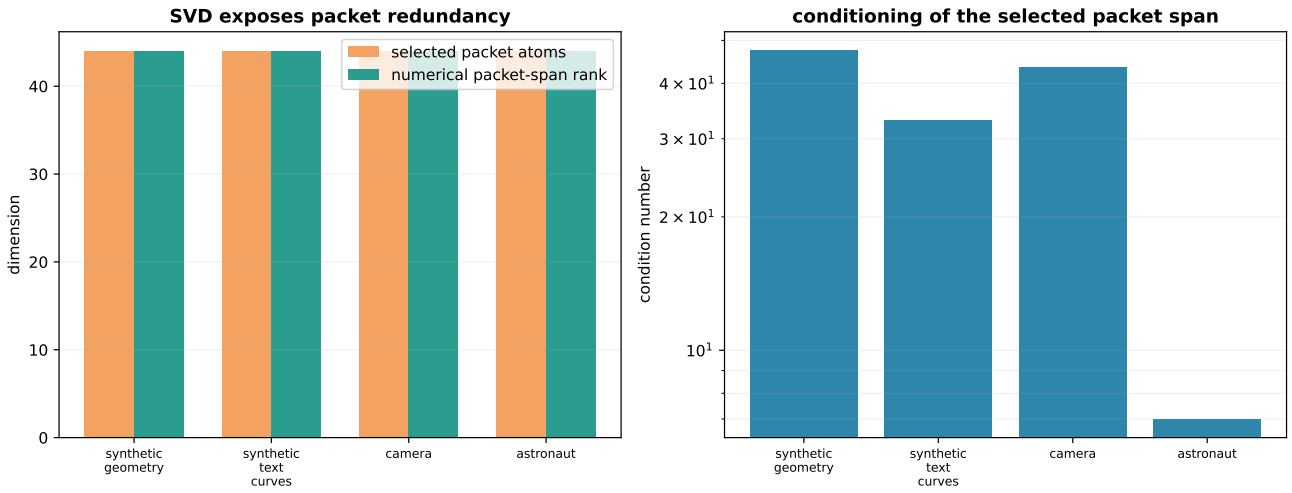


Figure 16: Rank and conditioning of the selected packet spans.

Reading the figure. Every final packet matrix has numerical rank forty-four. The singular spectrum nevertheless shows substantial anisotropy. Canonical singular packets make that anisotropy explicit and give a stable orthonormal synthesis basis.

Therefore the best m-coordinate truncation in this canonical packet basis keeps the m rows with largest $\|b_j\|_2^2$.

The first ten singular packets carry between 91.8% and 96.0% of packet-core energy in the four tests. Retaining 99% requires between seventeen and thirty singular packets. The selected packet matrices remain full rank, but their condition numbers range from about seven to forty-eight. SVD therefore exposes meaningful redundancy even when no atom is exactly dependent.

4.4 Global image SVD

For comparison, reshape the image as

$$F \in \mathbb{R}^{H \times (WC)}.$$

Its SVD is

$$F = U_F \Sigma_F V_F^{\mathsf{T}}.$$

The rank-r truncation

$$F_r = U_{F,r} \Sigma_{F,r} V_{F,r}^{\mathsf{T}}$$

is the best rank-r approximation in Frobenius norm for this matricization.

This is a strong and necessary baseline. It also has a different inductive bias.

- Global SVD modes are dense across the full image width and row direction.
- Packet atoms may be discontinuous, localized, compositional, and tied to explicit regions.
- A rank-one SVD mode stores roughly $H + WC + 1$ scalar values.
- A simple packet may need only a small geometry descriptor and one coefficient vector, though seed-mask storage must be counted in any genuine compression claim.

Claim boundary 4.3 (What SVD optimality means)**.** The best rank theorem makes global SVD optimal inside the chosen global matrix-rank class. It does not make that class optimal among nonlinear geometric packet models, local rank models, sparse transforms, or mixed representations.

4.5 Adaptive local singular packets

The residual layer now contains a second generative mechanism.

Let r_K be the residual after the geometric packet core. Let Ω be a dyadic cell or a bounding region inherited from a selected packet. Reshape the residual patch as

$$R_\Omega \in \mathbb{R}^{h_\Omega \times (w_\Omega C)}.$$

Take its leading SVD mode

$$R_\Omega = \sigma_1 u_1 v_1^\mathsf{T} + \text{remaining terms.}$$

Embed

$$s_\Omega = \sigma_1 u_1 v_1^\mathsf{T}$$

back into the full image and set it to zero outside Ω.

This embedded object is a local singular-packet proposal. It is generated from both a packet region and a data-adaptive rank-one mode.

Algorithm 4.4 (Adaptive singular-packet refinement)**.** At each iteration:

1. rank candidate regions by residual energy divided by a parameter-cost penalty
2. compute the leading local SVD mode in the strongest regions
3. orthogonalize each candidate against the current geometric and singular-packet span
4. select the best nondegenerate direction
5. append its normalized orthogonal component
6. update the exact global orthogonal projection

4.6 Monotone singular-packet gain

Let Q have orthonormal columns spanning the current reconstruction space, and let $r = (I - QQ^\mathsf{T})f$ be the residual. For a candidate direction d, define

$$q = (I - QQ^\mathsf{T})d.$$

If $q \neq 0$, append

$$\widetilde{q} = \frac{q}{\|q\|_2}.$$

The updated projection removes

$$\langle r, \widetilde{q}\rangle \widetilde{q}$$

from the residual.

Theorem 4.5 (Exact gain)**.** *The squared residual decreases by*

$$\boxed{\|r\|_2^2 - \|r_{\text{new}}\|_2^2 = \frac{|\langle r, q\rangle|^2}{\|q\|_2^2} \geq 0.}$$

The measured residual sequence was monotone in all four cases.

4.7 Pure SPD and hybrid SPD

Two models were tested.

1. **Pure SPD.** A constant packet is followed by twenty-two adaptive local singular packets.
2. **Hybrid PR-SPD.** The recorded forty-four-packet geometric core is followed by twenty-two adaptive local singular packets.

Pure SPD is directly comparable to a rank-twenty-two global SVD at the component level. Global SVD is better on all four tests, although the difference is small on the camera and astronaut images. This is consistent with global low-rank optimality.

The hybrid answers a different question. It asks whether an explicit geometric span followed by local singular modes can use its parameters more efficiently than a global residual SVD.

5 Exact finite-tree optimization and portfolio replacement

5.1 Why forward selection is insufficient

The first Singular Packet Decomposition adds one local rank-one mode at a time. That construction established that a local singular direction can be treated as a packet, orthogonalized against the full active span, and globally refitted without increasing the residual.

It also commits early. A mode that looks useful at iteration five may become redundant after iteration twenty. A fixed dyadic tree has the opposite strength. It can optimize a complete multiscale pruning, but it is restricted to one region hierarchy. The refined algorithm therefore carries two completed decompositions and allows either one to be partially reopened.

Definition 5.1 (Adaptive multiscale singular packet)**.** An adaptive multiscale singular packet is a tuple

$$\mathcal{P}_\nu = (\Omega_\nu, o_\nu, r_\nu, U_\nu, C_\nu, \pi_\nu),$$

where Ω_ν is its support, o_ν is the selected unfolding, r_ν is its local rank, U_ν is the embedded orthonormal packet subspace, C_ν is its declared parameter cost, and π_ν records its parent or residual source.

The finite reconstruction class is

$$\boxed{F = P_K A_K + \sum_{\nu \in \mathcal{T}} U_\nu \gamma_\nu + R_{K,\mathcal{T}}.}$$

The geometric packet core remains explicit. The singular packets carry local shading, texture, and unresolved correlation. The archive keeps everything not yet represented.

5.2 Finite AM-SPD model

AM-SPD is easiest to understand as a careful separation of three jobs.

1. The geometric atlas proposes supports and interpretable packets.
2. Local singular models choose the rank and orientation inside each support.
3. The residual archive keeps every value that the finite model has not yet explained.

For a selected tree $\mathcal{T}$, write

$$\widehat{F}_{\mathcal{T}} = P_K A_K + \sum_{\nu \in \mathcal{T}} M_{\Omega_\nu} \left(U_\nu \Sigma_\nu V_\nu^{\mathsf{T}} \right).$$

The finite approximation is allowed to be sparse or lossy. The complete image is not:

$$\boxed{F = \widehat{F}_{\mathcal{T}} + R_{\text{archive}}.}$$

This identity is the reconstruction contract carried through the whole method.

Definition 5.2 (Final singular packet)**.** A singular packet is a tuple

$$\mathcal{P}_\nu = (\Omega_\nu, o_\nu, r_\nu, Q_\nu, \sigma_\nu, C_\nu, \pi_\nu),$$

where Ω_ν is the support, o_ν is the unfolding, r_ν is the local rank, Q_ν is the embedded orthonormal packet basis, σ_ν stores the selected singular values, C_ν is the declared cost, and π_ν records its parent or residual source.

A packet is therefore more than one singular vector. It is a supported local subspace with its own scale, rank, coordinate orientation, cost, and genealogy.

5.3 Best local rank model

By the Eckart–Young theorem [1], let $A \in \mathbb{R}^{m\times n}$ be one unfolded residual patch with singular values

$$\sigma_1 \geq \sigma_2 \geq \cdots \geq 0.$$

Define

$$A_r = \sum_{j=1}^{r} \sigma_j u_j v_j^{\mathsf{T}}.$$

Theorem 5.3 (Best local rank-r packet)**.** *For every matrix B of rank at most r,*

$$\boxed{\|A - A_r\|_F^2 = \sum_{j>r} \sigma_j^2 \leq \|A - B\|_F^2.}$$

Proof. Let P be the orthogonal projector onto the column space of B. Since $PB = B$,

$$A - B = (I - P)A + (PA - B).$$

The two terms are orthogonal in Frobenius inner product, so

$$\|A - B\|_F^2 = \|(I - P)A\|_F^2 + \|PA - B\|_F^2 \geq \|(I - P)A\|_F^2.$$

Write $AA^{\mathsf{T}} = \sum_j \sigma_j^2 u_j u_j^{\mathsf{T}}$. Then

$$\|PA\|_F^2 = \operatorname{tr}(PAA^{\mathsf{T}}) = \sum_j \sigma_j^2 \|Pu_j\|_2^2.$$

The weights $\|Pu_j\|_2^2$ lie in $[0, 1]$ and sum to $\operatorname{rank} P \leq r$. Their weighted sum is largest when the first r weights are one and the rest are zero. Hence

$$\|PA\|_F^2 \leq \sum_{j=1}^{r} \sigma_j^2.$$

Therefore

$$\|(I - P)A\|_F^2 \geq \sum_{j>r} \sigma_j^2.$$

Taking P onto $\operatorname{span}\{u_1, \ldots, u_r\}$ and $B = A_r$ gives equality. □

This theorem is applied independently to the row, column, and channel unfoldings. The orientation is part of the packet search because the same local tensor can have very different ranks under different matricizations.

5.4 Orthogonal block gain

Let V be the current reconstruction space, let P_V be its orthogonal projector, and let

$$r = (I - P_V)F$$

be the current residual. A candidate block has raw direction matrix D. Remove its represented component:

$$D_\perp = (I - P_V)D.$$

Let Q have orthonormal columns spanning $\operatorname{range} D_\perp$.

Theorem 5.4 (Exact block gain)**.** *After enlarging the active space from V to $V \oplus \operatorname{range} Q$ and refitting globally,*

$$\boxed{\|r\|_2^2 - \|r_{\text{new}}\|_2^2 = \|Q^\mathsf{T} r\|_2^2.}$$

Proof. The columns of Q are orthonormal and orthogonal to V. Thus

$$r = QQ^\mathsf{T} r + (I - QQ^\mathsf{T})r$$

is an orthogonal decomposition. The second term is the new residual. Pythagoras gives

$$\|r\|_2^2 = \|Q^\mathsf{T} r\|_2^2 + \|r_{\text{new}}\|_2^2.$$

□

The gain is exact. The complexity exponent used to rank candidates is a search preference, not part of the theorem.

Corollary 5.5 (Monotone global refit)**.** *If the accepted active spaces satisfy*

$$V_0 \subseteq V_1 \subseteq \cdots,$$

then

$$\boxed{\|F - P_{V_{k+1}} F\|_2 \le \|F - P_{V_k} F\|_2.}$$

5.5 Integer-budget tree optimization

The earlier implementation swept a Lagrange multiplier. That traces useful rate–distortion points, but it can skip a Pareto state that matters at one exact integer budget. The final tree engine stores the nondominated cost–error frontier directly.

For a node ν and integer budget B, let $E_\nu(B)$ be the smallest additive local squared error achievable in the subtree rooted at ν. A zero-rank model is included with cost zero, so every budget is feasible.

If $\mathcal{M}_\nu$ is the finite set of stop models at ν, then

$$\boxed{E_\nu(B) = \min\left\{ \min_{\substack{m \in \mathcal{M}_\nu \\ C_m \le B}} E_m, \quad \min_{B_1 + \cdots + B_s \le B} \sum_{j=1}^{s} E_{\mu_j}(B_j) \right\},}$$

where $\mu_1, \ldots, \mu_s$ are the children of ν.

Theorem 5.6 (Exact Pareto tree theorem)**.** *For the declared finite tree, local unfolding family, rank range, additive squared-error model, and integer cost model, the recursion above returns the minimum additive local error at every budget, conditional on the computed local costs.*

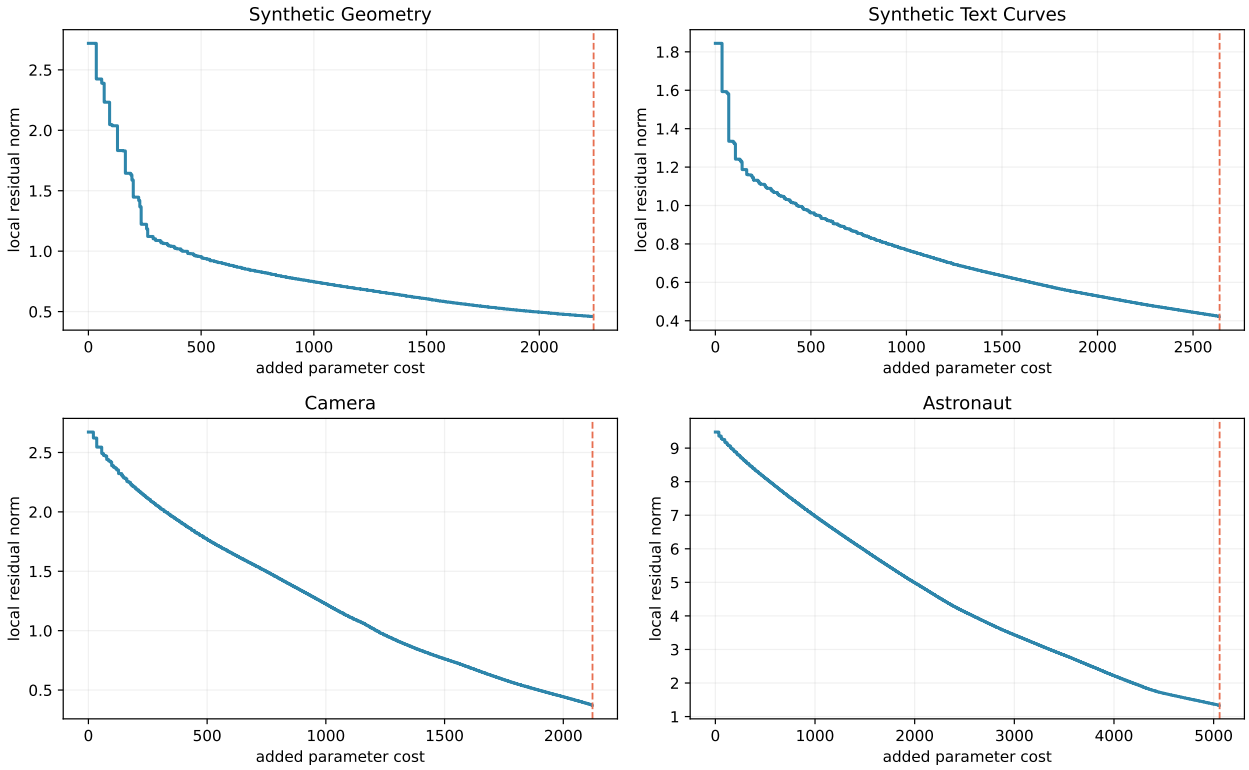


Figure 17: Exact Pareto frontiers for the four benchmark images.

Reading the figure. Each point is a nondominated local tree model. The dashed line is the inherited budget. The frontier records every cost level that can matter, rather than only the points exposed by a Lagrange multiplier.

Proof. At a leaf, every admissible representation is one local stop model, so the formula is exact. Assume it is exact for the children of an internal node. Any complete representation of the node has exactly two forms. It either stops at the node with one model, or it splits and represents every child. In the split case, the total cost and local squared error are sums of the child values. The induction hypothesis makes each child allocation optimal. Taking the minimum over the stop choices and every integer budget allocation therefore covers every admissible pruning and chooses the best one. □

The implementation stores only nondominated states. A state (C_2, E_2) is discarded when another state satisfies

$$C_1 \le C_2, \qquad E_1 \le E_2,$$

with at least one strict inequality. This pruning changes neither the answer nor the proof.

Remark 5.7 (Combinatorial and numerical exactness). The dynamic program is combinatorially exact for the declared finite state space. Its local errors are floating-point quantities, so equality comparisons and reported residuals remain subject to numerical tolerance.

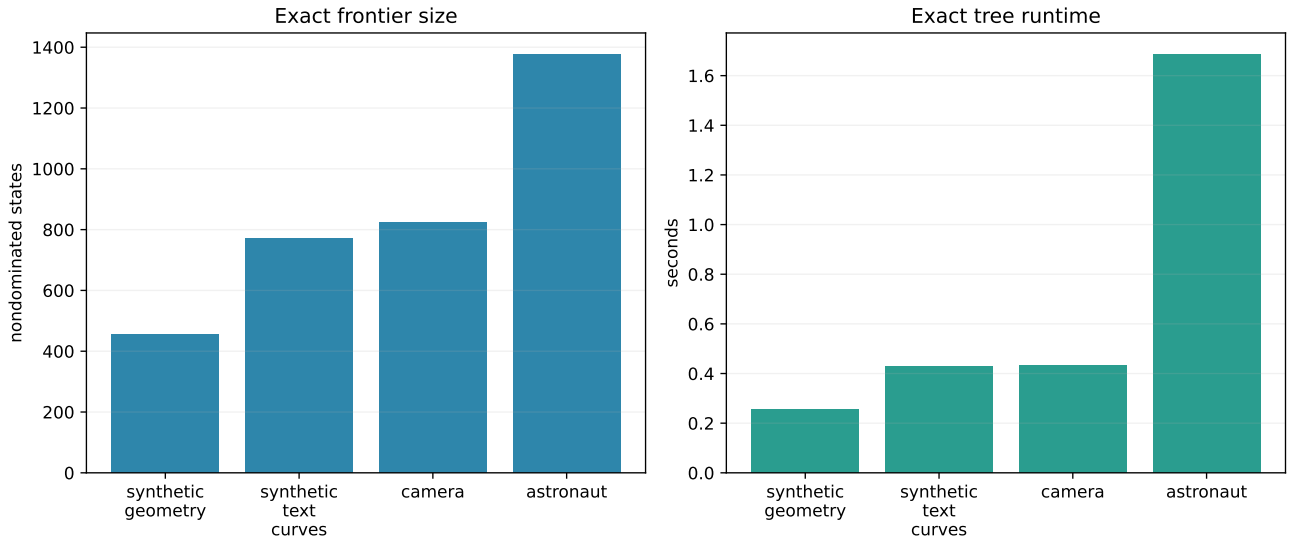


Figure 18: Frontier size and exact-tree runtime.

Reading the figure. The four exact frontiers contain between 457 and 1,378 states. Their measured construction time ranges from about a quarter second to under two seconds in the reported environment.

5.6 Local tree optimality and global refitting

The exact tree theorem concerns the additive local residual model. Once a pruning has been chosen, its directions are placed beside the inherited geometric core and globally orthogonalized. This second projection may improve the reconstruction because it can exploit interactions that the additive local objective intentionally ignores.

Proposition 5.8 (Global refit cannot hurt a selected tree)**.** *Let $\widehat{F}_{\mathrm{local}}$ be the additive tree reconstruction and let W be the span of the inherited core and every selected local direction. Then*

$$\boxed{\|F - P_W F\|_F \le \|F - \widehat{F}_{\mathrm{local}}\|_F.}$$

Proof. The additive reconstruction belongs to W. Orthogonal projection is the least-squares minimizer over W, so its error cannot exceed the error of any other member of W. □

5.7 Moving residual regions

The tree is not the whole atlas. During backward replacement, the finite frontier is regenerated from:

- dyadic rectangles
- packet bounding boxes
- connected components of large residual magnitude
- connected components of large residual gradient
- translated, expanded, and contracted descendants

A support can become useful only after neighboring packets have been removed. The moving stage lets the algorithm leave the initial tree.

5.8 Backward replacement

Suppose a seed decomposition contains directions q_j with costs C_j. Define the seed value

$$\eta_j = \frac{|\langle F, q_j \rangle|^2}{C_j}.$$

For a reopened fraction ρ, the algorithm removes the lowest-value directions until approximately ρ of the added budget has been released. It then regenerates residual regions, inserts new variable-rank blocks, and globally refits the complete surviving span.

The cycle is

$$\boxed{\text{forward decomposition} \longrightarrow \text{weak-mode deletion} \longrightarrow \text{moving support search} \longrightarrow \text{global refit.}}$$

The deletion order is a heuristic. Every completed candidate is judged by its exact residual.

5.9 Three-seed portfolio and nonregression

The reported portfolio contains three complete seeds:

1. the original greedy rank-one SPD path
2. the Lagrange-sweep tree path
3. the exact-budget Pareto tree path

Each unchanged seed is retained. Each is also tested after reopening several fractions of its added budget and refilling that budget with moving residual blocks.

Theorem 5.9 (Finite-portfolio nonregression)**.** *Let $\mathfrak{F}$ contain every unchanged seed and every completed replacement candidate. If*

$$\widehat{F}_* = \arg\min_{G \in \mathfrak{F}} \|F - G\|_F,$$

then $\widehat{F}_*$ is no worse than any seed included in $\mathfrak{F}$.

Proof. Every seed is an element of the set over which the minimum is taken. □

The selector is exact over this finite portfolio. The moving support proposals inside a replacement run remain greedy.

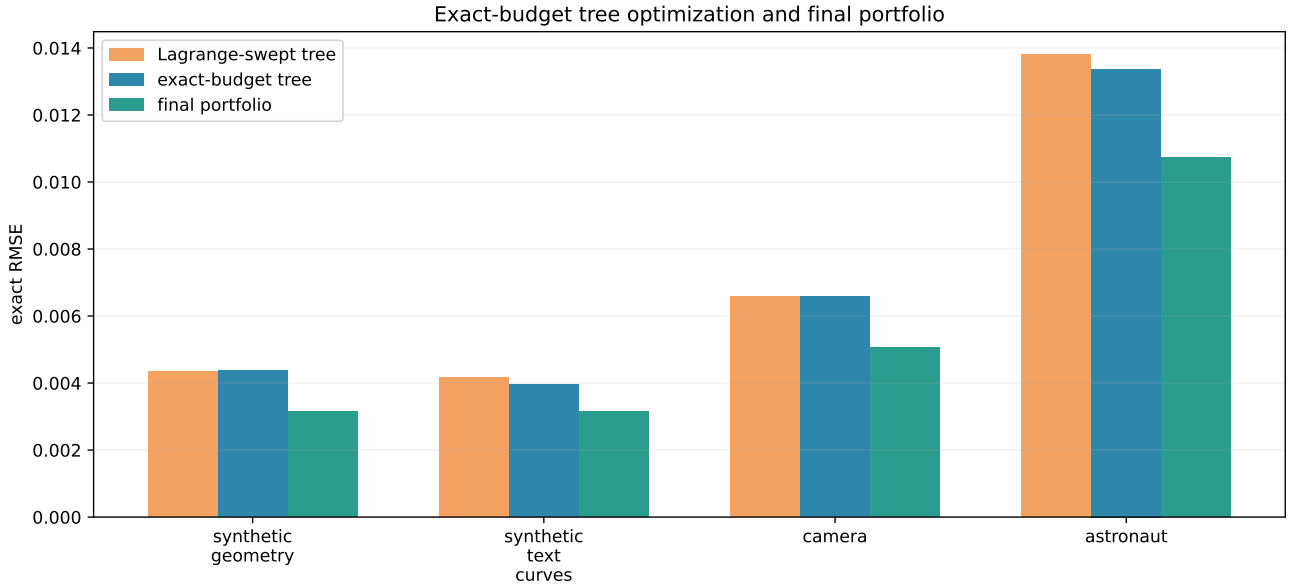


Figure 19: Exact-budget tree and final three-seed portfolio.

Reading the figure. The exact tree improves the Lagrange tree on the text-and-curves and astronaut images, while the two tree seeds tie on camera. The replacement portfolio lowers the final error again.

5.10 Singular-value stability

A packet rank should not change because of a perturbation smaller than the gap separating retained and discarded modes. This is a standard singular-value perturbation principle [12].

Lemma 5.10 (Singular values are Lipschitz)**.** *For matrices A and E of the same shape,*

$$\boxed{|\sigma_j(A+E) - \sigma_j(A)| \le \|E\|_2}$$

for every j.

Proof. Use the min–max formula

$$\sigma_j(A) = \min_{\dim S = n-j+1} \max_{x \in S,\ \|x\|_2 = 1} \|Ax\|_2.$$

For every unit vector,

$$\|(A+E)x\|_2 \le \|Ax\|_2 + \|E\|_2.$$

Taking the max and then the min gives

$$\sigma_j(A+E) \le \sigma_j(A) + \|E\|_2.$$

Swap A and $A+E$ to obtain the reverse inequality. □

Corollary 5.11 (Stable rank decision)**.** *If*

$$\gamma = \sigma_r(A) - \sigma_{r+1}(A) > 2\|E\|_2,$$

then the midpoint threshold between $\sigma_r(A)$ and $\sigma_{r+1}(A)$ selects the same rank for A and $A+E$.

The singular gap is therefore a natural confidence score for a local packet rank.

6 Experimental protocol and main results

6.1 Benchmark design and metric conventions

The principal benchmark uses four 56×56 images:

- two synthetic RGB images, one with piecewise-smooth geometric objects and one with text-like curves and narrow structures
- the grayscale camera image distributed with `scikit-image`
- the RGB astronaut image distributed with `scikit-image` [13]

The original packet-core experiment selects forty-four packets and then applies a twenty-two-coefficient residual DCT completion. The later SPD and AM-SPD experiments inherit comparable added-cost budgets from that baseline. Comparisons include global Fourier and DCT approximations, adaptive quadtree reconstruction, total-variation denoising, global SVD, residual SVD, and restricted packet-tree ablations.

Two error conventions occur in the recovered results. *Exact RMSE* is computed from the raw floating-point reconstruction. *Bounded RMSE* is computed after coordinatewise projection onto $[0, 1]$. The tables in the main text label these quantities separately. Parameter counts are scalar storage proxies, not encoded bit rates.

6.2 Benchmark replay and metric correction

The final packet histories were replayed exactly rather than rerunning the expensive frontier search. During this audit, the reconstruction metric was corrected. The earlier helper globally rescaled any reconstruction that left $[0, 1]$. That operation is not a decoder and can materially change the approximation. The revised metric uses bounded synthesis,

$$\widehat{f} \longmapsto \min(1, \max(0, \widehat{f})),$$

without min–max renormalization.

The replayed packet histories and the corrected bounded metric are the authoritative benchmark in this edition. This correction improves the reported packet-core results and makes the comparison reproducible from the stored history.

6.3 Four-image benchmark

The reported run retains three complete seed families: the original greedy SPD path, the Lagrange-sweep tree, and the exact-budget Pareto tree. Two images select the exact-budget seed and two select the Lagrange seed. The unchanged greedy seed remains in the finite portfolio but is not selected in the reported run.

The exact-RMSE reductions relative to the original SPD baseline are approximately

$$15.3\%, \quad 7.5\%, \quad 66.1\%, \quad 68.0\%.$$

Image	Original exact	Exact-tree exact	Final exact	Final bounded	Exact reduction
Synthetic geometry	0.003726	0.004365	0.003155	0.003155	15.3%
Synthetic text and curves	0.003394	0.003958	0.003141	0.003141	7.5%
Camera	0.014943	0.006578	0.005064	0.005064	66.1%
Astronaut	0.033553	0.013374	0.010742	0.010691	68.0%

Table 1: Four-image benchmark. Exact RMSE is computed before clipping. Bounded RMSE is computed after projection onto $[0, 1]$. The reduction column uses exact RMSE relative to the original SPD baseline.

Image	Budget	Final cost	Frontier states	Tree time	Selected seed
Synthetic geometry	2241	2224	457	0.255 s	exact budget tree
Synthetic text and curves	2639	2616	773	0.430 s	exact budget tree
Camera	2123	2120	823	0.435 s	lagrange tree
Astronaut	5060	5017	1378	1.688 s	lagrange tree

Table 2: Declared budget, selected model cost, exact-tree frontier size, measured tree-construction time, and winning seed. The cost is a scalar storage proxy rather than an entropy-coded bit rate.

The corresponding bounded RMSE values are

$$0.003155, \quad 0.003141, \quad 0.005064, \quad 0.010691.$$

The astronaut bounded-RMSE reduction is approximately 67.6%. The mean exact-RMSE reduction across the four cases is 39.2%.

6.4 Ten-case stress suite

The additional suite uses five synthetic images and five natural images at 32×32 under one common parameter budget. It compares:

- one global SVD unfolding
- a row-only rank-one tree
- a row/column rank-one tree
- the full row/column/channel variable-rank tree

The full model improves on both rank-one tree ablations in all ten cases. It beats the parameter-matched global SVD in seven of ten cases. The median error ratio is

$$\frac{\mathrm{RMSE}_{\text{AM-SPD}}}{\mathrm{RMSE}_{\text{global SVD}}} = 0.947,$$

and the median ratio to the row-only rank-one tree is

$$0.624.$$

Case	Global SVD	Row rank 1	Row+column rank 1	Full AM-SPD
Diagonal Edge	0.031678	0.023702	0.022897	0.012927
Checkerboard	0.000000	0.000000	0.000000	0.000000
Concentric Rings	0.090688	0.197907	0.197907	0.137076
Oriented Texture	0.126959	0.113413	0.098770	0.075295
Color Blocks	0.012400	0.046092	0.038465	0.000000
Camera	0.042280	0.067144	0.067144	0.041128
Coins	0.038827	0.073104	0.073104	0.039887
Moon	0.010254	0.010146	0.010146	0.006192
Page	0.018040	0.028005	0.028005	0.017787
Astronaut	0.093798	0.115350	0.112969	0.086405

Table 3: Ten-case stress suite at a common parameter budget. The full local model is not uniformly superior to global SVD.

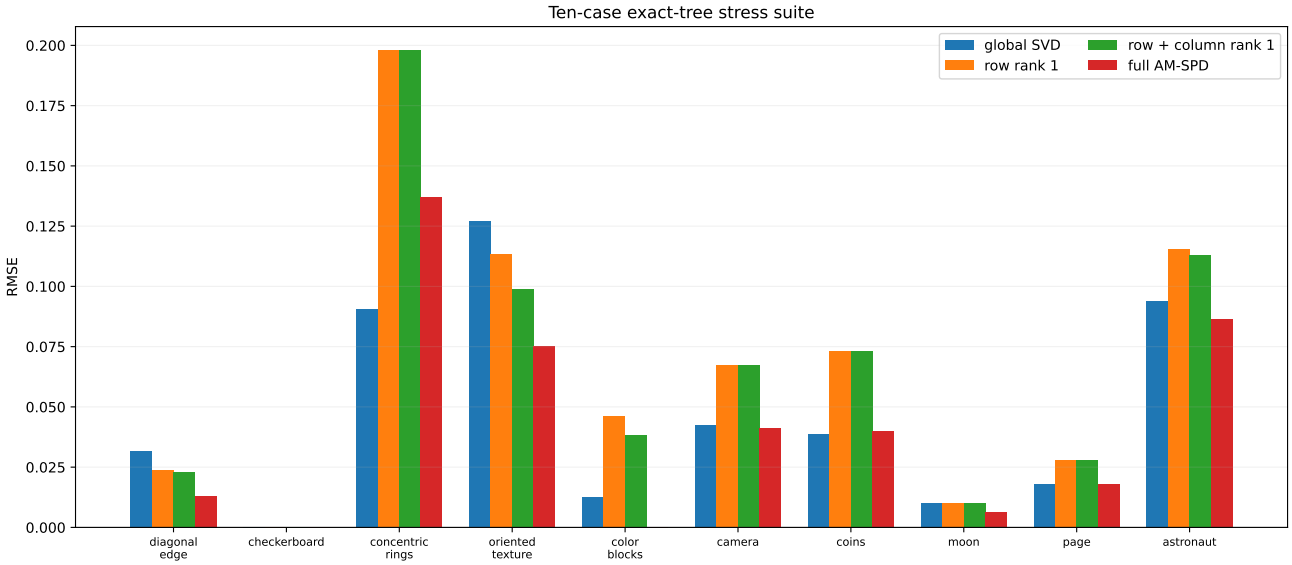


Figure 20: Ten-case stress comparison.

Reading the figure. No single model wins every image. The full local model is strongest on seven cases, ties the exact low-rank checkerboard case, and loses to global SVD on two highly global textures.

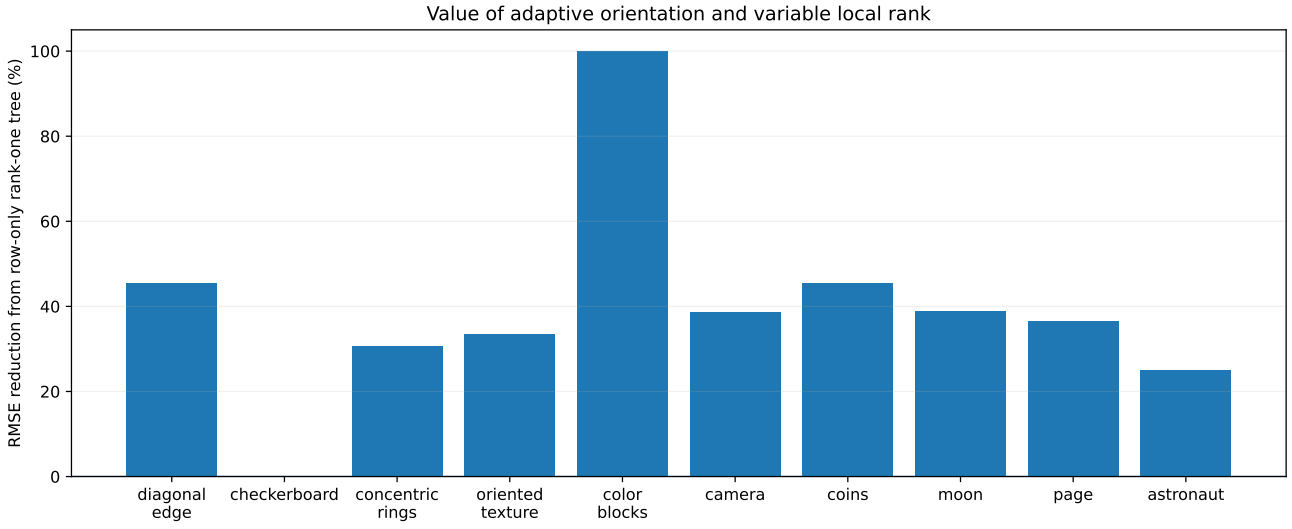


Figure 21: Ablation gain from orientation and rank adaptation.

Reading the figure. Every case benefits or ties when column, channel, and higher-rank local packets are added to the row-only rank-one tree.

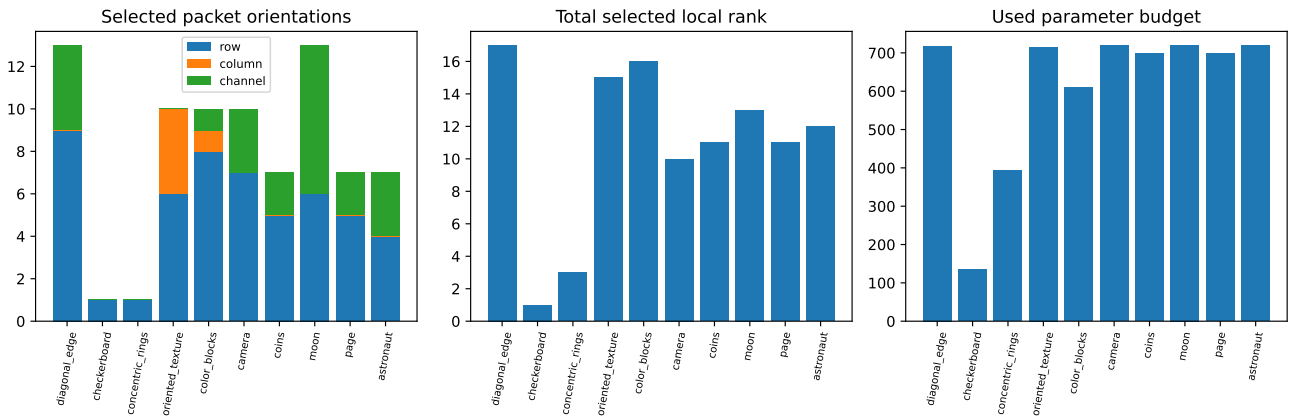


Figure 22: Selected orientations, ranks, and budgets.

Reading the figure. The optimizer uses different local structures for different images. Channel packets become important for color blocks, the moon, and natural images.

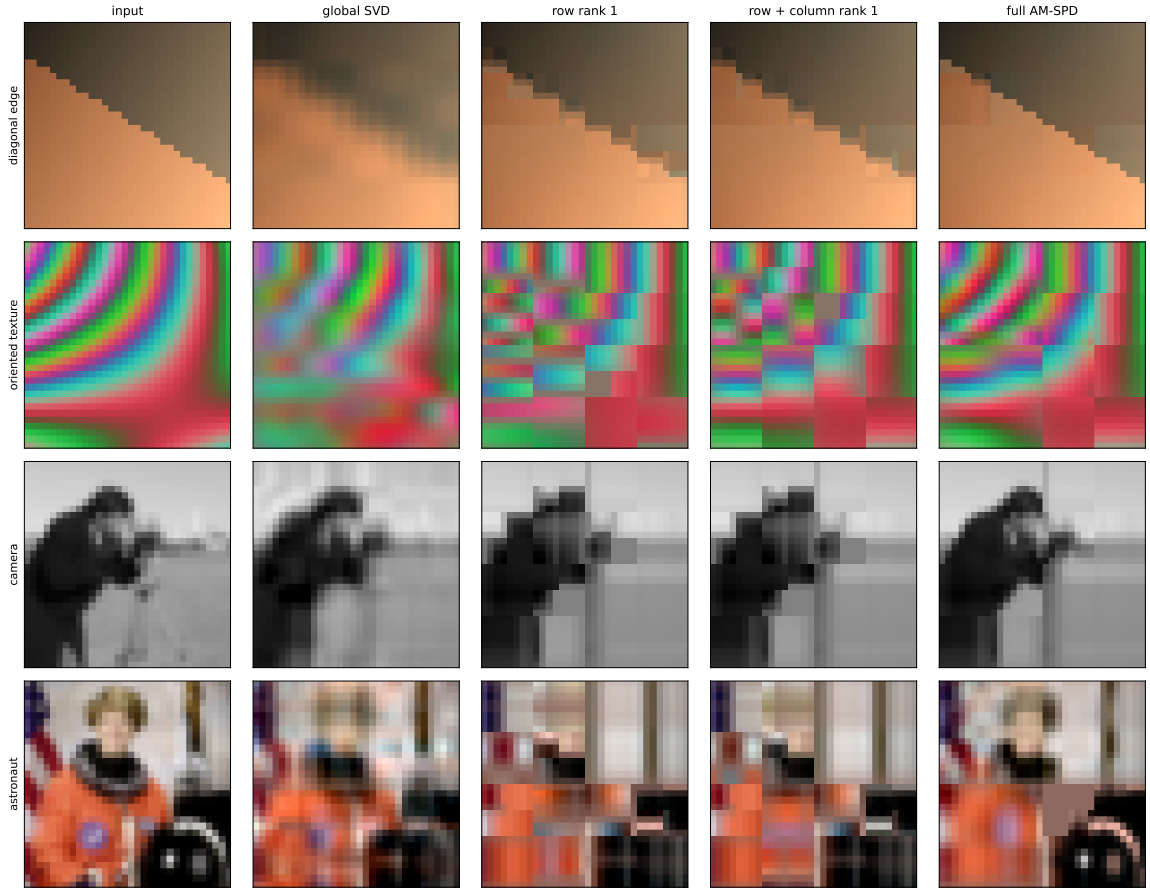


Figure 23: Representative stress-suite reconstructions.

Reading the figure. The diagonal edge benefits from local support. The oriented texture remains difficult. Natural images show the tradeoff between one global low-rank model and many local low-rank packets.

6.5 Noise study

A fixed-budget approximation can denoise when its model bias is smaller than the noise being removed. It can also be worse than the noisy input when the clean-image approximation error already exceeds the noise level.

The final noise stress contains 36 noisy scenarios across four images, three noise levels, and three random seeds. The tree is closer to the clean image than the noisy input in nine scenarios. The mean selection Jaccard score is 0.622. The diagonal-edge model denoises reliably at the two larger noise levels. The oriented texture and astronaut remain approximation limited at this budget.

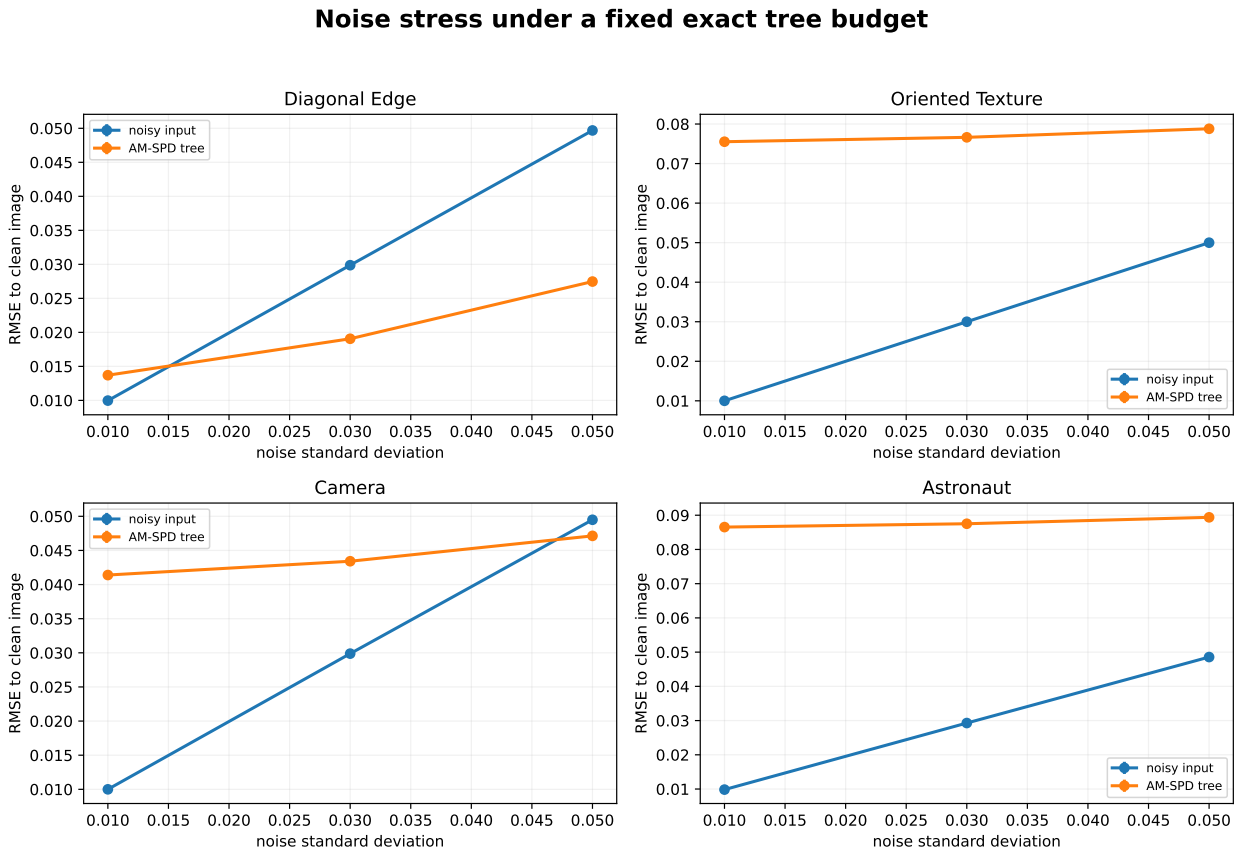


Figure 24: Noise stress under one fixed budget.

Reading the figure. The result is intentionally mixed. The tree suppresses larger noise on the simpler edge image and on the camera at the highest noise level. It is not a universal denoiser.

Noise-study quantity	Value
Scenarios	36
Positive denoising scenarios	9
Mean denoising gain	-0.027500
Mean support Jaccard score	0.6223

Table 4: Noise-study summary. The fixed-budget deterministic approximation rule is not an automatic denoiser.

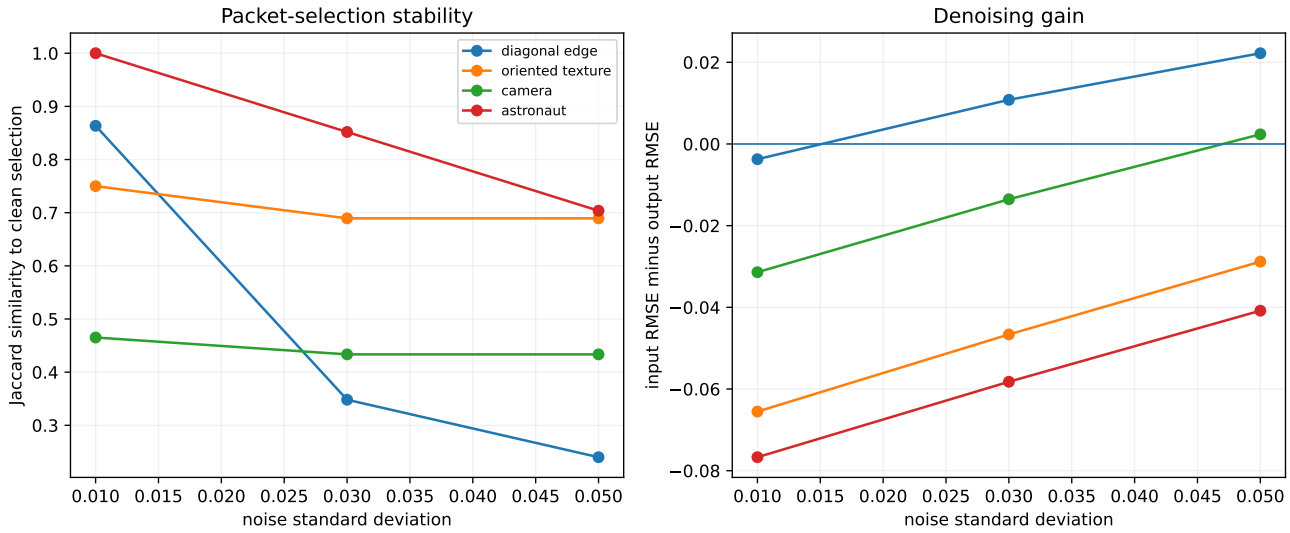


Figure 25: Selection stability and denoising gain.

Reading the figure. Support choices remain moderately stable, but the clean support is not frozen. A noise-aware objective would need an explicit statistical penalty rather than the deterministic approximation objective used here.

6.6 Interpretation of the SVD comparison

The comparison leads to a sharper architecture.

1. Use the perturbation grammar to discover discontinuities, objects, and local coordinate neighborhoods.
2. Replace the selected nonorthogonal packet atoms by their canonical singular packets for stable synthesis and energy ordering.
3. Generate local singular modes from residual regions where the semantic grammar is incomplete.
4. Keep DCT or wavelet coordinates for very low-cost residual updates.
5. Preserve the exact residual archive and the pixel-depth closure path.

The resulting model is neither a conventional global SVD nor a fixed packet dictionary. It is a moving union of geometry-aware subspaces, each equipped with its own singular coordinates.

Definition 6.1 (Projector-Resolved Singular Packet Decomposition)**.** PR-SPD is the decomposition obtained by selecting a finite geometry-aware packet span, replacing that span by its canonical singular coordinates, and extending it with adaptively localized singular modes drawn from the current residual.

7 Verification and formal claim boundaries

7.1 Computational verification

The implementation verification suite covers exact matricization round trips, local SVD optimality, orthogonal-complement construction, exact block gain, monotone refitting, exact budget dynamic programming, spatial partitioning, singular-gap stability, archive identities, portfolio nonregression, and consistency of every reported benchmark.

Across all groups,

$$\boxed{12{,}154/12{,}154}$$

Verification group	Scenarios	Failures	Maximum residual
patch roundtrip	90	0	0.000e+00
local svd optimality	791	0	9.592e-14
orthogonal complement	60	0	3.662e-14
exact block gain	80	0	3.819e-14
monotone global refit	320	0	0.000e+00
exact budget tree dp	9960	0	2.220e-16
region partition	394	0	0.000e+00
singular gap stability	70	0	0.000e+00
exact archive identity	60	0	5.684e-14
portfolio nonregression	300	0	0.000e+00
reported results consistency	29	0	0.000e+00
Total	12154	0	–

Table 5: Implementation verification ledger. These tests establish consistency of the finite implementation and do not replace the analytic proofs.

implementation scenarios passed. These checks support the code and reported tables, while the analytic statements are justified by the proofs in the text. The exact tree recursion alone was compared with brute-force enumeration at every budget in 120 randomized trees, giving 9,960 exact budget checks.

The earlier layers also remain intact:

- sixteen core packet-atlas groups passed
- eight Singular Packet Decomposition groups passed
- fifteen AM-SPD portfolio groups passed
- eleven final implementation and stress groups passed

The largest residual in the implementation verification suite was below 10^{-13}.

7.2 Formal guarantees and heuristic components

The reported formulation has a clean boundary.

- The local rank-r model is exact for its declared unfolding and support.
- The integer-budget tree frontier is exact for the declared finite tree and additive local objective.
- Global refitting is exact least squares on the selected span.
- The finite three-seed portfolio selector is exact over the candidates that were actually completed.
- Backward deletion order and moving residual-region proposals remain heuristic.
- The scalar parameter count remains a storage proxy rather than a complete entropy-coded rate.
- Exact image closure still comes from the residual archive or complete finite coordinates.

These are the finite guarantees established by the present implementation. It is strong enough to prove the optimization statements actually used by the code, while keeping open-ended geometry learning separate from those proofs.

8 Scope, limitations, and extensions

8.1 Unbounded generation depth

The atlas is endless in three senses.

1. **Parameter depth.** Dyadic translations, rotations, scales, boundary amplitudes, and phases continue to finer resolutions.
2. **Compositional depth.** Every selected combination can become a new parent.
3. **Spatial depth.** The quadtree continues until singleton pixels.

No finite computation visits all packets. The algorithm remains an anytime best-first traversal of a countably infinite tree.

8.2 Implications for unknown structures

A fixed dictionary fails when the true object lies between templates. The perturbation atlas replaces nearest-template selection by a local coordinate search. Residual seeds allow the search to leave its original neighborhoods. Binary composition allows several local pieces to become one global packet. The dyadic fallback prevents a permanently unexplained region from becoming unreachable.

This changes the reconstruction problem from

"Which stored packet is the object?"

to

"Which finite path through the generative atlas explains the object to the required tolerance?"

8.3 Limitations and research directions

The current implementation is not yet a universal geometry learner. Several important directions remain.

- Learn boundary deformations from residual gradients rather than evaluating a finite perturbation frontier.
- Introduce spline, level-set, and neural implicit packets while keeping exact residual accounting.
- Use multiscale branch-and-bound to certify that no unvisited packet can beat the current candidate gain.
- Extend spatial masks to video tubes and three-dimensional surfaces.
- Replace repeated normal-equation solves with incremental QR or Cholesky updates.
- Develop a rate-distortion theory for the semantic packet forest and residual archive.
- Extend singular packets from local matrix rank to multiscale tensor and video-tube ranks.
- Account for seed-mask transmission cost in a complete rate-distortion code.

Claim boundary 8.1 (No universal finite-budget sparsity claim)**.** The existence of a countably infinite dense generator and an exact pixel fallback does not imply that every image has a short representation in the semantic grammar. The astronaut experiment demonstrates this boundary directly.

Claim boundary 8.2 (No automatic unknown-boundary theorem)**.** Residual-driven perturbation can improve an approximate geometry, but exact unknown continuous-boundary recovery still requires identifiability, sampling, noise, and stability hypotheses. When those hypotheses fail, the residual remains the correct certificate.

8.4 Reconstruction contract

The full-image architecture is

$$f = \underbrace{\sum_{k=1}^{K} p_k \gamma_k}_{\text{generated geometric and local-field core}} + \underbrace{S_M r_K}_{\text{DCT, wavelet, or singular residual packets}} + \underbrace{r_{K,L}}_{\text{exact remaining archive}}.$$

At any time, the last term may be preserved exactly, further modeled, or closed by complete finite coordinates. The calculus therefore does not have to choose between semantic approximation and lossless reconstruction. It records both.

9 Conclusion

The present AM-SPD formulation is finite, explicit, and auditable.

moving geometric atlas $\longrightarrow$ three complete singular seeds
exact integer-budget frontiers $\longrightarrow$ backward budget reopening
moving variable-rank replacement $\longrightarrow$ exact finite portfolio selection
residual archive or complete finite closure

The central improvement is not simply a lower error number. Early decisions are no longer permanent. The tree can choose mixed depth and mixed rank under an exact budget. The portfolio can keep a strong seed intact or reopen only the part that has become inefficient. Every accepted span is refitted globally. Every finite reconstruction still carries its exact residual.

The reported bounded RMSE values are

$$0.003155, \quad 0.003141, \quad 0.005064, \quad 0.010691.$$

They improve on the original SPD by roughly 15.3%, 7.5%, 66.1%, and 67.6%. The ten-case stress suite shows where the model helps and where a single global SVD remains stronger. The noise study shows that deterministic approximation is not automatically denoising. Those negative results are part of the reported evidence rather than exceptions omitted from it.

The exact claims are restricted to their proper domains. The local rank theorem is exact for one support and unfolding. The budget recursion is exact for the declared finite tree and additive local objective. The final selector is exact over the completed finite portfolio. Moving region generation remains heuristic. The parameter count remains a proxy until a full entropy code is specified.

The reconstruction identity does not change:

$$F = \widehat{F} + R_{\text{archive}}.$$

The finite model explains as much structure as its budget allows. The archive preserves the rest without ambiguity.

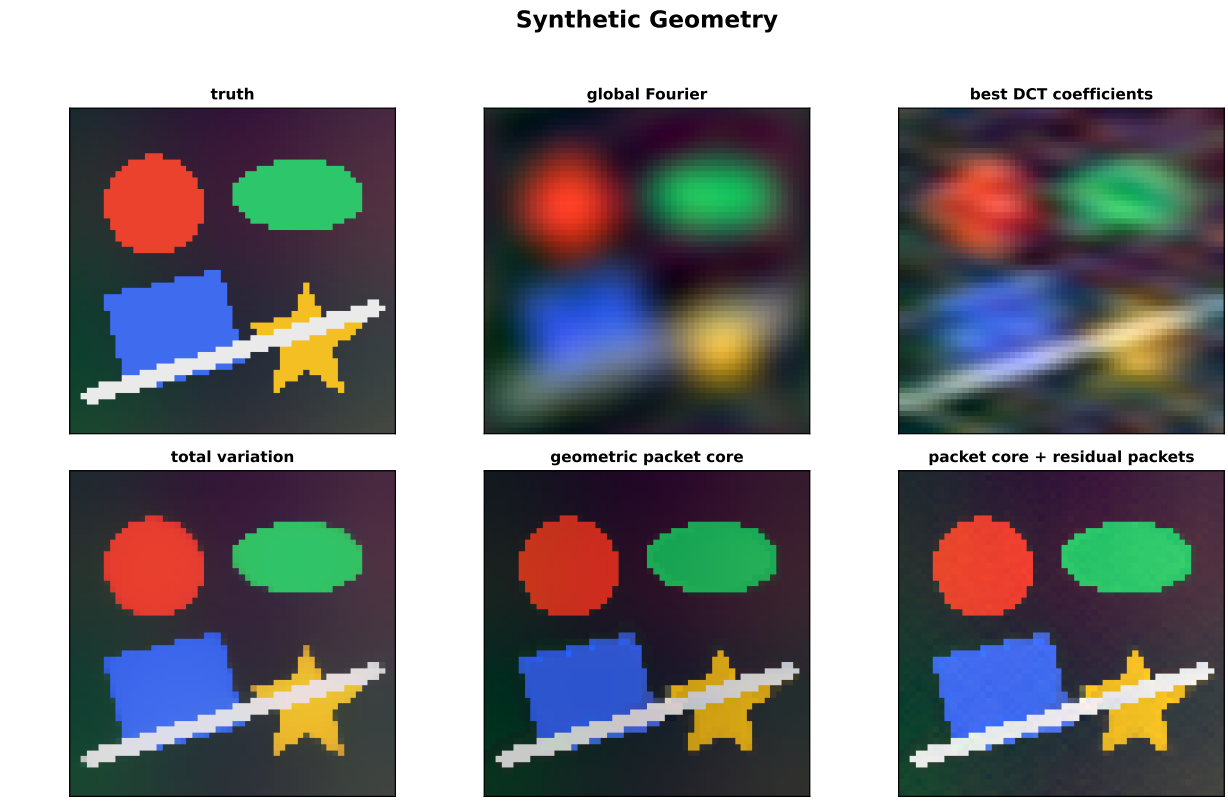


Figure 26: Reconstruction methods on the synthetic geometry image.

Reading the figure. The packet core captures regions and boundary structure. Residual completion reduces the remaining field error. Total variation remains slightly more accurate in RMSE, while the packet representation exposes an explicit geometric decomposition.

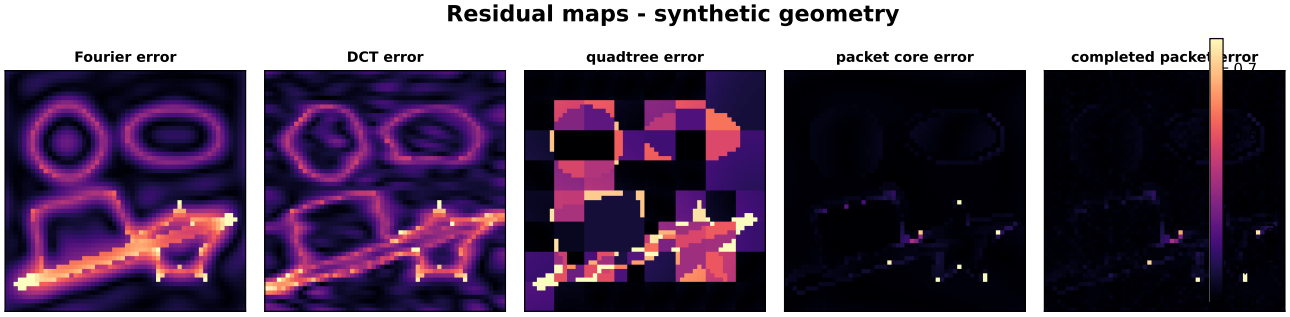


Figure 27: Residual maps for the synthetic geometry image.

Reading the figure. Global spectral errors are distributed around boundaries. The completed packet residual is smaller and more localized.

A Initial packet-core benchmark

Historical baseline. This appendix preserves the first four-image packet-core experiment. Its numerical results are not the final AM-SPD results in Section 6.

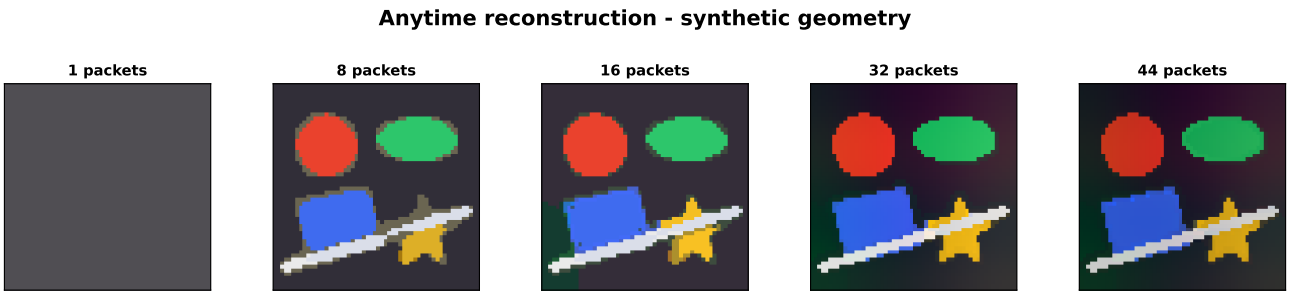


Figure 28: Anytime packet reconstruction of the synthetic geometry image.

Reading the figure. The image becomes recognizable before the full packet budget is reached. Later packets refine geometry and local field behavior rather than rebuilding the entire image from scratch.

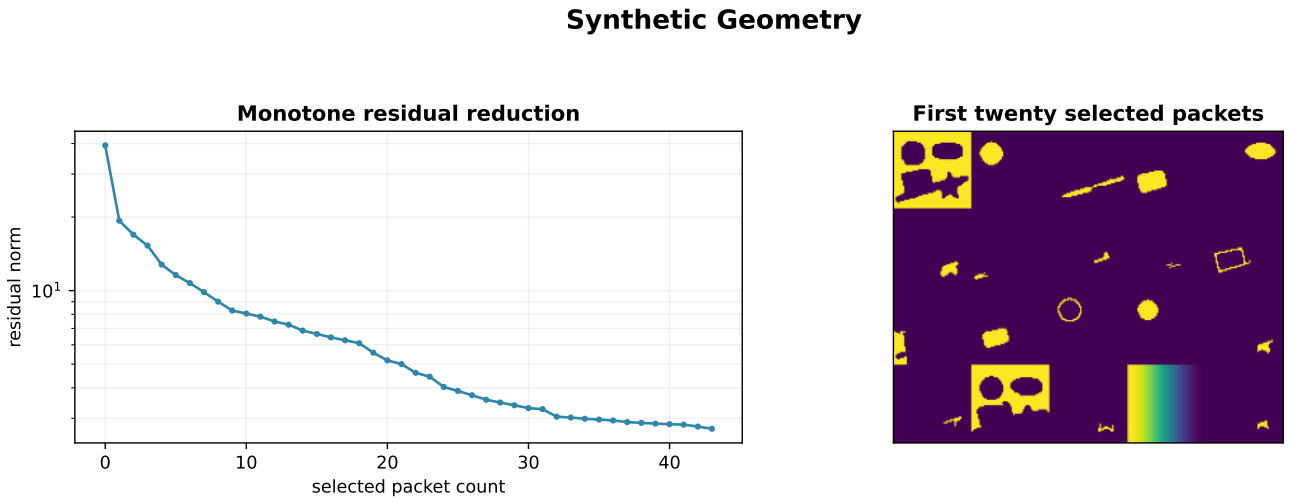


Figure 29: Convergence and selected packets for the synthetic geometry image.

Reading the figure. The final forest includes ordinary masks, morphology, local cosine and polynomial packets, DCT packets, and four binary composition packets.

A.1 Synthetic geometry baseline

The completed packet reconstruction has

$$\mathrm{RMSE} = 0.026957, \qquad \mathrm{SSIM} = 0.9645.$$

Its RMSE is approximately 17% of the Fourier error and 20% of the DCT error.

A.2 Synthetic text and curves baseline

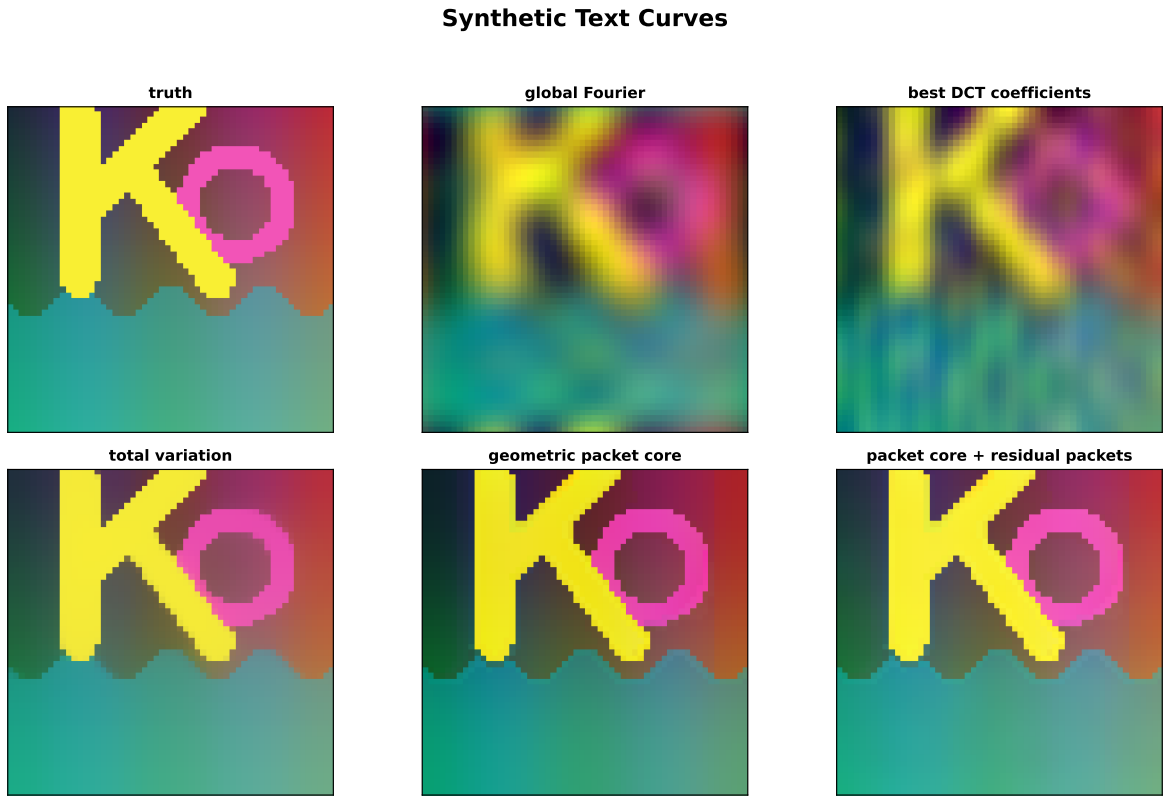


Figure 30: Reconstruction methods on the text-and-curves image.

Reading the figure. Thin and curved structures require several local masks, field packets, and one affine refinement. The completed packet reconstruction slightly outperforms the tested total-variation setting in RMSE and SSIM.

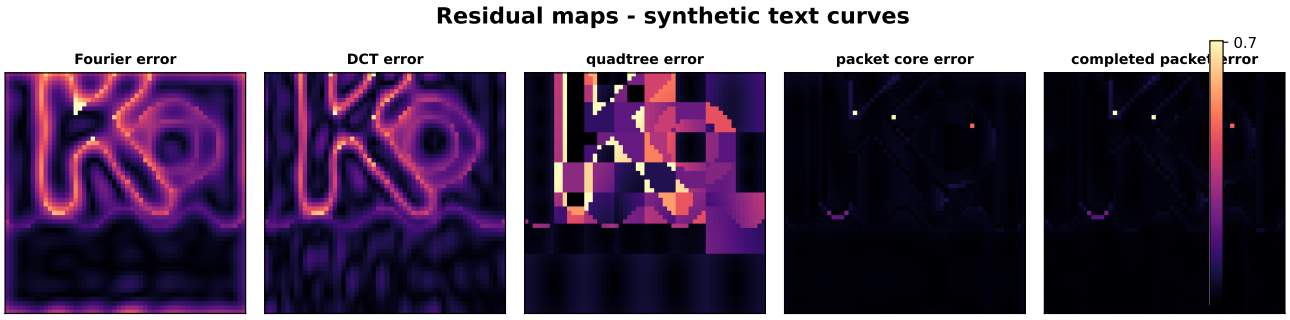


Figure 31: Residual maps for text and curves.

Reading the figure. Most of the remaining packet error lies on the finest strokes and local intensity variations.

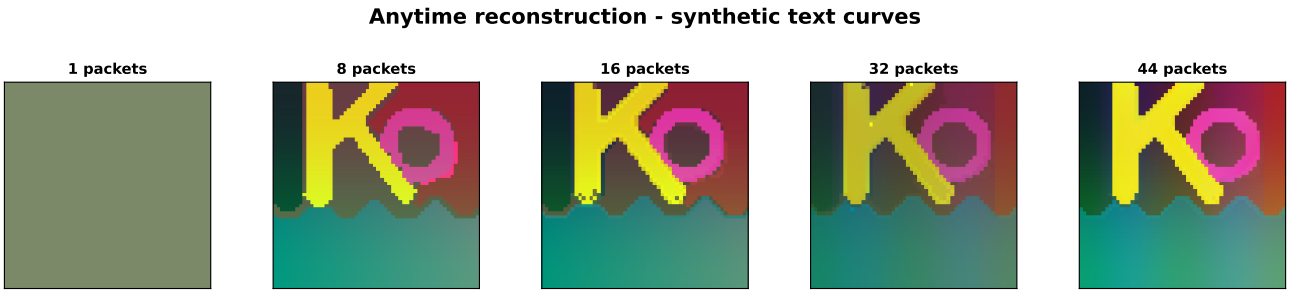


Figure 32: Anytime reconstruction of text and curves.

Reading the figure. The active atlas grows from broad region proposals toward local stroke and field corrections.

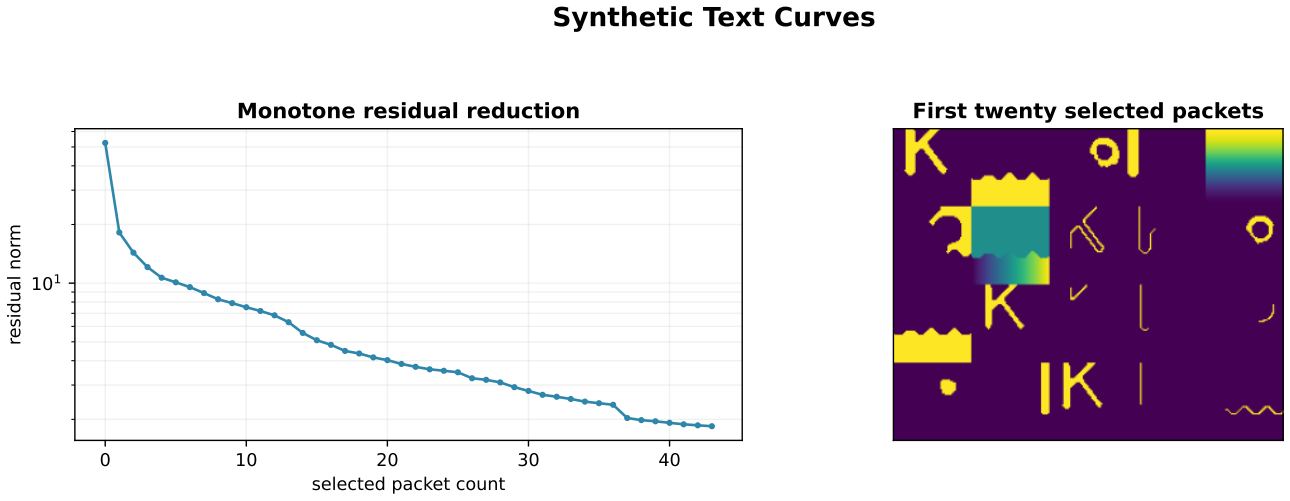


Figure 33: Convergence and selected packets for text and curves.

Reading the figure. The completed packet method reaches RMSE 0.017660 and SSIM 0.9809 under the reported budget.

A.3 Camera image



Figure 34: Reconstruction methods on the camera image.

Reading the figure. The packet method improves strongly over Fourier, DCT, and quadtree baselines, but the tested total-variation method remains more accurate. Natural textures and gradual illumination expose limits of the present geometry grammar.

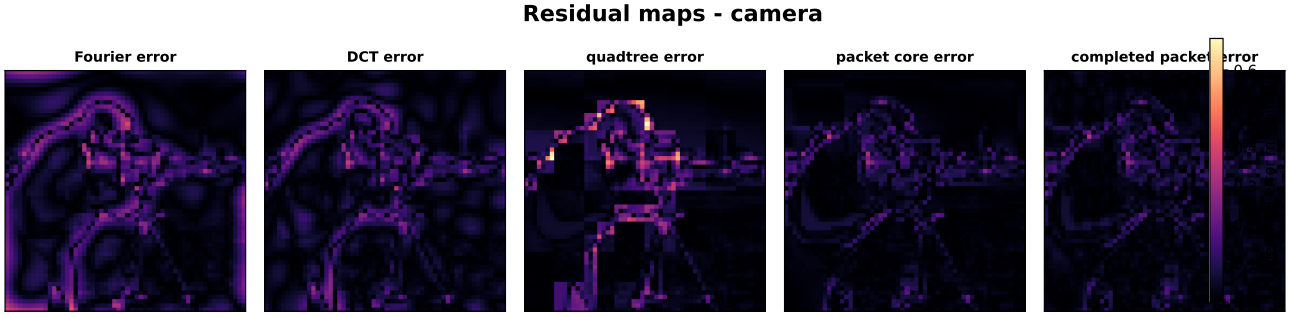


Figure 35: Camera residual maps.

Reading the figure. The geometric forest captures major regions. Fine texture and intensity variation remain the dominant residual.

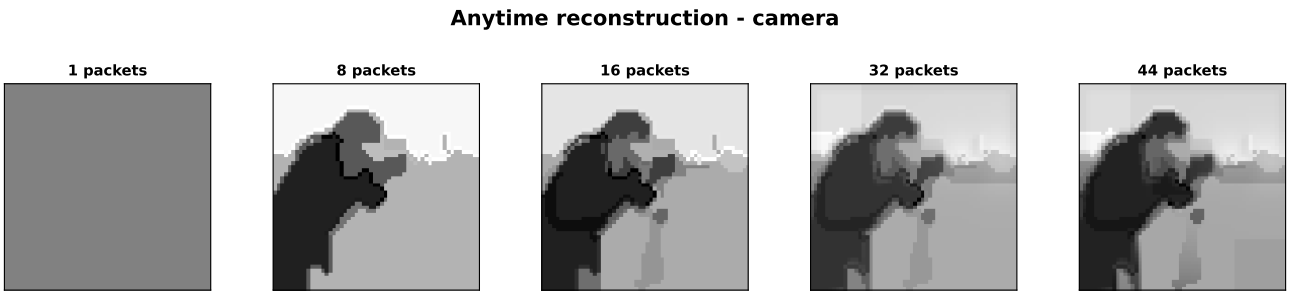


Figure 36: Anytime camera reconstruction.

Reading the figure. The global subject and high-contrast boundaries emerge early. Later refinements concentrate on local detail.

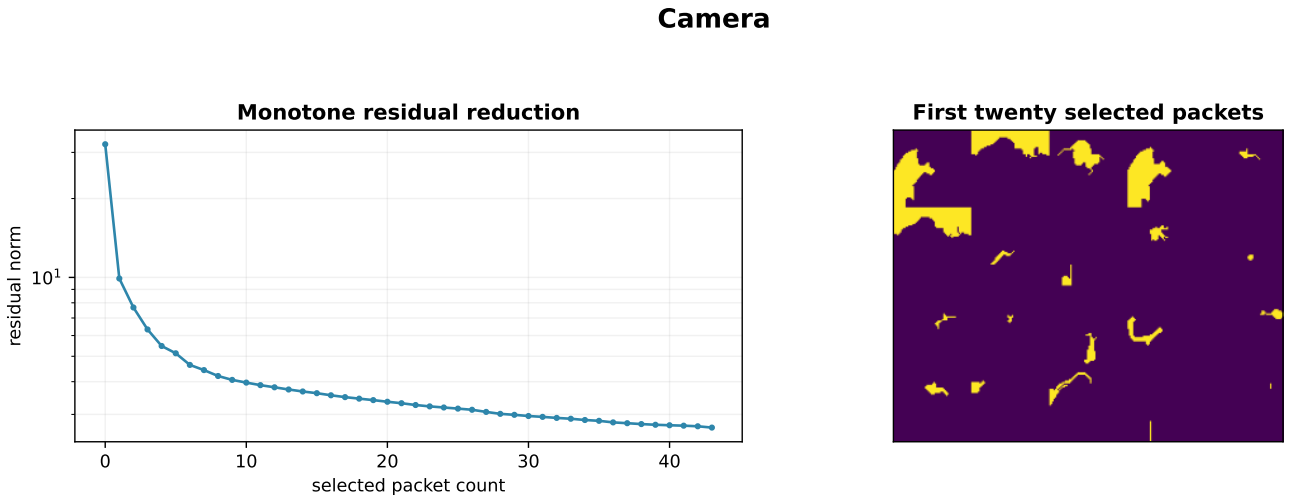


Figure 37: Camera convergence and packet examples.

Reading the figure. The selected forest includes affine, radial, soft-boundary, local DCT, polynomial, morphology, and quadtree packets.

A.4 Astronaut image



Figure 38: Reconstruction methods on the astronaut image.

Reading the figure. The packet completion roughly halves the Fourier RMSE but does not match total variation. The image has dense texture, shading, facial structure, and many interacting boundaries.

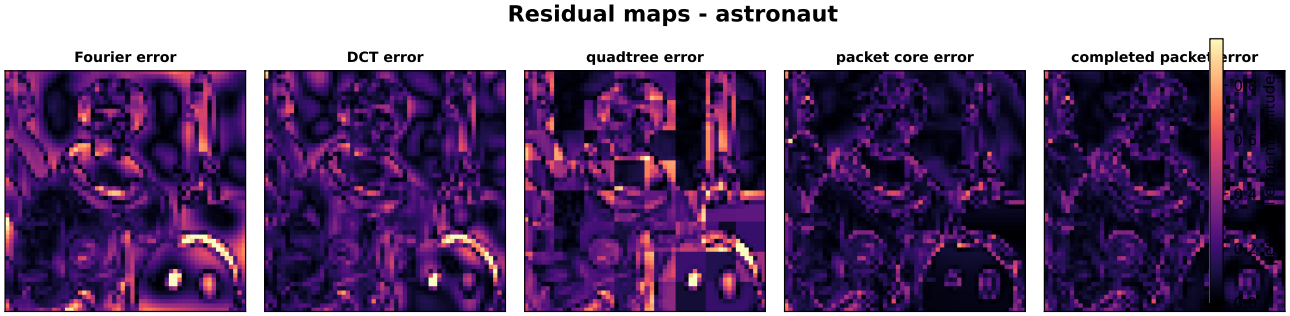


Figure 39: Astronaut residual maps.

Reading the figure. The residual identifies the structures missing from the current grammar, especially fine texture and complex local illumination.



Figure 40: Anytime astronaut reconstruction.

Reading the figure. Broad object masks appear first. The finite budget is not enough to recover the full natural-image detail.

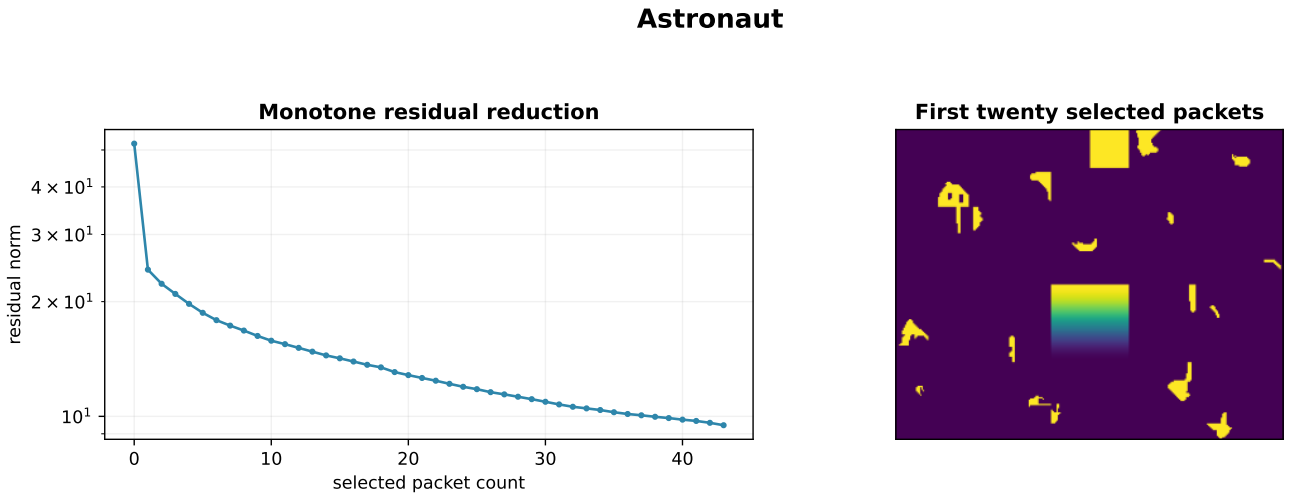


Figure 41: Astronaut convergence and packet examples.

Reading the figure. The example is a useful negative control. An endless atlas can continue refining, but a fixed budget still imposes a real approximation limit.

A.5 Global metric comparison

Table 6: Benchmark metrics for every case and method.

Case	Method	RMSE	PSNR	SSIM	Edge RMSE
synthetic geometry	fourier	0.141263	16.999	0.4787	0.225829
synthetic geometry	dct	0.124429	18.102	0.4812	0.187015
synthetic geometry	quadtree	0.157849	16.035	0.5156	0.237191
synthetic geometry	tv	0.019953	34.000	0.9664	0.031600
synthetic geometry	packet core	0.027841	31.106	0.9712	0.049632
synthetic geometry	packet completed	0.026957	31.387	0.9645	0.047588
synthetic text curves	fourier	0.116976	18.638	0.5586	0.183483
synthetic text curves	dct	0.093288	20.604	0.6009	0.148784
synthetic text curves	quadtree	0.130630	17.679	0.5051	0.214813
synthetic text curves	tv	0.019793	34.070	0.9709	0.029843
synthetic text curves	packet core	0.018993	34.428	0.9773	0.030976
synthetic text curves	packet completed	0.017660	35.060	0.9809	0.030076
camera	fourier	0.096803	20.282	0.5406	0.142583
camera	dct	0.072395	22.806	0.6094	0.116351
camera	quadtree	0.091801	20.743	0.5955	0.158815

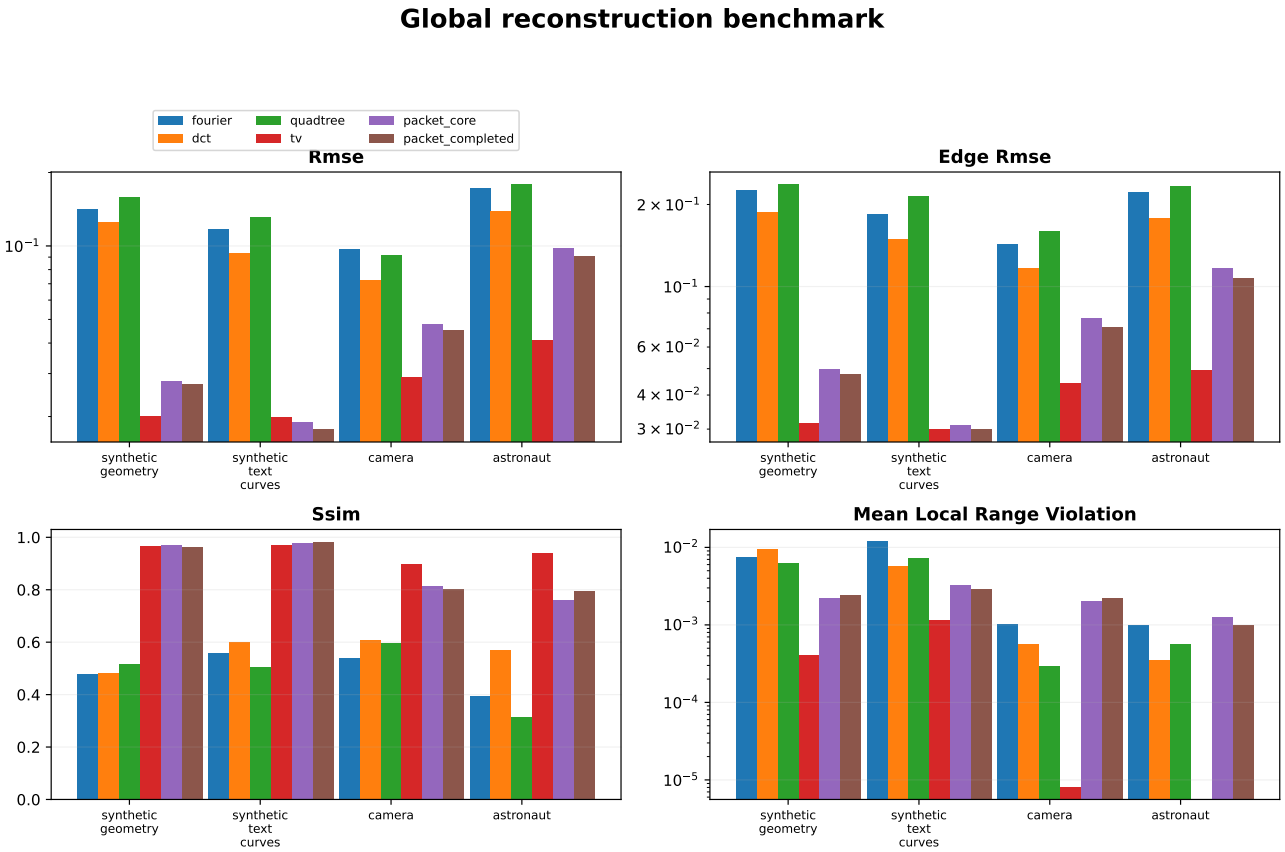


Figure 42: RMSE and SSIM across the four cases.

Reading the figure. The packet completion consistently improves on global Fourier and DCT reconstruction. Its strongest relative performance is on images whose boundaries and local fields are well matched by the generated grammar.

Case	Method	RMSE	PSNR	SSIM	Edge RMSE
camera	tv	0.028897	30.783	0.8989	0.044118
camera	packet core	0.047631	26.442	0.8149	0.076694
camera	packet completed	0.045070	26.922	0.8033	0.070887
astronaut	fourier	0.172403	15.269	0.3948	0.222275
astronaut	dct	0.138125	17.195	0.5696	0.178496
astronaut	quadtree	0.178929	14.946	0.3153	0.232900
astronaut	tv	0.041007	27.743	0.9392	0.049107
astronaut	packet core	0.097366	20.232	0.7603	0.116895
astronaut	packet completed	0.090254	20.891	0.7940	0.107059

Table 7: Selected packet types in the four image reconstructions.

Case	Packet type	Count
astronaut	dct	1
astronaut	mask	28
astronaut	masked poly	1
astronaut	morph	11
astronaut	quadtree	1
astronaut	soft mask	1
camera	affine	3
camera	mask	22
camera	masked dct	5
camera	masked poly	1
camera	morph	8
camera	quadtree	2
camera	radial	1
camera	soft mask	1
synthetic geometry	combine	4
synthetic geometry	dct	3
synthetic geometry	mask	19
synthetic geometry	masked dct	5
synthetic geometry	masked poly	4
synthetic geometry	morph	8
synthetic text curves	affine	1
synthetic text curves	dct	4
synthetic text curves	mask	22
synthetic text curves	masked dct	4
synthetic text curves	masked poly	6
synthetic text curves	morph	6

Table 8: Earlier budget sweep for the same packet-selection architecture.

Packet budget	Residual budget	Core RMSE	Completed RMSE	Seconds	Selected
8	4	0.098988	0.091417	0.162	8
16	8	0.083805	0.077948	0.399	16
24	12	0.054446	0.050016	0.560	24
32	16	0.042908	0.040863	0.894	32
48	24	0.027053	0.025120	1.513	48

Table 9: Earlier fixed-budget scaling sweep.

Side length	Pixels	Seconds	Core RMSE	Packets
32	1024	0.481	0.033871	28
48	2304	0.626	0.032763	28
64	4096	1.445	0.047798	28
80	6400	1.894	0.034348	28

Table 10: Verification ledger.

Check	Recorded value
archive identity	5.551115e-17
monotone residual max increase	0.000000e+00
pixel completion residual	5.796179e-10
native step residual	2.220446e-15
native step range violation	4.385101e-16
affine perturbation distance	8.124038e+00
radial perturbation distance	4.789757e+00
dct finite	True
fourier finite	True
global refit stationarity	2.120856e-08
bounded synthesis nonexpansive ratio	7.080496e-01
residual completion ratio	7.413943e-01
shared mask color residual	5.551115e-16
distinct descendants from one packet	29
dyadic translation step ratio	7.812500e-03
soft boundary finite	True
compositional packet finite	True
initial image derived seed masks	60
test groups	16

A.6 Budget and scaling behavior

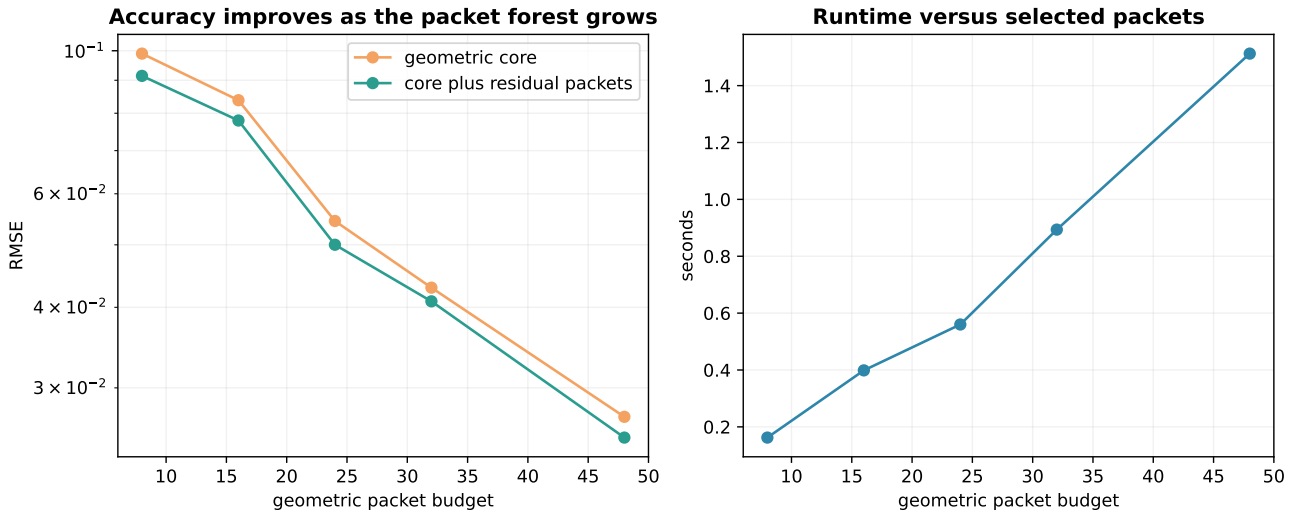


Figure 43: Earlier packet-budget sweep.

Reading the figure. Error decreases as the selected forest grows. The geometric core and residual completion are separately visible. The sweep predates the final dyadic-dense grammar but uses the same global selection and refitting architecture.

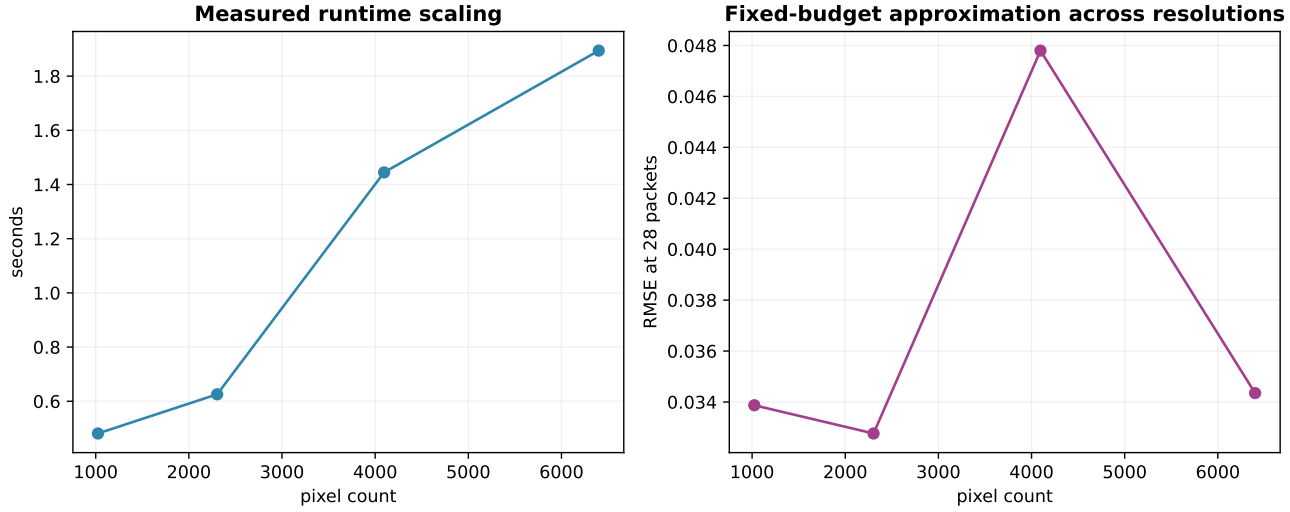


Figure 44: Earlier fixed-budget resolution sweep.

Reading the figure. Runtime grows with pixel count under a fixed selected-packet budget. Approximation error is not monotone in resolution because the image geometry and finite candidate frontier change with raster size.

B Singular Packet Decomposition baseline

Historical baseline. This appendix records the global-SVD and first-generation SPD comparison from which AM-SPD was developed. The principal results in Section 6 supersede these values.

B.1 Direct comparison with SVD

The main comparison uses twenty-two global SVD modes or twenty-two adaptive singular packets. A second comparison matches approximate added scalar-parameter counts. One joint rank-one SVD mode for a

56×56 RGB image uses about

$$56 + 3 \cdot 56 + 1 = 225$$

scalars. A local singular packet uses

$$h_\Omega + Cw_\Omega + 6$$

scalars under the descriptor used in the benchmark.

Table 11: Direct SVD and Singular Packet Decomposition comparison.

Case	Method	RMSE	SSIM	Edge RMSE	Added parameters
synthetic geometry	global SVD, rank 22	0.036736	0.8505	0.045686	4950
synthetic geometry	pure singular packets, 22	0.044286	0.8148	0.056642	4391
synthetic geometry	packet core, 44	0.027841	0.9712	0.049632	–
synthetic geometry	packet core + 22 DCT coordinates	0.026957	0.9645	0.047588	110
synthetic geometry	packet core + parameter-matched residual SVD	0.007631	0.9907	0.010610	2025
synthetic geometry	packet core + 22 adaptive singular packets	0.003726	0.9974	0.005311	2241
synthetic text curves	global SVD, rank 22	0.021527	0.9660	0.031816	4950
synthetic text curves	pure singular packets, 22	0.027079	0.9452	0.039884	4399
synthetic text curves	packet core, 44	0.018993	0.9773	0.030976	–
synthetic text curves	packet core + 22 DCT coordinates	0.017660	0.9809	0.030076	110
synthetic text curves	packet core + parameter-matched residual SVD	0.005063	0.9958	0.007231	2475
synthetic text curves	packet core + 22 adaptive singular packets	0.003394	0.9981	0.004501	2639
camera	global SVD, rank 22	0.009686	0.9748	0.012173	2486
camera	pure singular packets, 22	0.011713	0.9681	0.015185	2541
camera	packet core, 44	0.047631	0.8149	0.076694	–
camera	packet core + 22 DCT coordinates	0.045070	0.8033	0.070887	66
camera	packet core + parameter-matched residual SVD	0.015359	0.9444	0.019775	2034
camera	packet core + 22 adaptive singular packets	0.014921	0.9458	0.019076	2123
astronaut	global SVD, rank 22	0.024875	0.9767	0.027807	4950
astronaut	pure singular packets, 22	0.025464	0.9757	0.028095	5063
astronaut	packet core, 44	0.097366	0.7603	0.116895	–
astronaut	packet core + 22 DCT coordinates	0.090254	0.7940	0.107059	110
astronaut	packet core + parameter-matched residual SVD	0.033127	0.9571	0.036047	4950
astronaut	packet core + 22 adaptive singular packets	0.032976	0.9567	0.035665	5060

Table 12: Canonical SVD of the selected packet span.

Case	Packets	Rank	Condition	99% energy	99.9% energy	Top-10 energy
synthetic geometry	44	44	47.53	26	38	0.923
synthetic text curves	44	44	33.14	22	36	0.956
camera	44	44	43.53	17	27	0.960
astronaut	44	44	6.99	30	41	0.918

Table 13: Parameter-matched residual comparison.

Case	SPD modes	SPD parameters	SVD rank	SVD parameters	RMSE: SPD / SVD
synthetic geometry	22	2241	9	2025	0.003726 / 0.007631
synthetic text curves	22	2639	11	2475	0.003394 / 0.005063
camera	22	2123	18	2034	0.014921 / 0.015359
astronaut	22	5060	22	4950	0.032976 / 0.033127

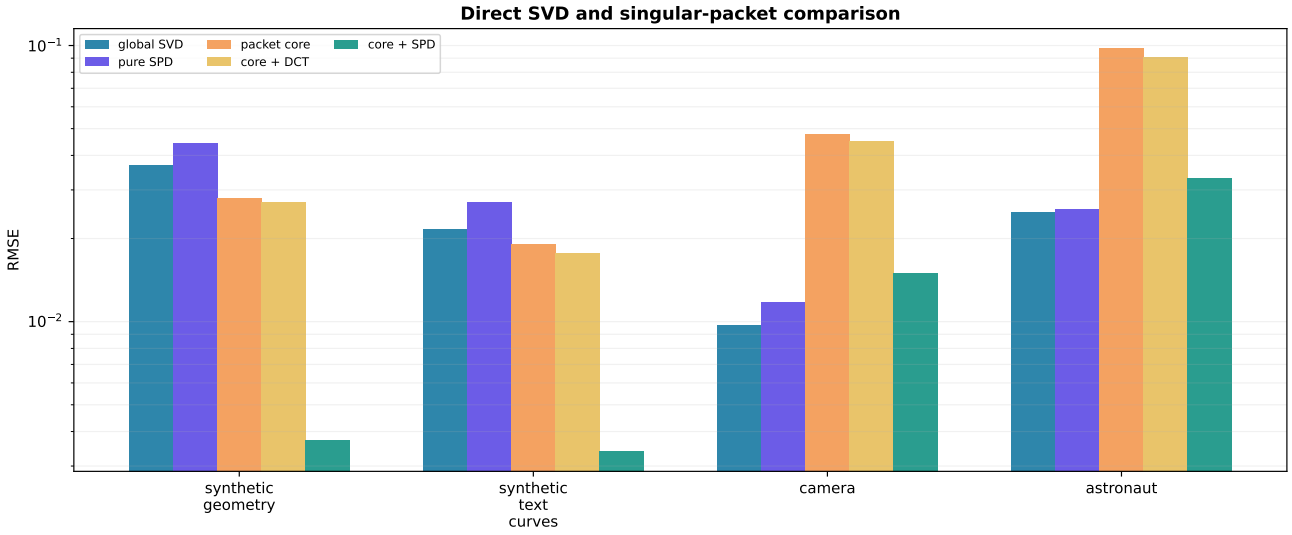


Figure 45: Global comparison of SVD and packet methods.

Reading the figure. Global SVD is a strong pure low-rank model. The hybrid singular-packet method is strongest on the structured synthetic images and becomes competitive with global SVD on the natural images after the geometry core is supplemented by local singular modes.

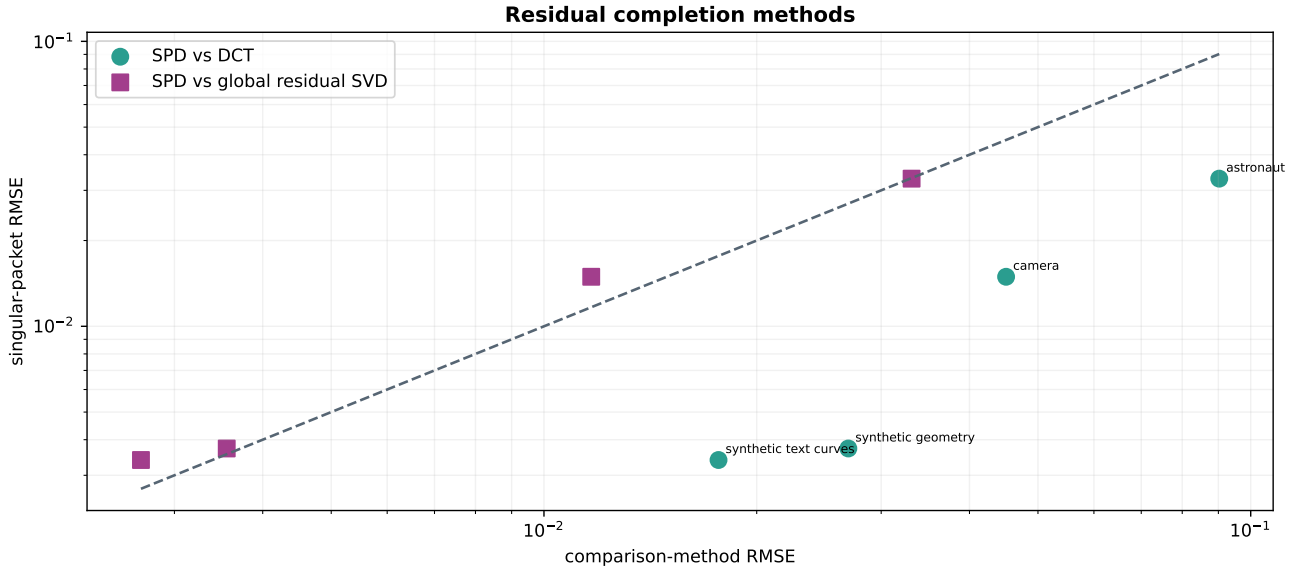


Figure 46: Residual completion methods.

Reading the figure. At similar added parameter counts, adaptive singular packets improve on parameter-matched global residual SVD in all four cases. At the same component count, global residual SVD remains slightly better in two cases because each global rank-one component carries more parameters.

B.2 Synthetic geometry under SVD and SPD

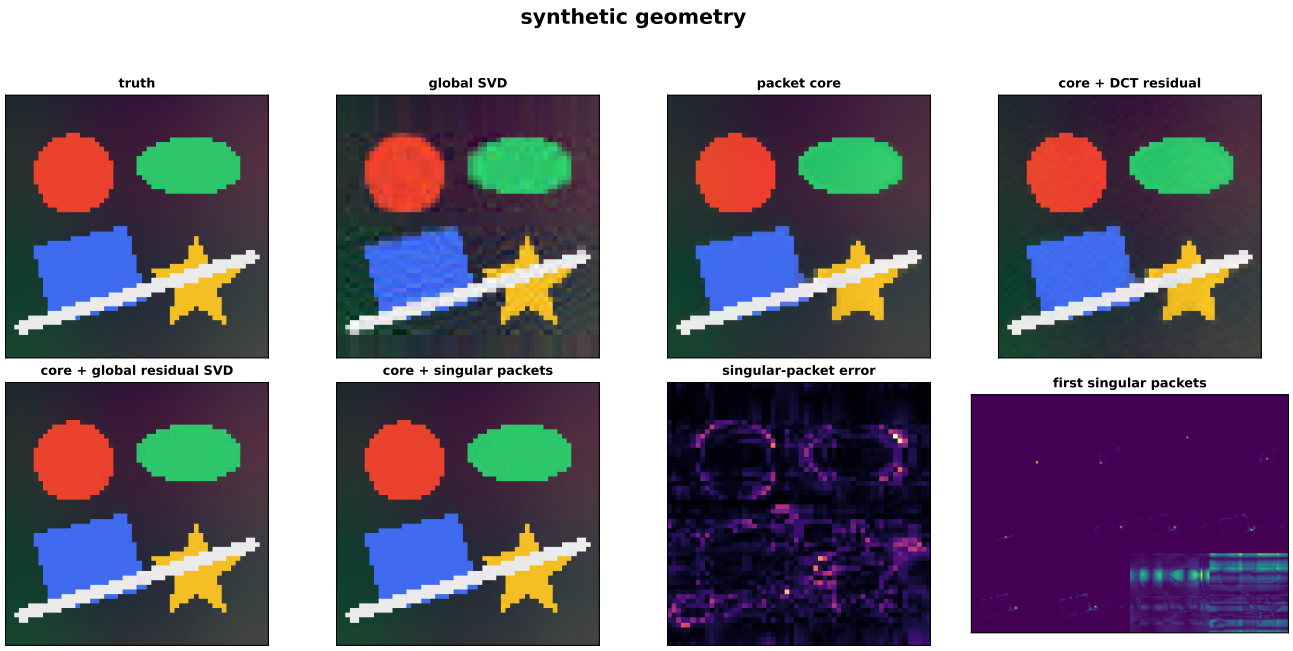


Figure 47: SVD and singular-packet reconstructions of synthetic geometry.

Reading the figure. The geometric core already resolves the discontinuities. Local singular packets then model smooth fields and remaining local structure. The hybrid result reaches RMSE 0.003726, compared with 0.036736 for rank-twenty-two global SVD.

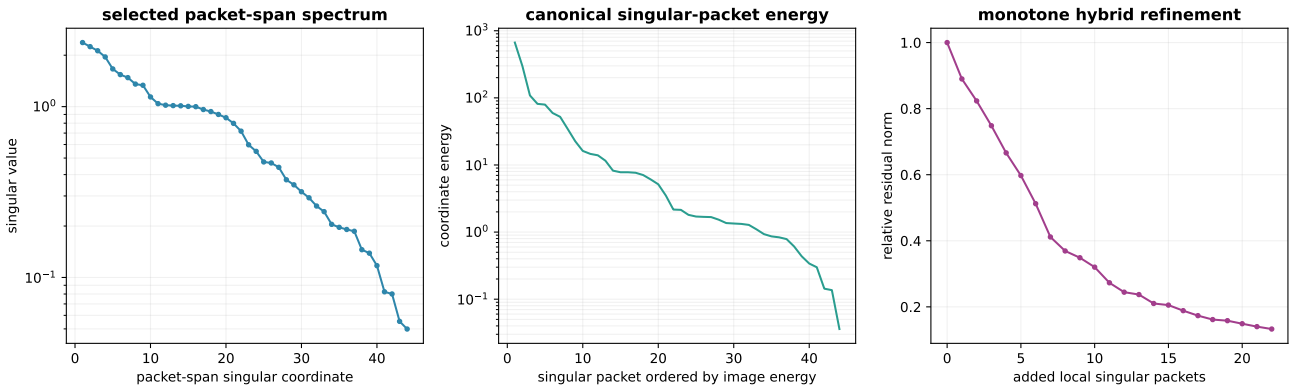


Figure 48: Packet singular spectrum and hybrid convergence.

Reading the figure. The packet-span SVD is exact. The local singular-packet residual falls monotonically over twenty-two updates.

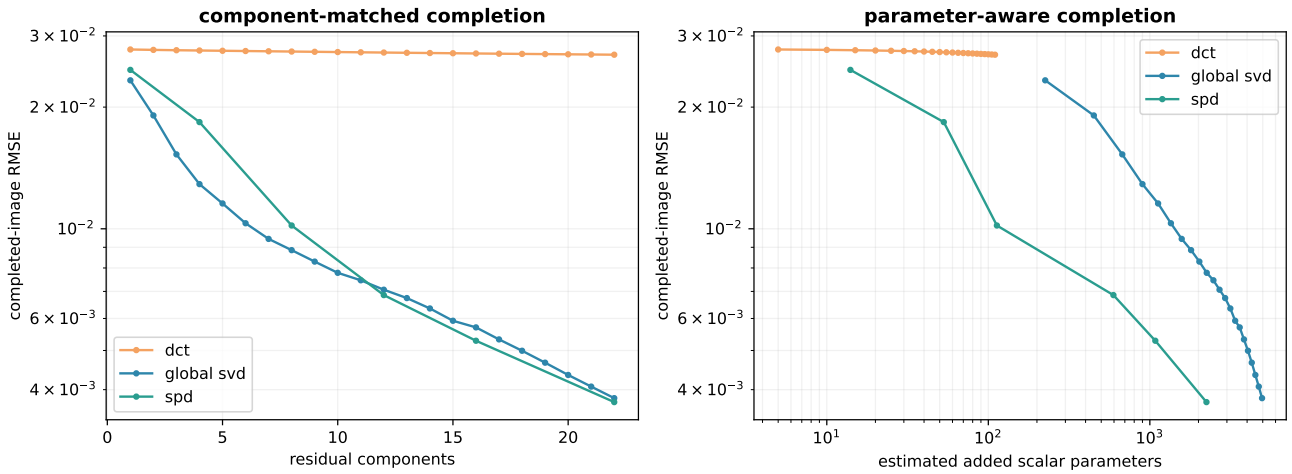


Figure 49: Residual rate-distortion comparison for synthetic geometry.

Reading the figure. Global residual SVD is strongest at equal component count. Singular packets are stronger at approximately matched added parameter count because their modes concentrate on selected regions.



Figure 50: Adaptive singular-packet supports for synthetic geometry.

Reading the figure. The selected rectangles move between broad field corrections and localized boundary or object regions. The singular vectors inside each region remain data adaptive.

B.3 Synthetic text and curves under SVD and SPD

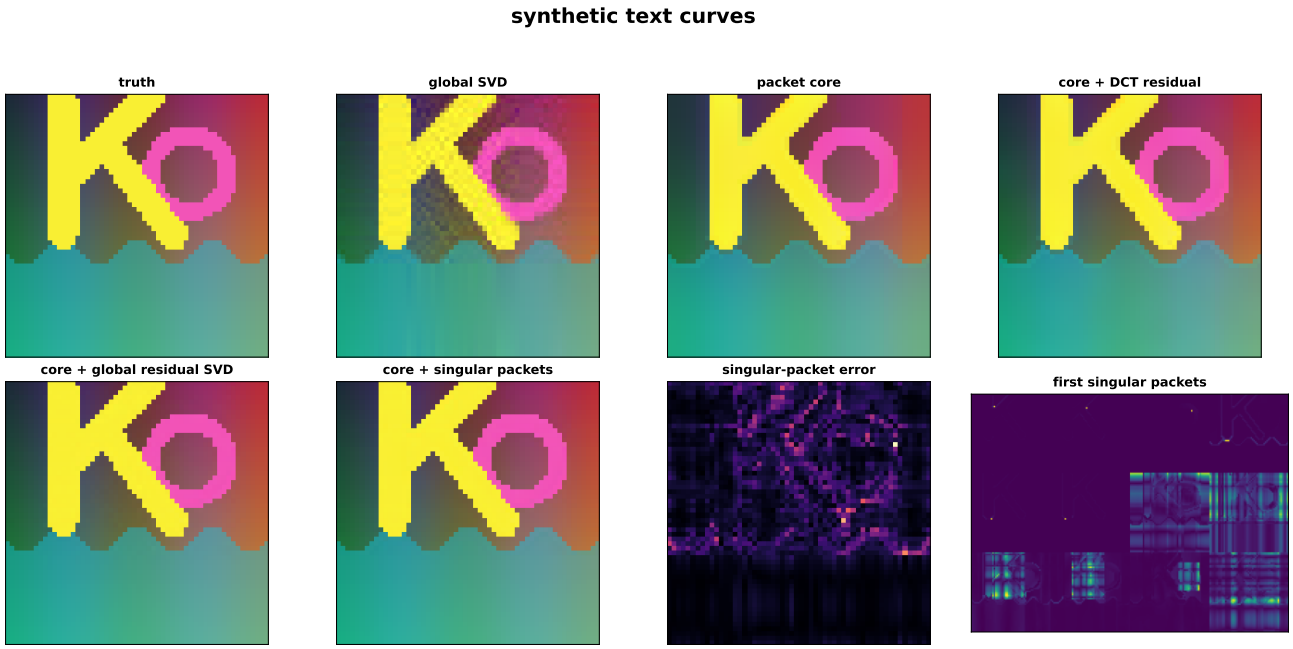


Figure 51: SVD and singular-packet reconstructions of text and curves.

Reading the figure. The hybrid result reaches RMSE 0.003394. Thin structures benefit from geometry packets before low-rank residual completion.

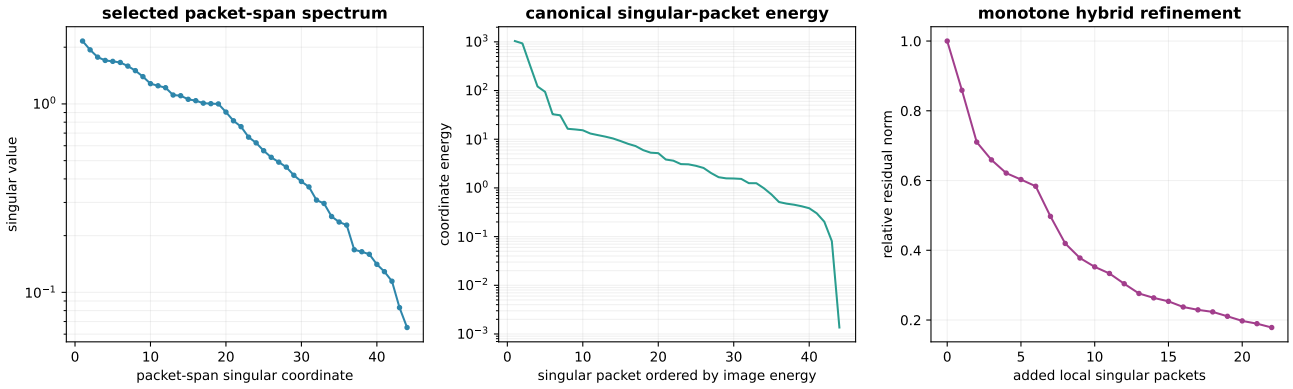


Figure 52: Packet singular spectrum and convergence for text and curves.

Reading the figure. Twenty-two singular packet coordinates retain 99% of packet-core energy. The residual refinement remains monotone.

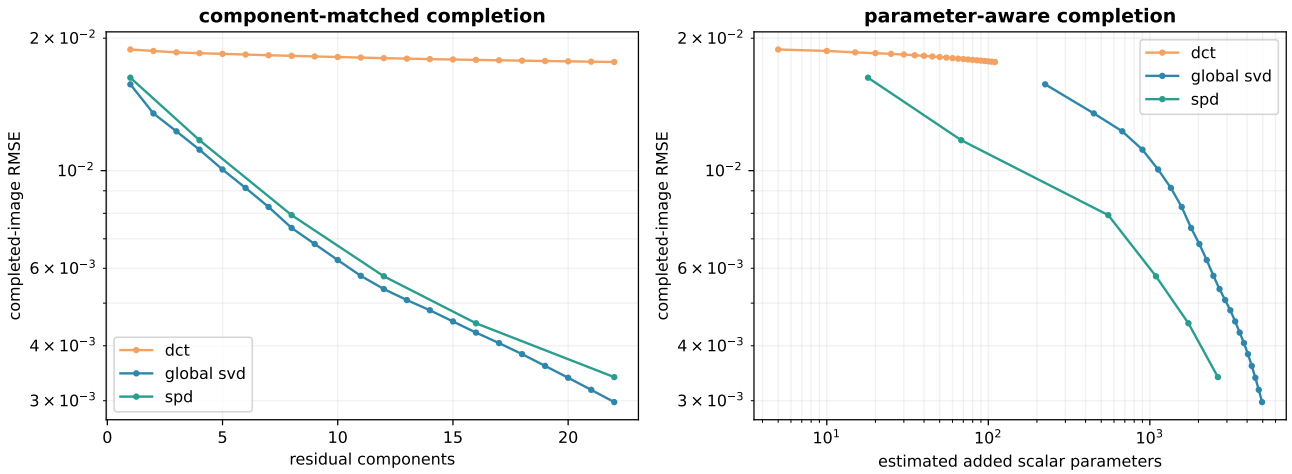


Figure 53: Residual rate-distortion comparison for text and curves.

Reading the figure. The global residual SVD has the lowest error at twenty-two full components. The local singular packets achieve lower error when the comparison is adjusted to similar parameter counts.

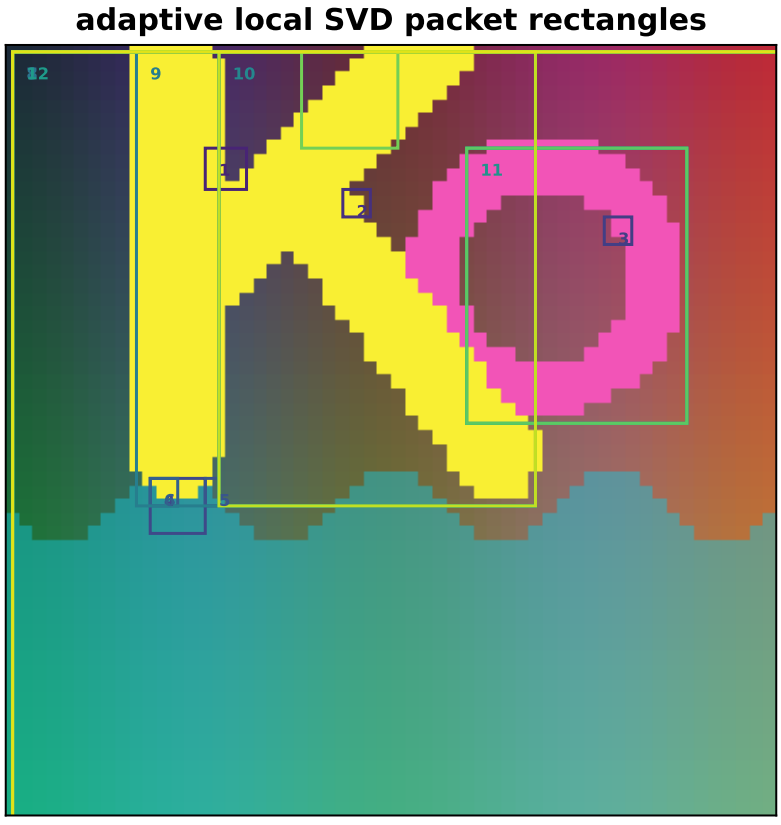


Figure 54: Adaptive singular supports for text and curves.

Reading the figure. Several selected regions follow narrow strokes and curved boundaries rather than spanning the entire image.

B.4 Camera under SVD and SPD

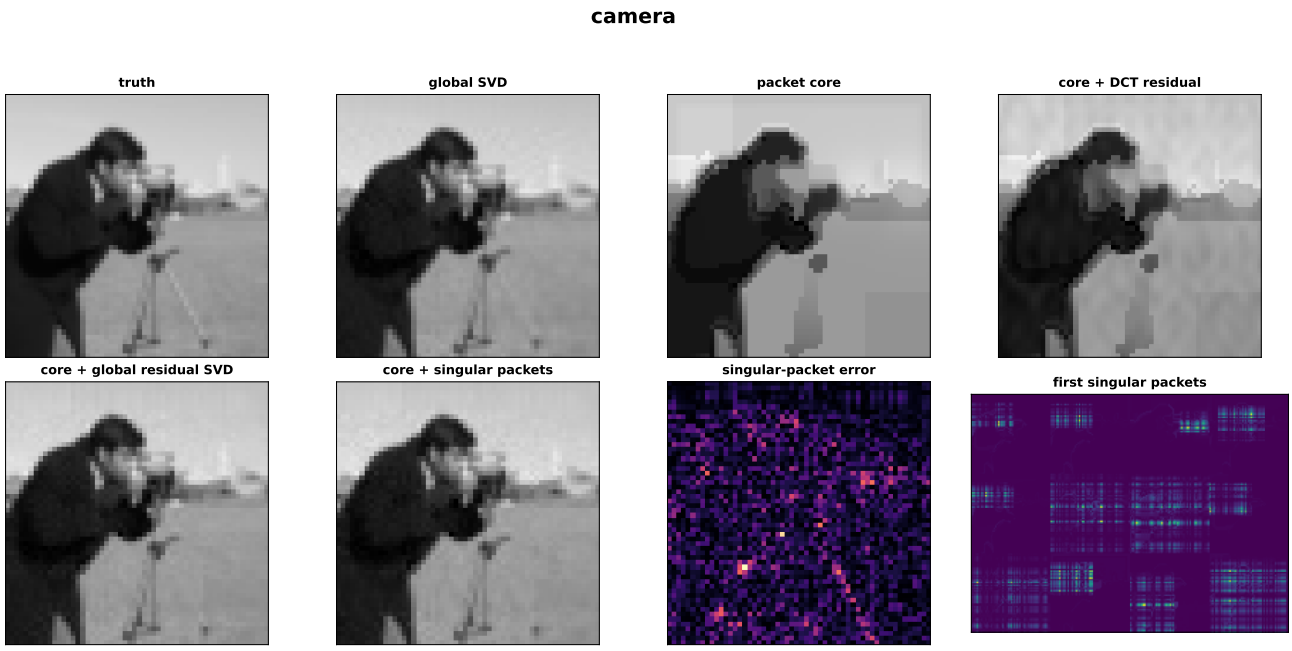


Figure 55: SVD and singular-packet reconstructions of the camera image.

Reading the figure. Global SVD remains the strongest pure model. Hybrid SPD reduces packet-core RMSE from 0.047631 to 0.014921 and approaches the parameter-matched residual SVD result 0.015359.

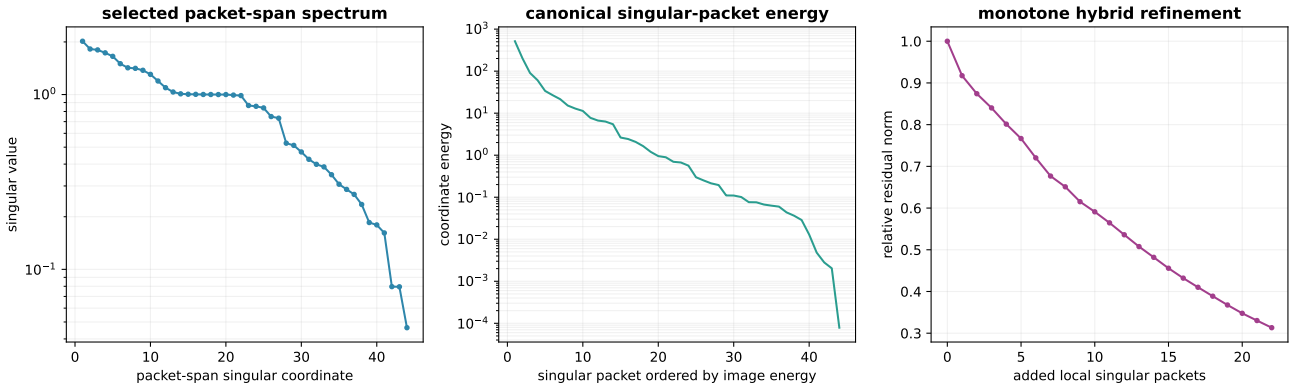


Figure 56: Camera packet spectrum and singular refinement.

Reading the figure. The packet core has a moderately ill-conditioned span. Its top ten singular coordinates carry about 96% of core energy.

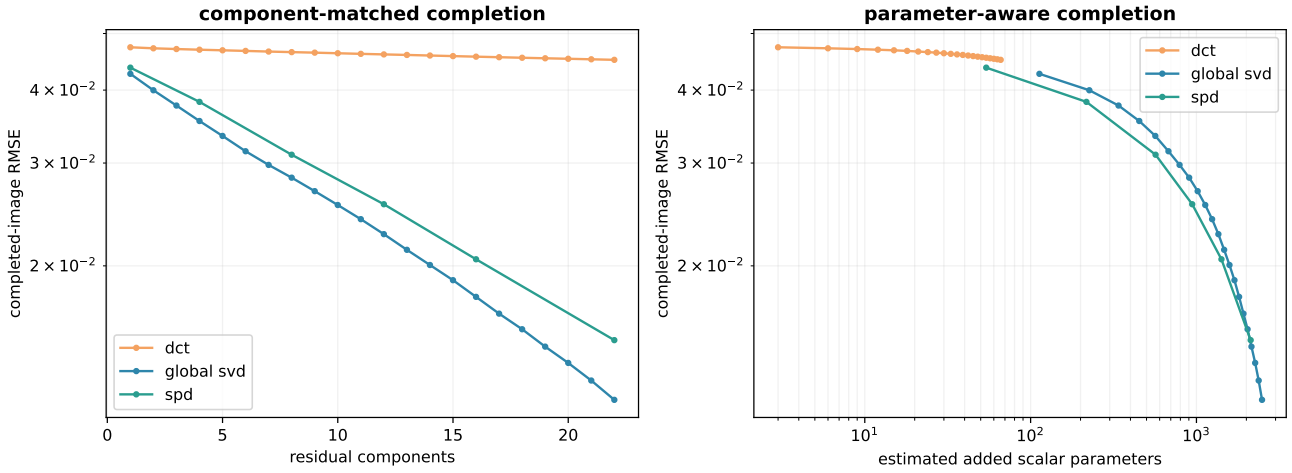


Figure 57: Camera residual rate-distortion comparison.

Reading the figure. DCT residual coordinates are inexpensive but weak at this small budget. Local and global singular modes capture natural-image variation more effectively.

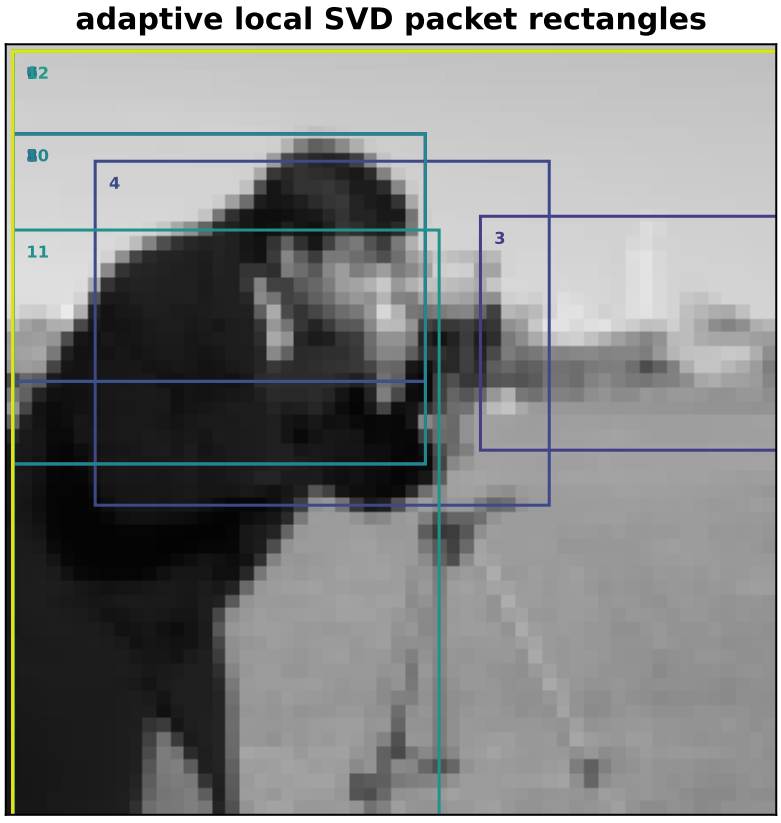


Figure 58: Adaptive camera singular-packet supports.

Reading the figure. The local rank modes repeatedly select high-variance subject and background regions.

B.5 Astronaut under SVD and SPD

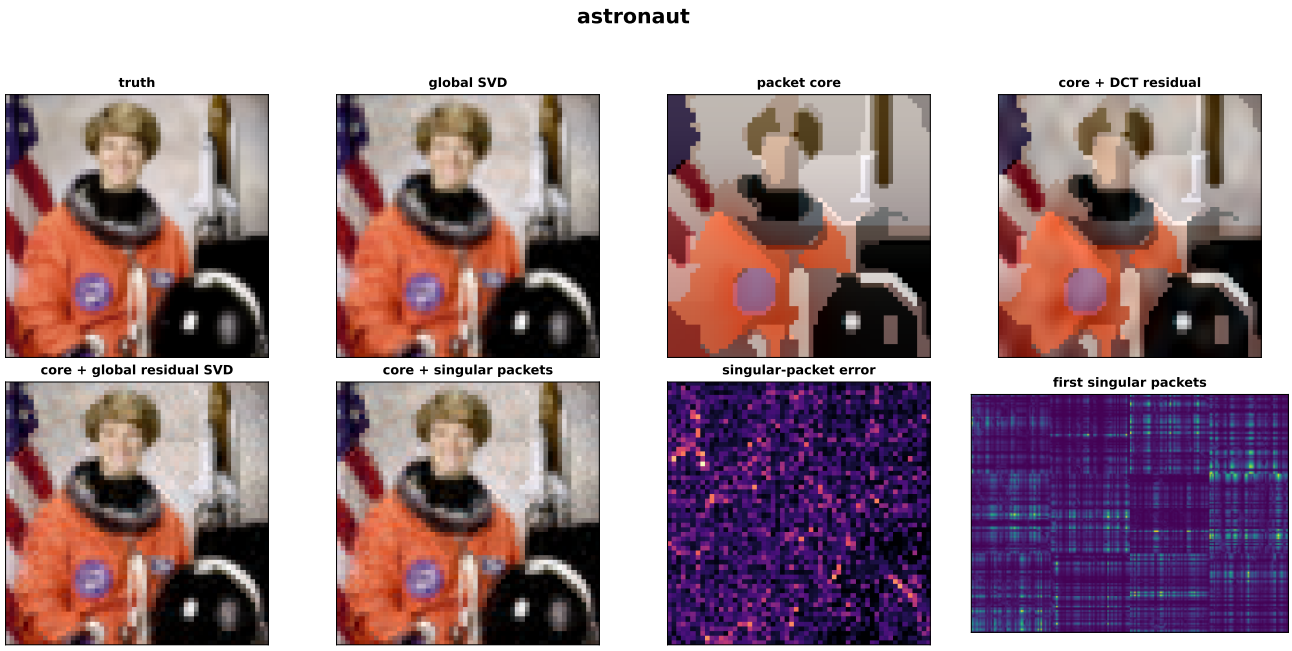


Figure 59: SVD and singular-packet reconstructions of the astronaut image.

Reading the figure. Pure SPD and global SVD are close. The hybrid reaches RMSE 0.032976, slightly below the globally refitted residual-SVD result 0.033127 at a similar parameter count.

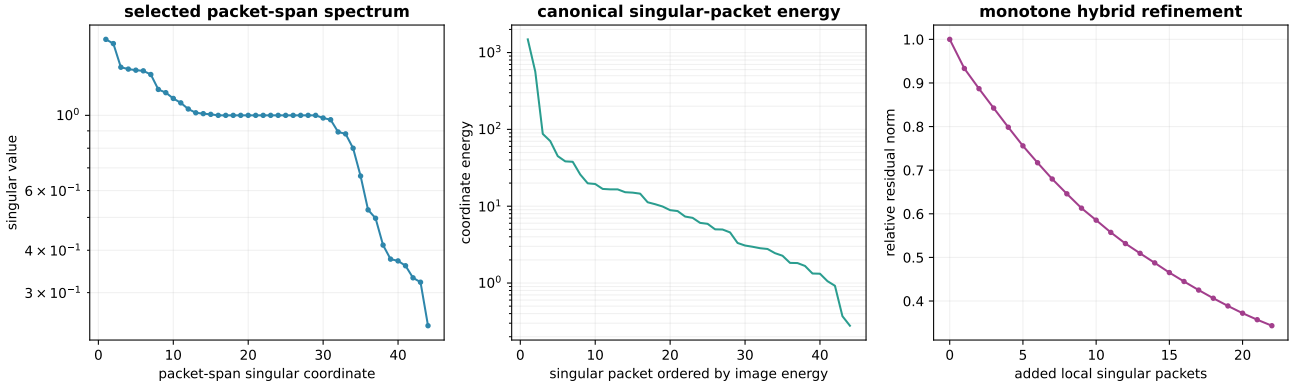


Figure 60: Astronaut packet spectrum and convergence.

Reading the figure. The packet span is better conditioned than in the other tests, but more singular coordinates are needed to retain the same energy because the image contains denser structure.

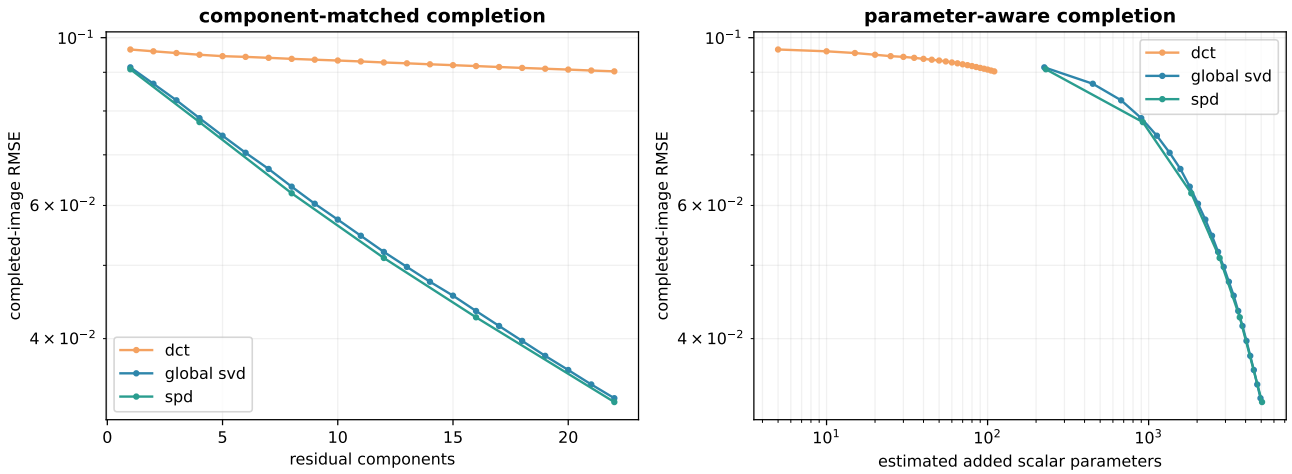


Figure 61: Astronaut residual rate-distortion comparison.

Reading the figure. Global and local singular modes have nearly identical parameter-aware performance at the final budget.

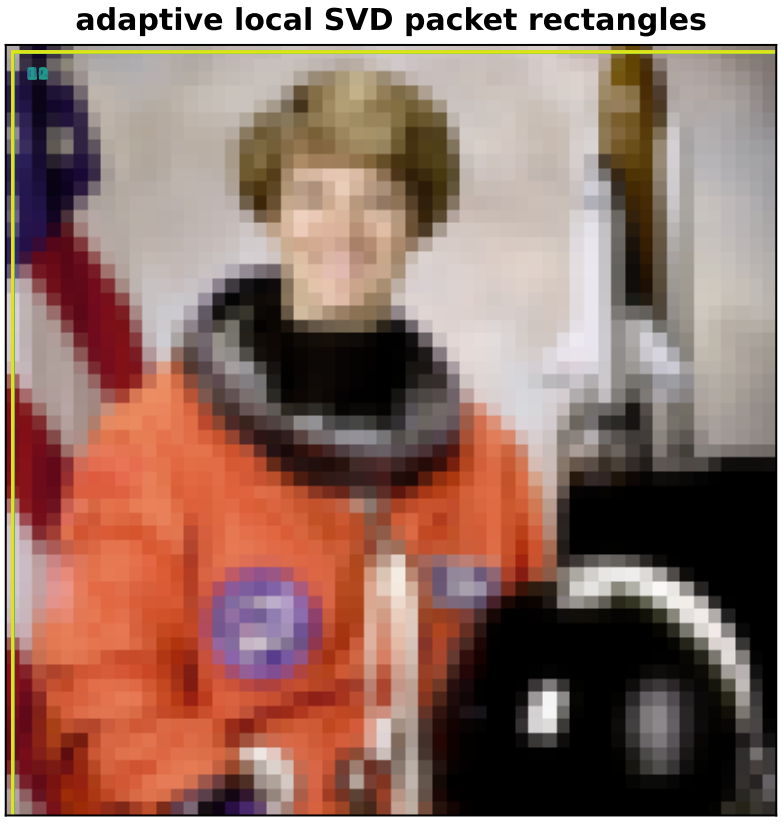


Figure 62: Adaptive astronaut singular-packet supports.

Reading the figure. The support path remains broader than in the synthetic cases because texture, shading, and object boundaries overlap throughout the image.

C Intermediate two-seed development study

Development record. The two-seed experiment below preceded the exact-budget third seed. It is retained to show which empirical observation motivated the final portfolio, not as the definitive benchmark.

C.1 Local singular subspaces

A residual patch has shape $h \times w \times C$. The algorithm tests

$$R_\Omega^{\text{row}} \in \mathbb{R}^{h \times wC}, \qquad R_\Omega^{\text{column}} \in \mathbb{R}^{w \times hC}, \qquad R_\Omega^{\text{channel}} \in \mathbb{R}^{C \times hw}.$$

For orientation o and rank r,

$$R_{\Omega,o,r} = \sum_{j=1}^{r} \sigma_j u_j v_j^{\mathsf{T}}.$$

The best local rank theorem proved in Part VIII makes this the optimal rank-r model for that local unfolding. The algorithm compares several unfoldings because low-rank truncation depends on how the local tensor is matricized.

For the row unfolding, the declared scalar parameter count is

$$C_{\Omega,\text{row},r} = 6 + r(h + wC + 1).$$

The other orientations use the corresponding matrix dimensions. This is a consistent parameter model, not a complete entropy code.

C.2 Exact best basis on a finite tree

A dyadic tree is built over the residual. Each node can stop with one local singular model or split into all of its children. For rate multiplier λ,

$$J_{\text{keep}}(\nu, o, r) = E_{\nu,o,r} + \lambda C_{\nu,o,r}.$$

Theorem C.1 (Finite-tree recursion)**.** *For a declared finite tree and additive local costs,*

$$J^*(\nu) = \min \left\{ \min_{o,r} J_{\text{keep}}(\nu, o, r), \sum_{\mu \in \text{ch}(\nu)} J^*(\mu) \right\}$$

returns a globally optimal pruning for the chosen λ.

Proof. Every representation of a leaf is a stop model. Every representation of an internal node either stops at that node or represents every child. The induction hypothesis supplies the optimal value in each child subtree. Taking the smaller complete alternative proves the recursion. □

The implementation follows the finite rate–distortion path, keeps each distinct pruning that appears, and chooses the smallest exact residual that fits the inherited budget. The theorem applies to this declared tree only.

C.3 Exact post-projection gain

Let B be the active orthonormal basis and let D contain the raw directions of one candidate block. Remove the represented span:

$$D_\perp = (I - BB^{\mathsf{T}})D.$$

Let Q be an orthonormal basis of range $D_\perp$. For current residual r,

$$\boxed{G(D \mid B) = \|Q^{\mathsf{T}} r\|_2^2.}$$

Proposition C.2 (Block improvement)**.** *After adding Q and globally refitting,*

$$\|r_{\text{new}}\|_2^2 = \|r\|_2^2 - G(D \mid B).$$

Proof. The new basis directions are orthonormal and orthogonal to the old span. Pythagoras gives the exact reduction. □

Candidate ranking uses

$$\text{score}(D) = \frac{G(D \mid B)}{C(D)^\alpha}.$$

The gain is exact. The cost exponent is a declared search preference.

C.4 Two seed families

The first seed is the original greedy rank-one SPD. It remains effective when a small number of isolated modes explain the residual.

The second seed is the exact finite-tree best basis. It is stronger when broad and local structures should be resolved at different depths before the moving search begins.

Each seed is retained unchanged and is also tested at several reopened fractions.

Theorem C.3 (Intermediate two-seed nonregression)**.** *Let $\mathfrak{F}$ contain the unchanged greedy seed, the unchanged tree seed, and all declared backward-replacement candidates. If*

$$\widehat{F}_* = \arg\min_{\widehat{F}\in\mathfrak{F}} \|F - \widehat{F}\|_F,$$

then

$$\|F - \widehat{F}_*\|_F \le \min\{\|F - \widehat{F}_{\text{greedy}}\|_F, \|F - \widehat{F}_{\text{tree}}\|_F\}.$$

Proof. Both unchanged seeds belong to $\mathfrak{F}$. The minimum over the portfolio cannot exceed either included value. □

C.5 Two-seed benchmark that led to the final model

This intermediate benchmark retains the four 56×56 images and the added-parameter budgets from the earlier SPD comparison. It records the two-seed stage that motivated the exact-budget and three-seed final formulation in the next part.

Table 14: Final bounded RMSE under the inherited added-parameter budgets.

Image	Original SPD	Tree seed	Matched residual SVD	Final AM-SPD	Parameters
Synthetic geometry	0.003726	0.004663	0.007631	**0.003470**	2218/2241
Synthetic text and curves	0.003394	0.004160	0.005063	**0.003260**	2602/2639
Camera	0.014921	0.007865	0.015359	**0.005738**	2121/2123
Astronaut	0.032976	0.013749	0.033127	**0.010691**	5017/5060

Table 15: Winning seed and backward-replacement structure.

Image	Seed	Reopened	Kept	Removed	Blocks	New modes	Tree leaves
Synthetic geometry	Greedy	35.0%	18	4	5	6	28
Synthetic text and curves	Tree	25.0%	38	15	4	7	31
Camera	Tree	12.0%	35	5	6	7	31
Astronaut	Tree	12.0%	57	9	4	5	43

Table 16: Final improvement and image-quality checks.

Image	RMSE reduction	Final SSIM	Exact RMSE	Runtime (s)	Matched SVD beaten
Synthetic geometry	6.9%	0.997798	0.003470	12.10	Yes
Synthetic text and curves	4.0%	0.998061	0.003260	11.38	Yes
Camera	61.5%	0.986003	0.005738	0.28	Yes
Astronaut	67.6%	0.994331	0.010742	1.60	Yes

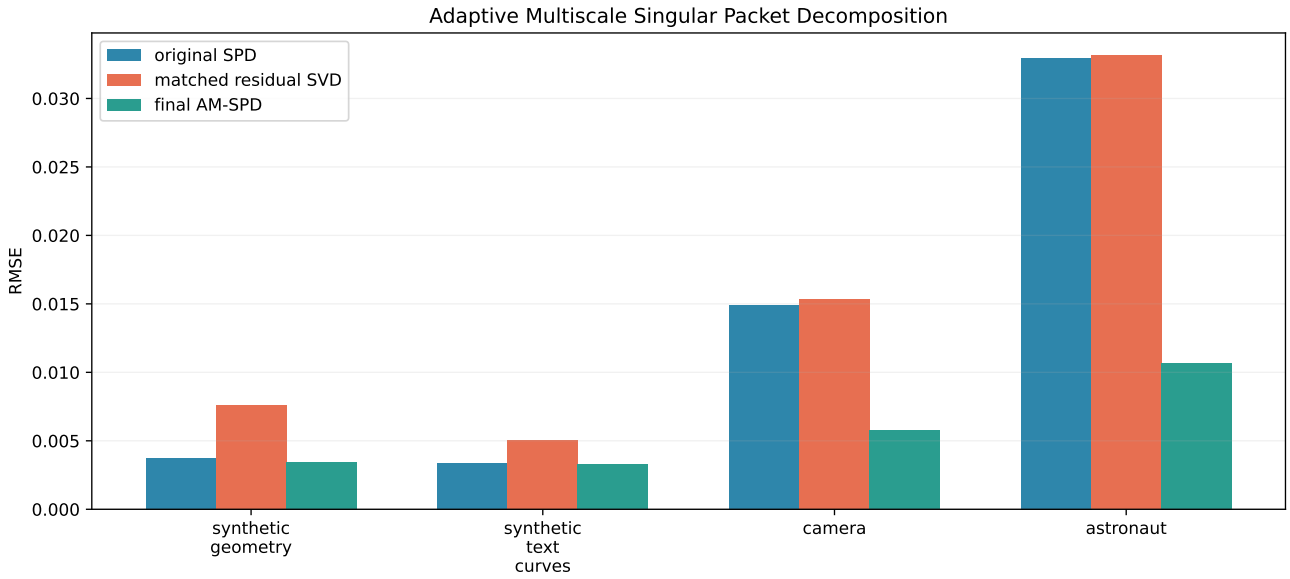


Figure 63: Final RMSE comparison.

Reading the figure. AM-SPD improves on the original SPD and the approximately parameter-matched residual SVD in every case. The largest gains occur on the two natural images.

Table 17: Aggregate finite verification.

Check	Result
Images tested	4
Exact nonregression certificates	4/4
Final representations within budget	4/4
Strict exact-RMSE improvements	4/4
Strict bounded-RMSE improvements	4/4
Matched residual SVD beaten	4/4
Greedy seed selected	1/4
Tree seed selected	3/4
Final new figure pairs	23

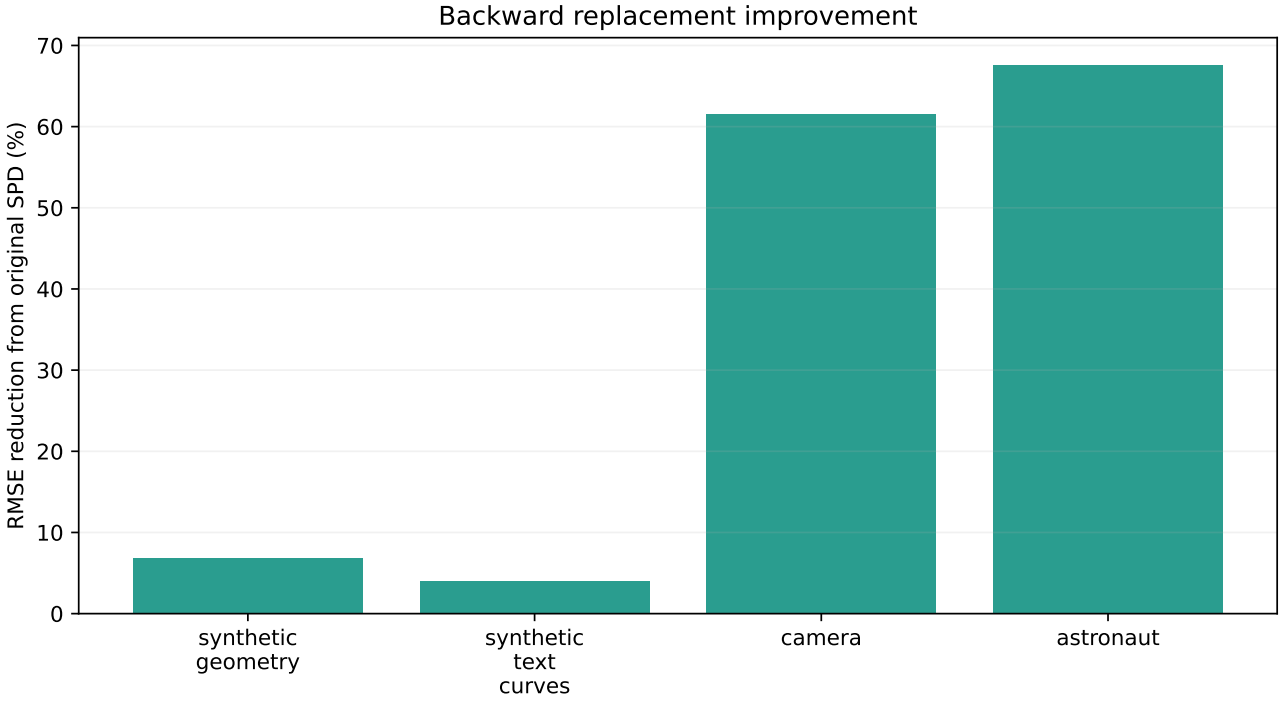


Figure 64: Improvement relative to the original SPD.

Reading the figure. Backward replacement reduces bounded RMSE by 6.9%, 4.0%, 61.5%, and 67.6% on the four images.

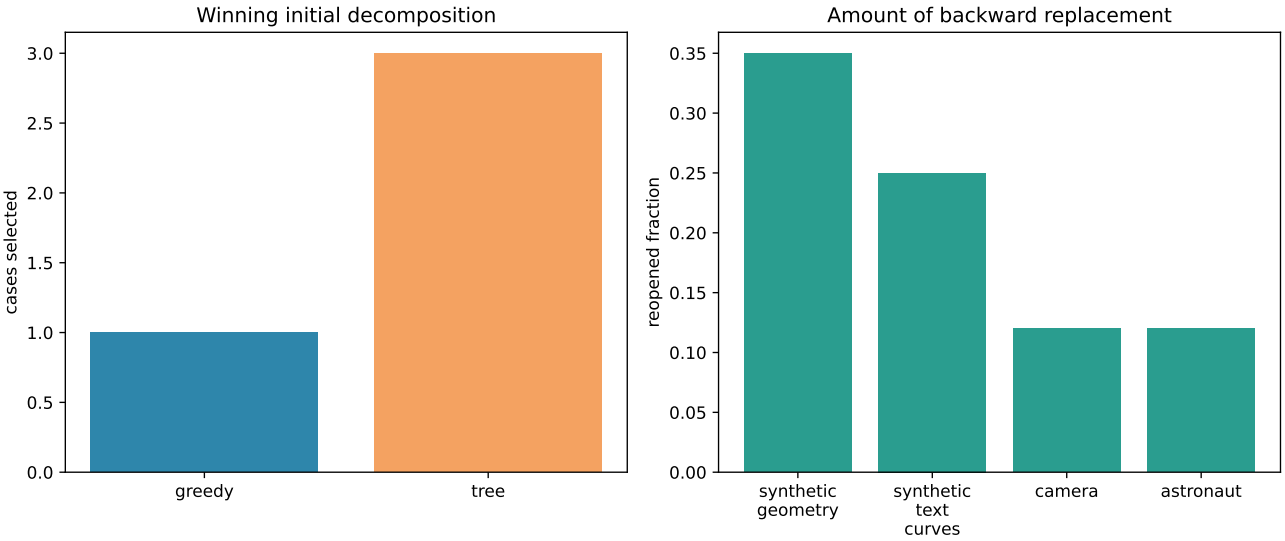


Figure 65: Winning seeds and reopened fractions.

Reading the figure. Synthetic geometry selects the greedy seed. The other three images select the tree seed. The reopened fraction ranges from 12% to 35%.

C.6 Synthetic geometry two-seed result

The synthetic-geometry result starts from the greedy seed. Four directions are removed. Five blocks contribute six replacement directions. The final model uses 2218 of 2241 parameters.

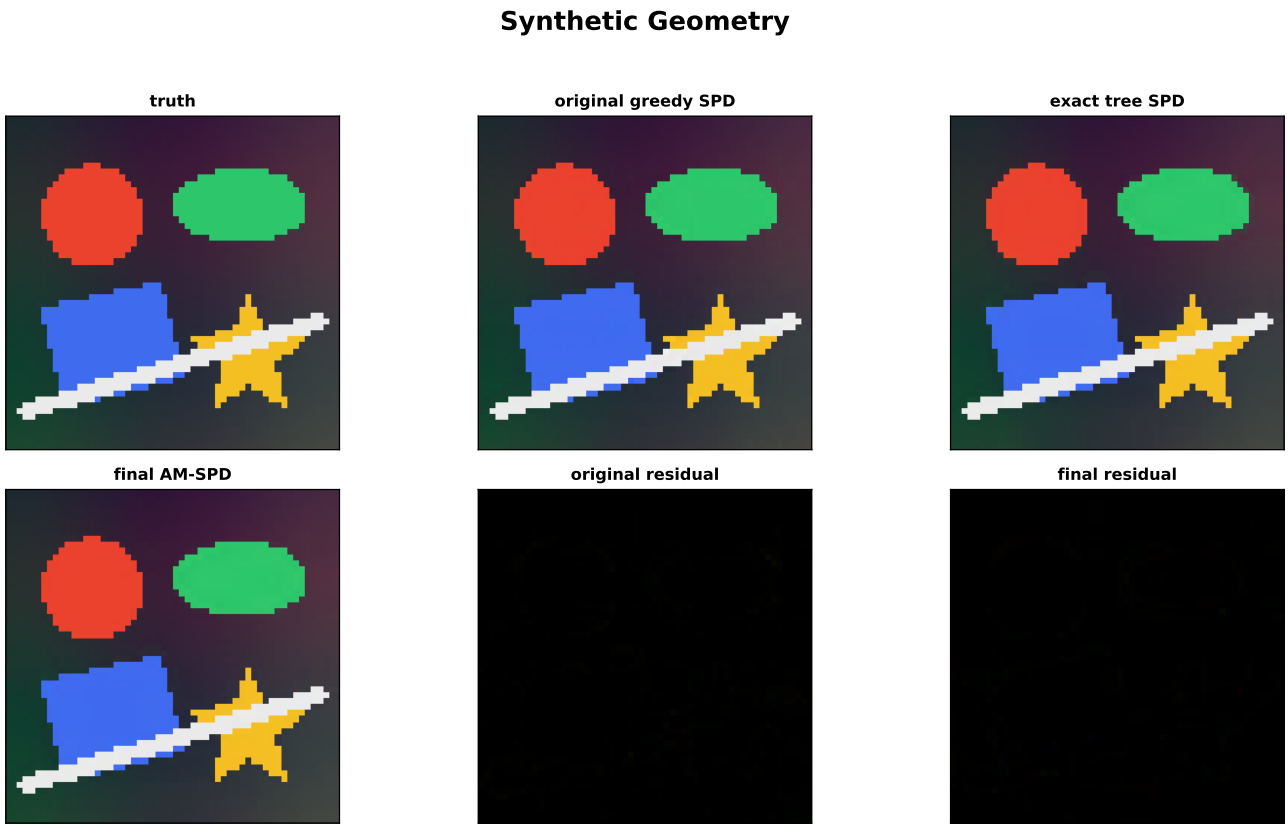


Figure 66: Final synthetic-geometry reconstruction.

Reading the figure. The exact RMSE falls from 0.003726 to 0.003470. The residual panels show the structure recovered by the backward step.

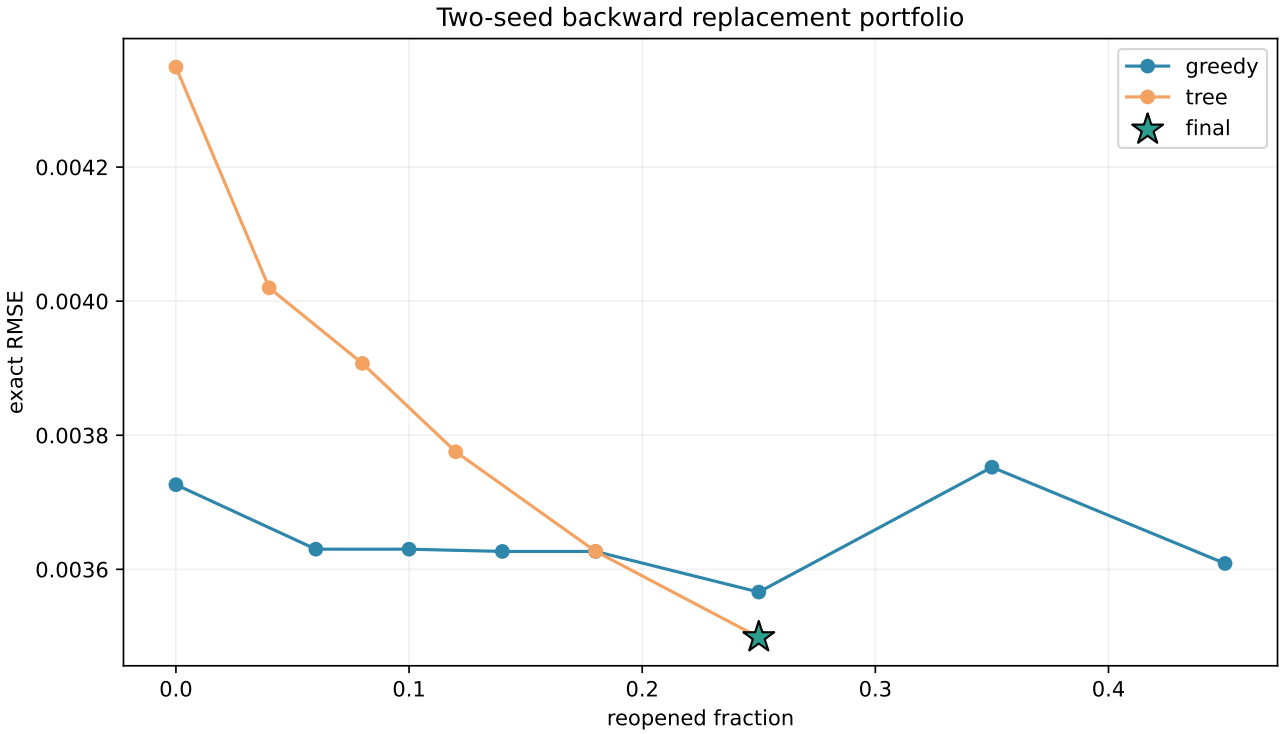


Figure 67: Synthetic-geometry portfolio path.

Reading the figure. The greedy seed remains stronger than the tree seed. Reopening 35% of its added budget gives the best final candidate.

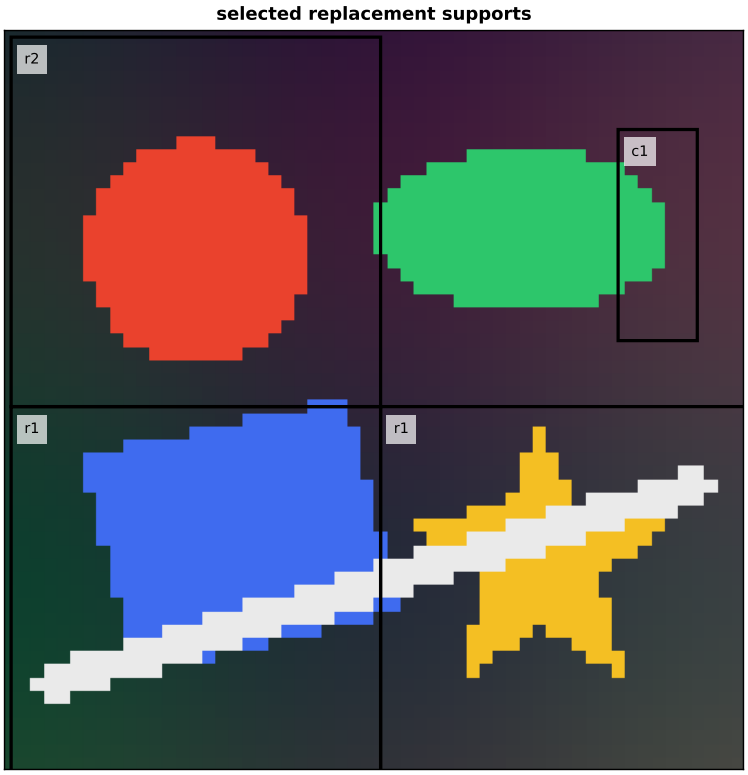


Figure 68: Synthetic-geometry replacement supports.

Reading the figure. The selected blocks revisit supports that were not retained by the original forward path.

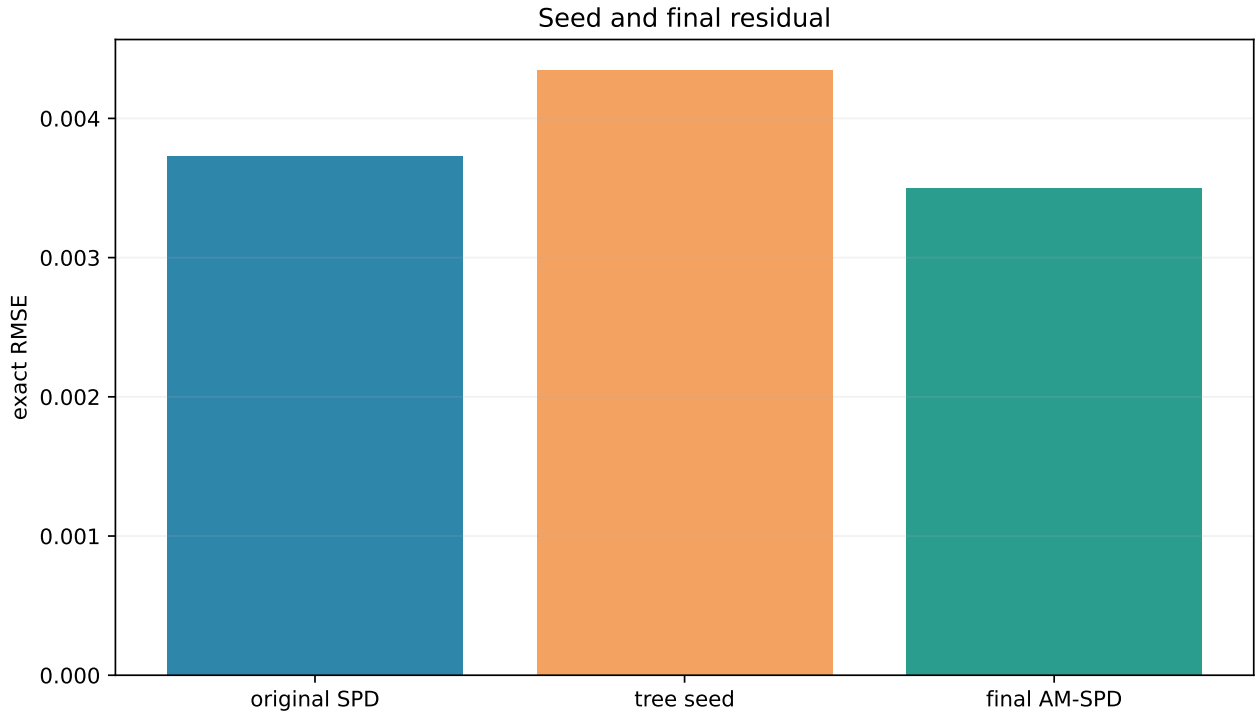


Figure 69: Synthetic-geometry seed and final error.

Reading the figure. The final bar records a strict improvement while staying below the inherited parameter ceiling.

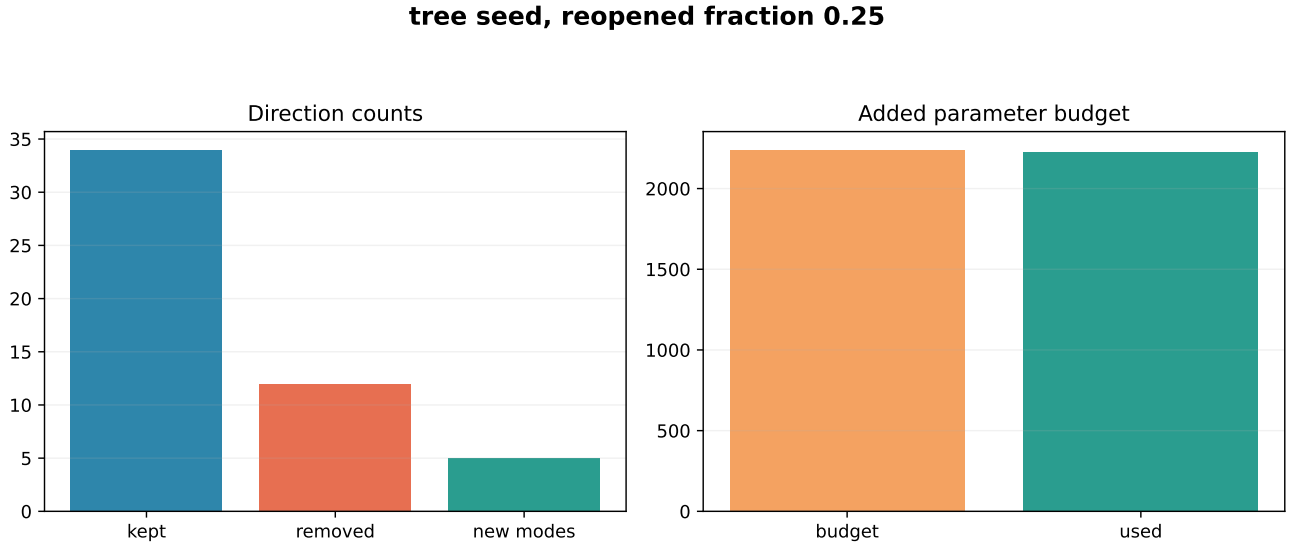


Figure 70: Synthetic-geometry action summary.

Reading the figure. The final representation combines eighteen retained seed directions with six new local components.

C.7 Synthetic text and curves two-seed result

The text-and-curves result starts from the tree seed. Fifteen directions are reopened, and four blocks contribute seven replacement components. The final model uses 2602 of 2639 parameters.

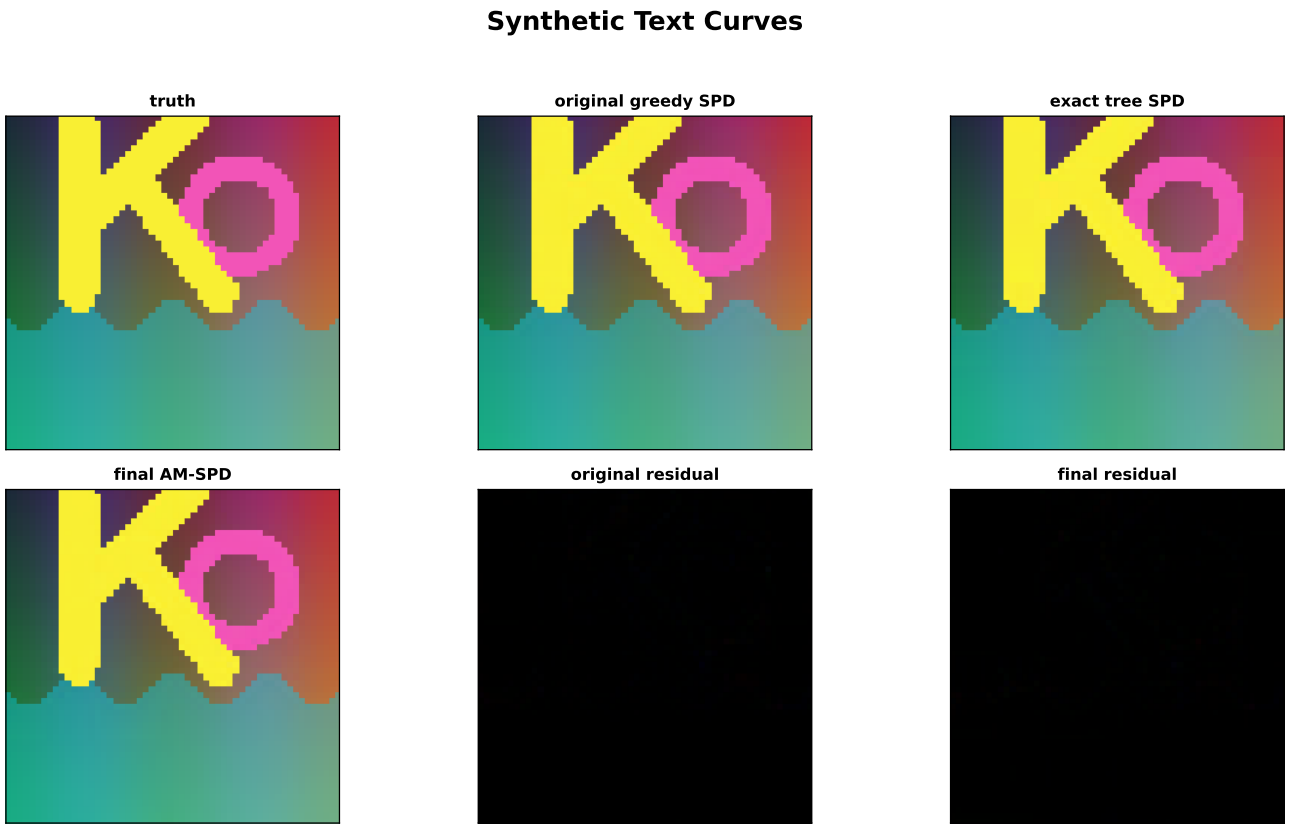


Figure 71: Final text-and-curves reconstruction.

Reading the figure. The exact RMSE falls from 0.003394 to 0.003260. The improvement is smaller than on the natural images because the original geometry-aware representation was already strong.

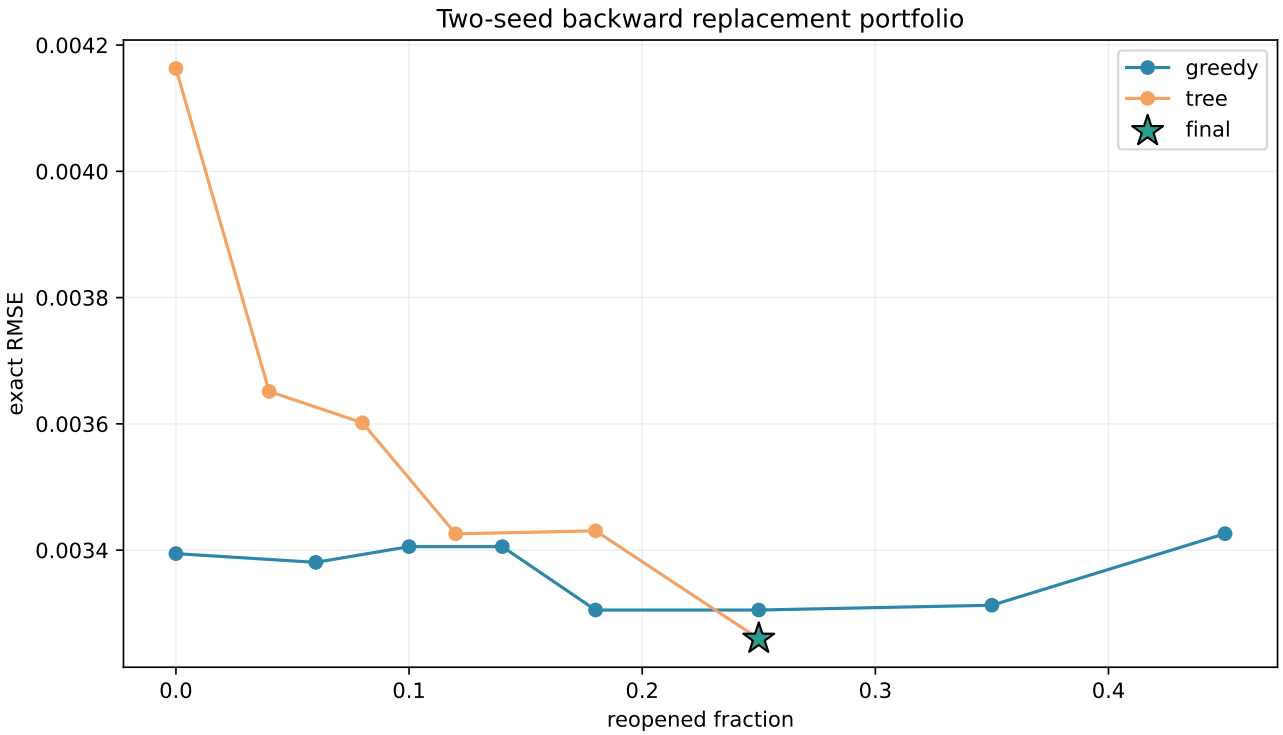


Figure 72: Text-and-curves portfolio path.

Reading the figure. The tree family becomes best after reopening 25% of its added budget.

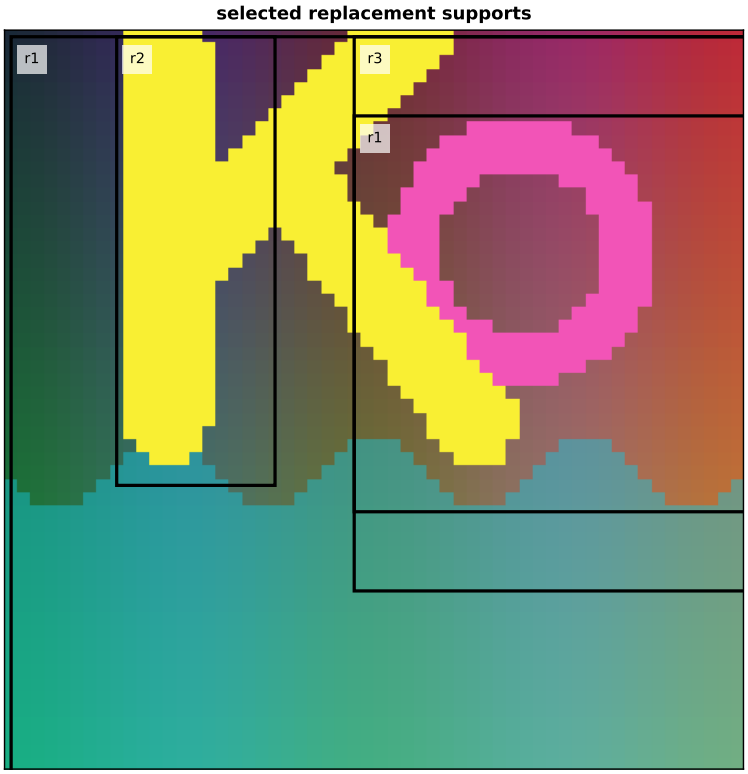


Figure 73: Text-and-curves replacement supports.

Reading the figure. Residual-derived regions concentrate on strokes, curved boundaries, and unresolved local fields.

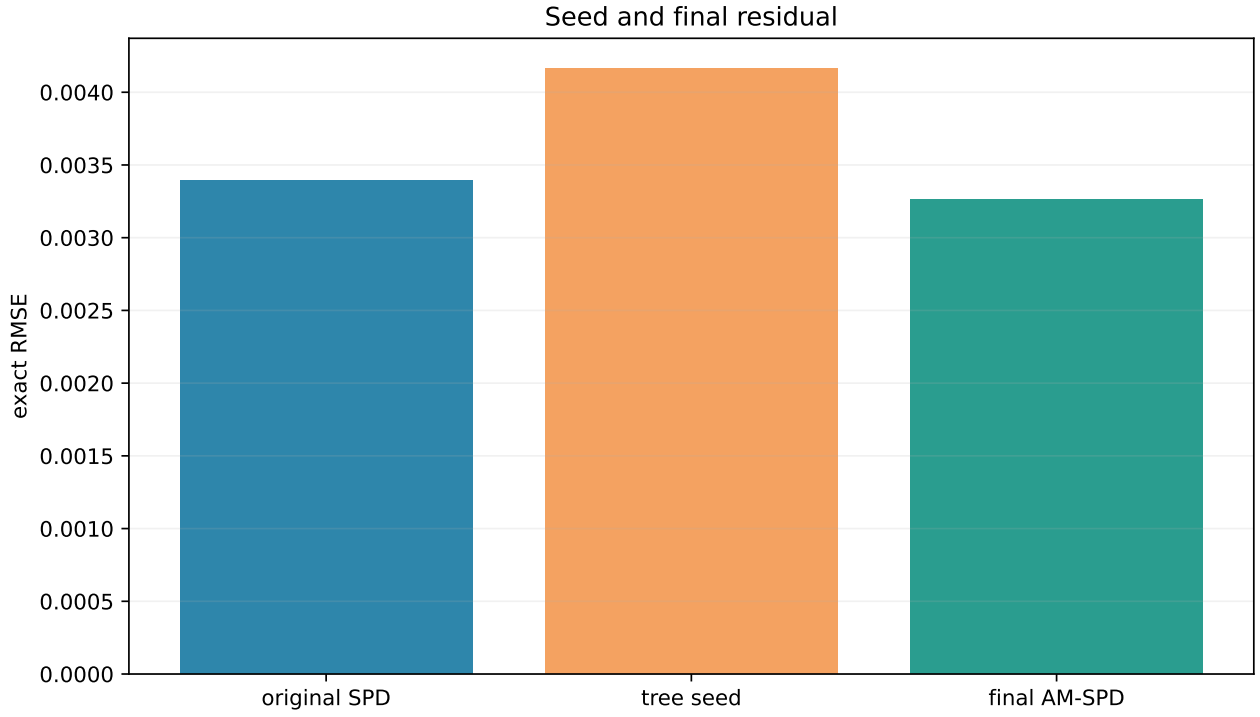


Figure 74: Text-and-curves seed and final error.

Reading the figure. The backward-forward step produces a strict reduction under the original budget.

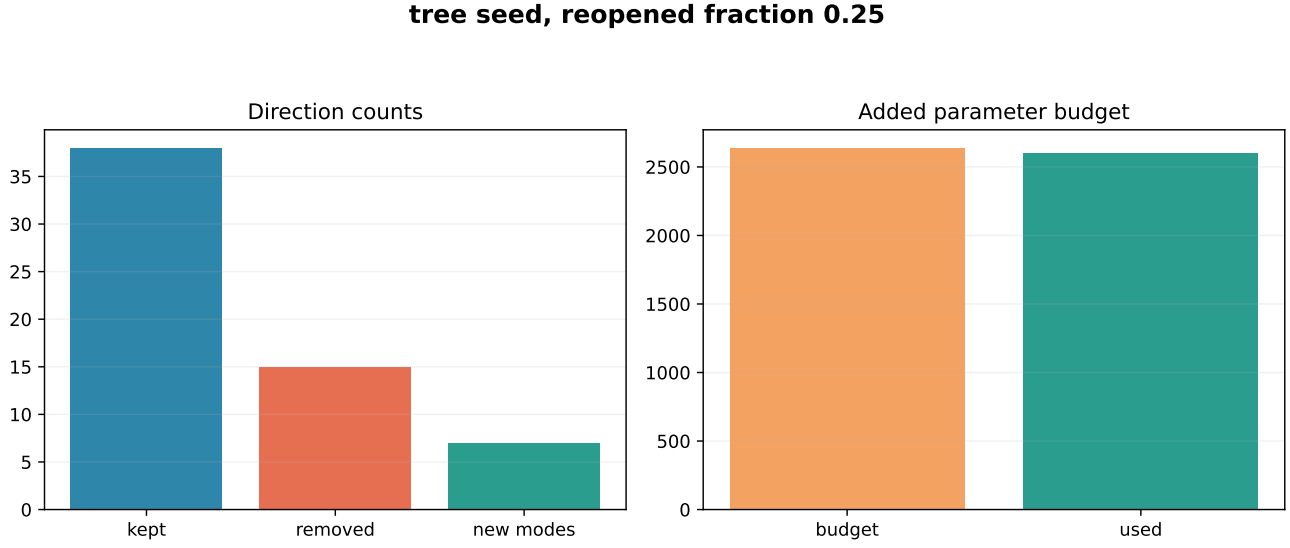


Figure 75: Text-and-curves action summary.

Reading the figure. The final candidate retains thirty-eight tree directions and adds seven replacement components.

C.8 Camera two-seed result

The camera result starts from the tree seed and reopens 12% of the budget. Five directions are removed. Six moving blocks contribute seven new components. The final model uses 2121 of 2123 parameters.



Figure 76: Final camera reconstruction.

Reading the figure. The tree seed lowers RMSE substantially. Backward replacement lowers it again to 0.005738 and raises SSIM to 0.986003.

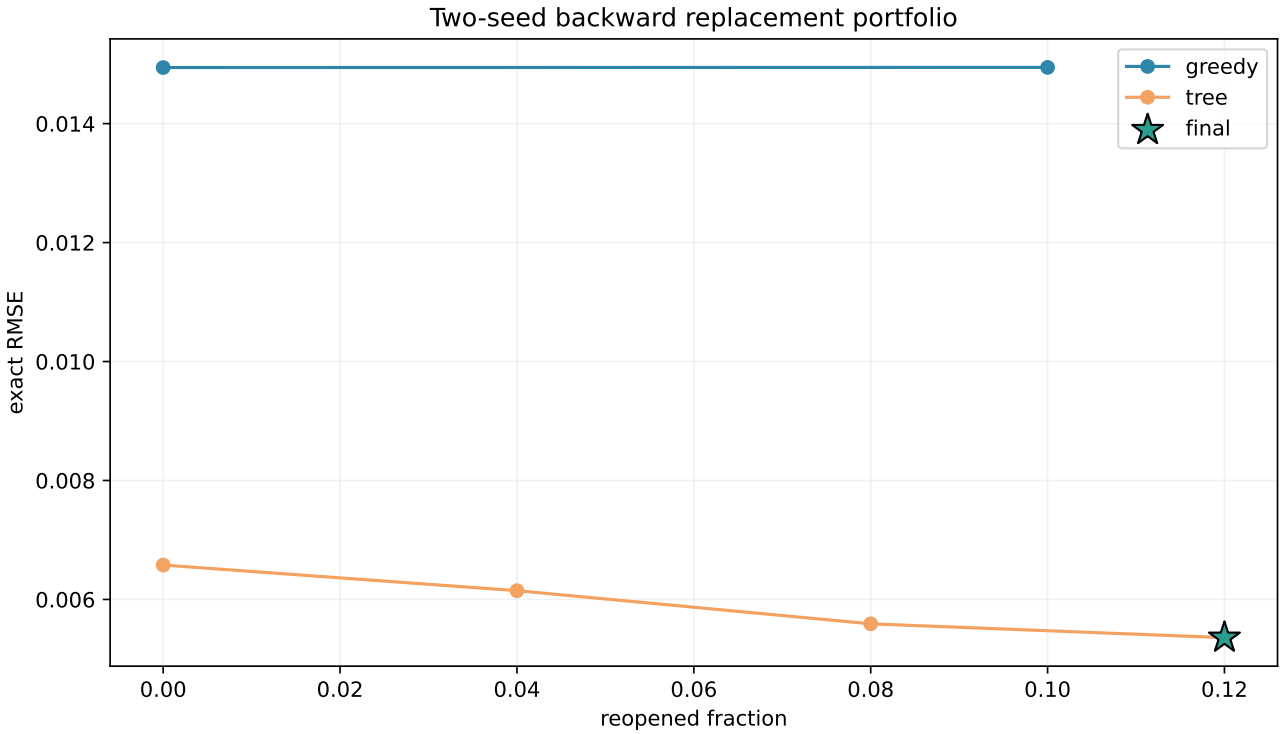


Figure 77: Camera portfolio path.

Reading the figure. The tree family dominates the greedy family throughout the tested range. The best point occurs after a moderate reopening of the tree budget.

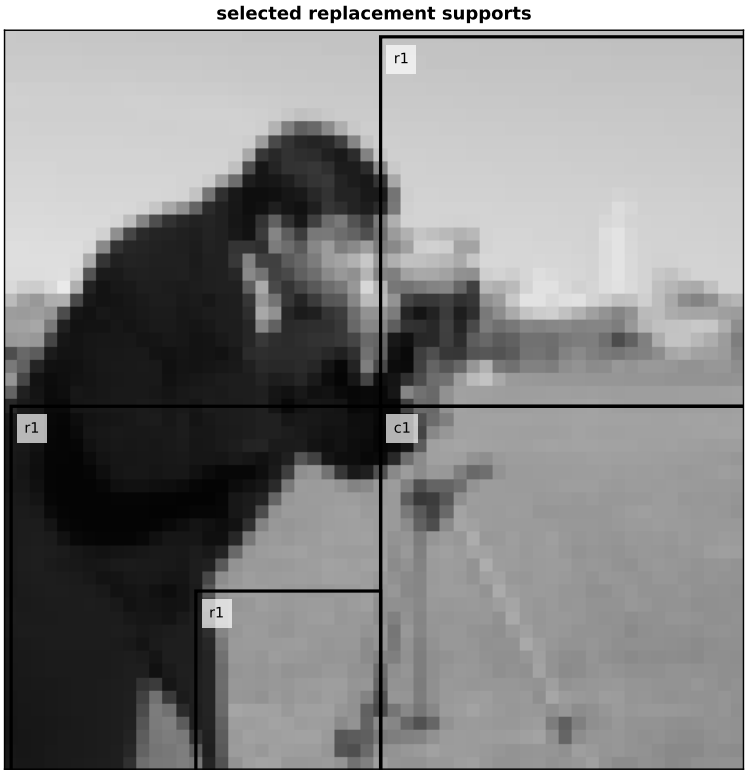


Figure 78: Camera replacement supports.

Reading the figure. The new blocks target residual regions rather than rebuilding the whole image.

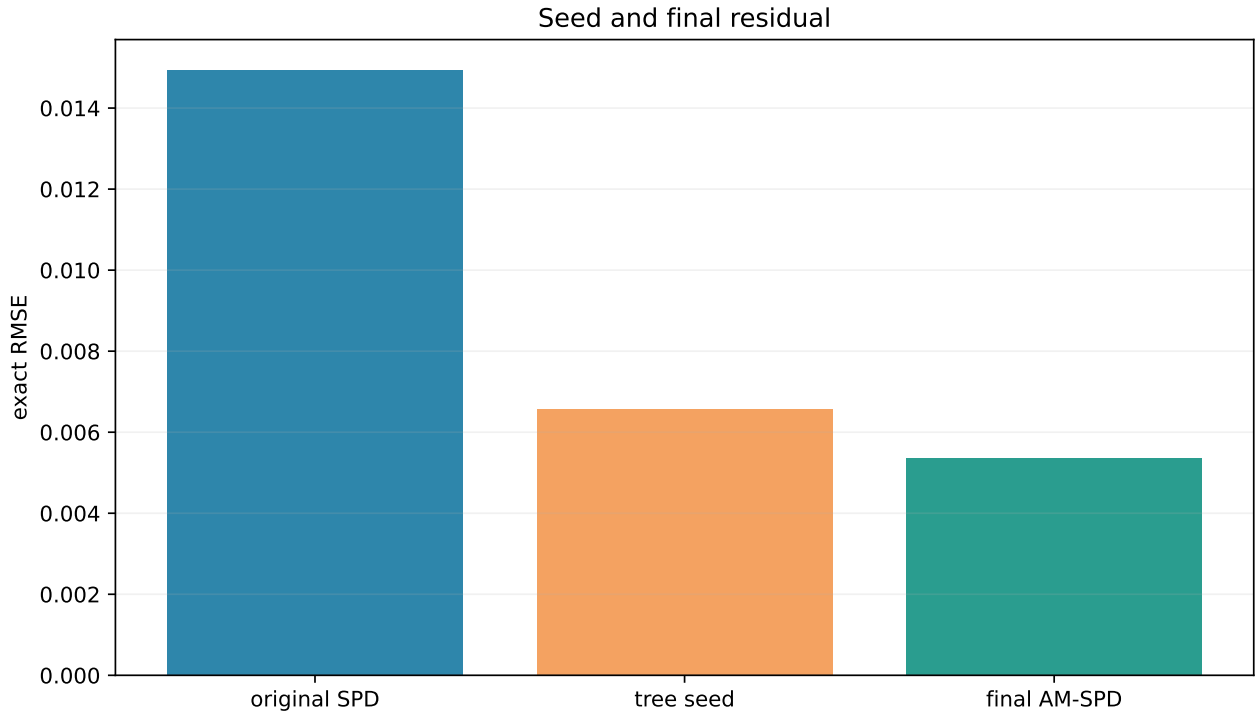


Figure 79: Camera seed and final error.

Reading the figure. The final error is below the original SPD, the tree seed, and the matched residual SVD.

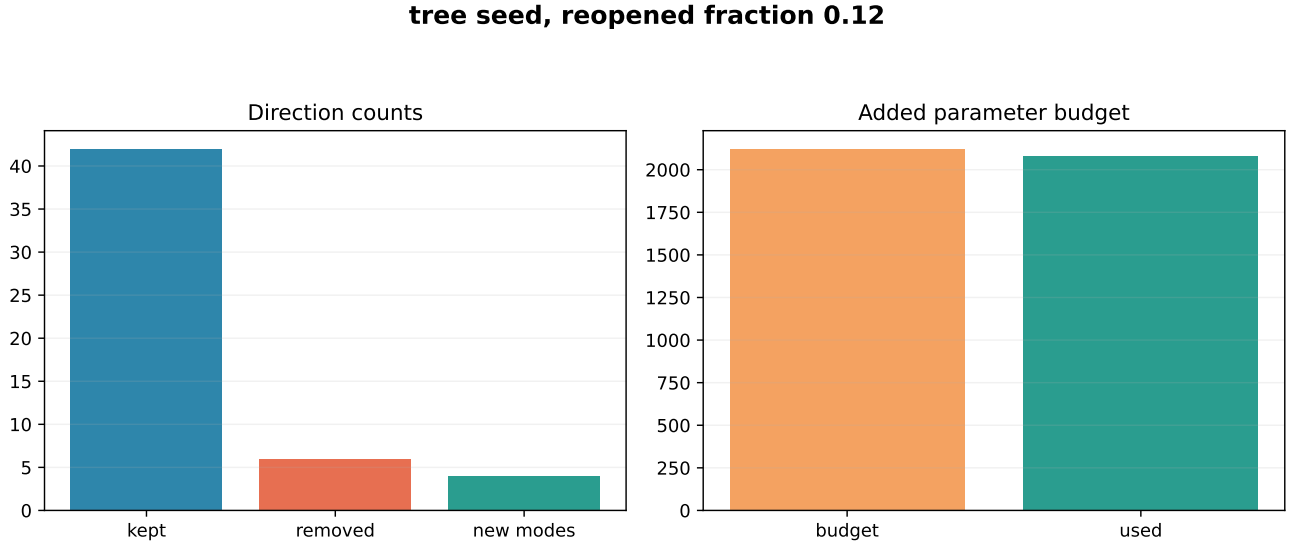


Figure 80: Camera action summary.

Reading the figure. Most tree directions survive. A small number of new local modes produces the final reduction.

C.9 Astronaut two-seed result

The astronaut result also begins from the tree seed and reopens 12% of the budget. Nine directions are removed. Four moving blocks add five components. The final representation uses 5017 of 5060 parameters.

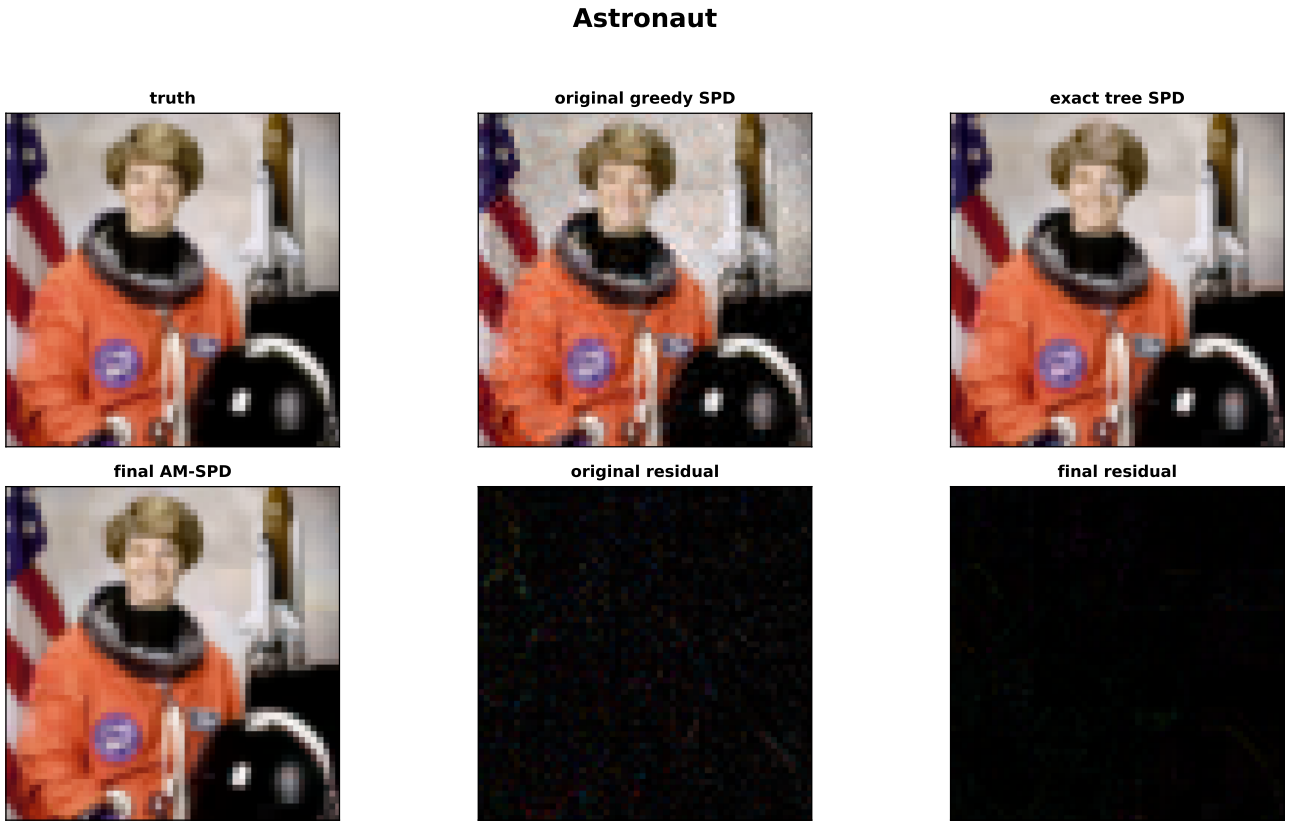


Figure 81: Final astronaut reconstruction.

Reading the figure. Bounded RMSE falls from 0.032976 to 0.010691. The final SSIM is 0.994331.

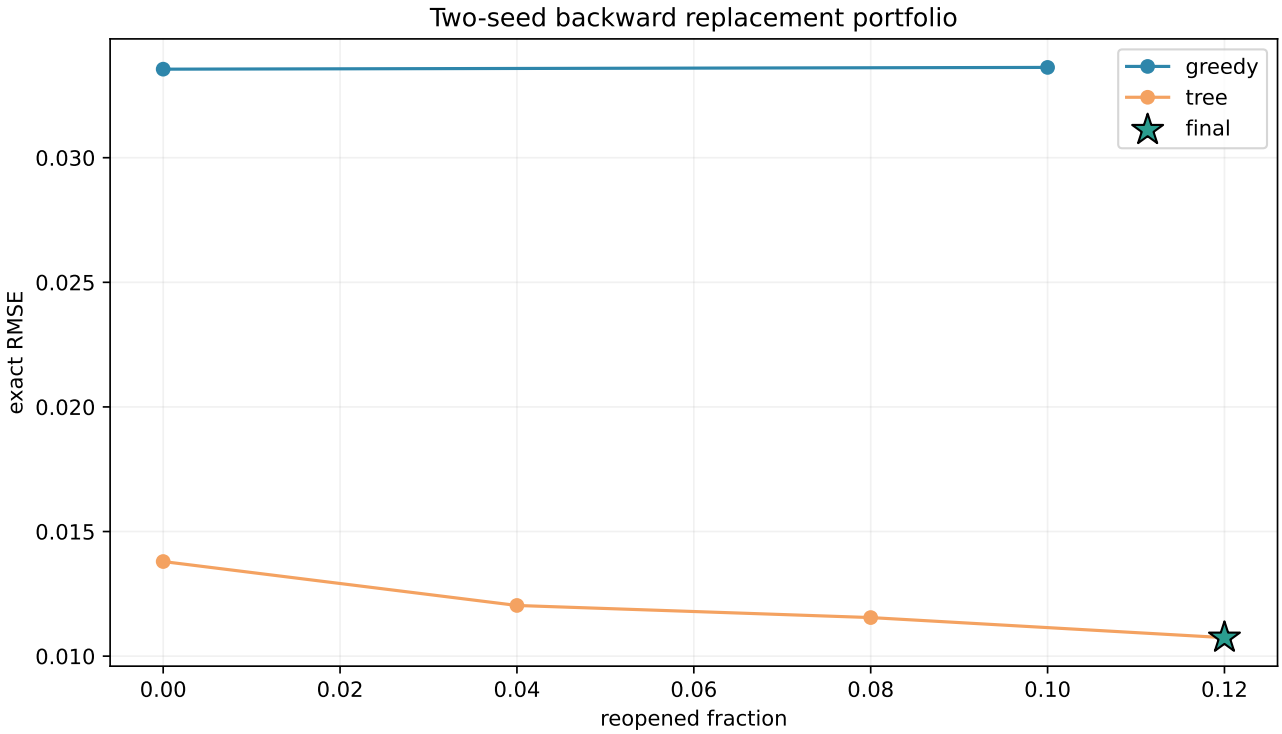


Figure 82: Astronaut portfolio path.

Reading the figure. The exact tree is the stronger initial representation. Backward replacement produces a further large improvement.

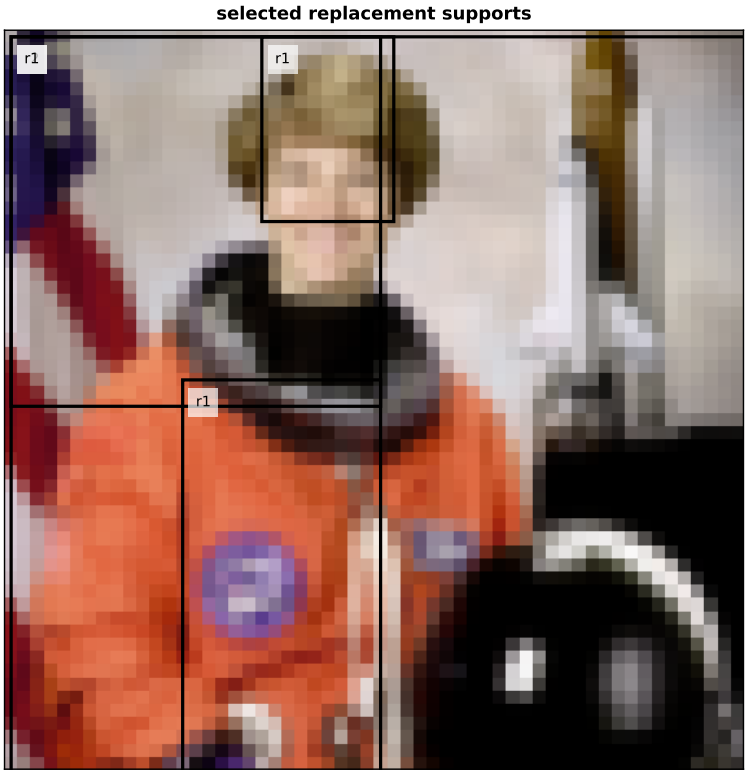


Figure 83: Astronaut replacement supports.

Reading the figure. The selected supports are mixed scale and residual driven.

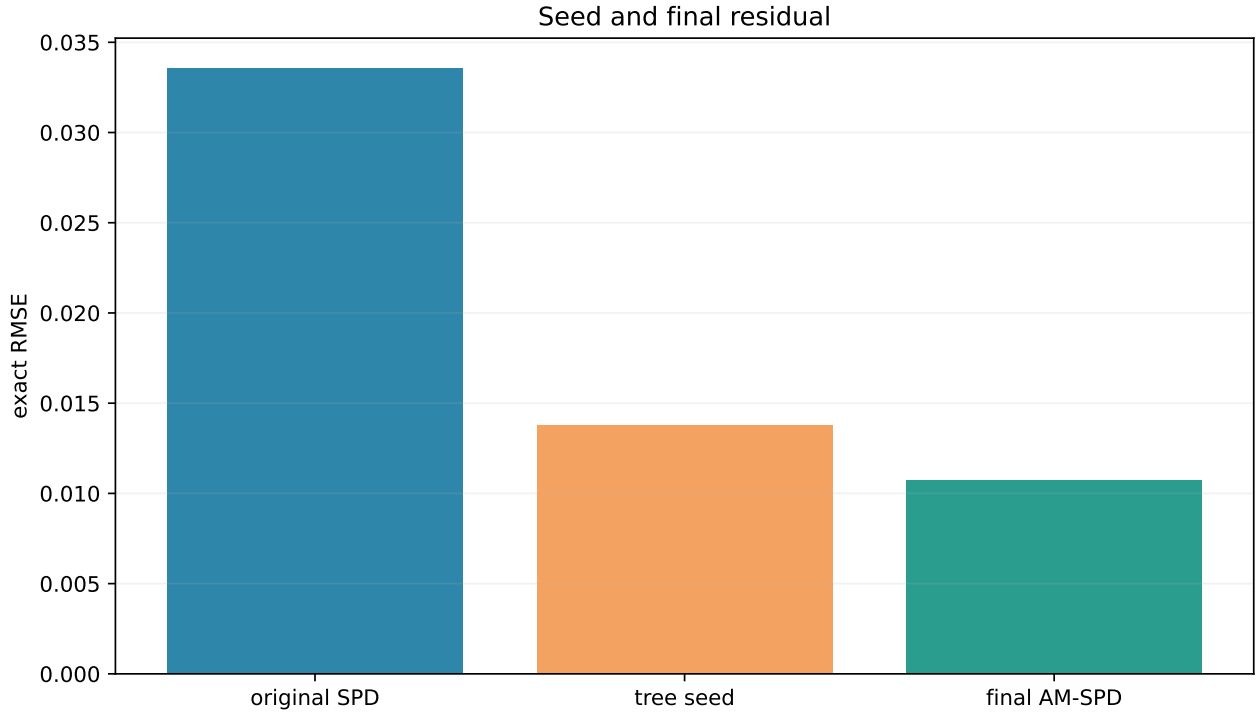


Figure 84: Astronaut seed and final error.

Reading the figure. The final model reduces bounded RMSE by 67.6% relative to the original SPD.

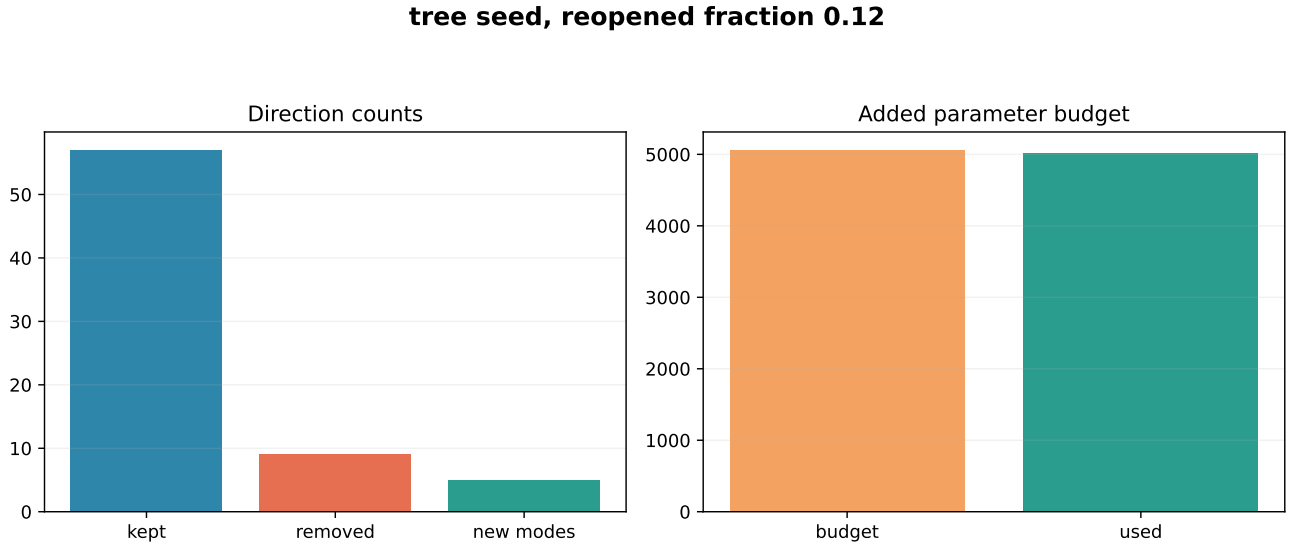


Figure 85: Astronaut action summary.

Reading the figure. The final candidate preserves fifty-seven tree directions and spends the reopened budget on five new components.

C.10 Interpretation of the two-seed stage

The reported portfolio beats the approximately parameter-matched residual SVD in all four reported cases. This does not contradict global SVD optimality. The residual SVD is optimal in one fixed global unfolding and rank. AM-SPD searches a finite union of supports, depths, unfoldings, ranks, seed families, and backward-replacement paths.

The exact tree recursion is globally optimal only for the declared tree. The final selector is globally exact only over the declared finite portfolio. The moving replacement stage remains greedy.

Claim boundary C.4 (Coding boundary)**.** The reported parameter count measures the added singular representation consistently across methods. It does not yet include a complete entropy-coded cost for the inherited geometric packet core, the seed identifier, or every support descriptor.

Claim boundary C.5 (Closure boundary)**.** At every stage,

$$F = \widehat{F} + R_{\text{archive}}.$$

Sparse approximation and exact finite closure remain separate claims.

D Reproducibility and source layout

D.1 Repository organization

The finalized manuscript is distributed with the executable package rather than with source-code listings embedded in the paper. The principal locations are:

- `src/am_spd/`: stable Python package and command-line interface
- `tests/`: package tests for the public API, exact tree, input/output, and command-line interface
- `research/original/`: byte-preserved recovered research snapshot
- `docs/paper/final/`: this manuscript, figures, tables, build scripts, and audit notes

D.2 Reproduction commands

A lightweight package check can be run from the repository root:

```
python -m pip install -e ".[dev,research]"
python -m pytest
am-spd verify
am-spd demo --output-dir demo --size 32
```

The recovered full research ledger can be replayed through the supplied scripts:

```
python research/original/verify_am_spd_final.py
python research/original/generate_am_spd_final_tables.py
python research/original/generate_am_spd_final_figures.py
```

The complete benchmark is more expensive than the verification suite because it repeatedly generates candidate frontiers and globally refits several reconstructions.

D.3 Implementation verification ledger

The principal verification summary appears in Table 5. The complete row-level outputs remain in the repository.

D.4 Interpretation of verification

The analytic proofs establish statements for ideal finite-dimensional least-squares projectors, declared tree recursions, and explicit finite portfolios. The computational suite checks the recovered implementation, its floating-point realization, exact archive identity to tolerance, exhaustive tree cases, and consistency of the reported result files. Neither layer should be substituted for the other.

D.5 Data and artifact availability

The benchmark images are either generated procedurally or loaded from `scikit-image`. Machine-readable JSON and CSV outputs are retained in `research/original/results/` and `research/original/tables/`. Vector figures are generated by Python and retained as PDF assets. The source bundle includes a SHA-256 manifest for the finalized manuscript files.

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